

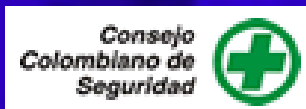


Generator Protection Setting Criteria

Juan M. Gers



Prestación de
los servicios
de Diseño y
estudios
asociados a
sistemas
eléctricos
Certificado
No. 637-1



Content

- ❖ Concepts and protective relaying evolution
- ❖ Functions required in the protection of generators
- ❖ Types of Generator Grounding
- ❖ Schemes for generator protection
- ❖ Setting criteria of generator protection
- ❖ Examples
- ❖ Handling of alarms and oscillographs

Preliminary

- Faults in power systems occur due to a high number of reasons such as:
 - Lightning
 - Aging of insulation
 - Equipment failure
 - Animal presence
 - Rough environmental conditions
 - Branch fall
 - Improper design, maintenance or operation
- The occurrence of faults is not the responsibility of poor protection systems. Protective devices are essential in Power Systems to detect fault conditions, clear them and restore the healthy portion of the systems.

Preliminary

- Protection relays sense any change in the signal which they are receiving, which could be of electrical or mechanical nature.
- Typical electrical protection relays include those that monitor parameters such as voltage, current, impedance, frequency, power, power direction or a ratio of any of the above.
- Typical mechanical protection relays include those that monitor parameters such as speed, temperature, pressure and flow among others.

Teaching Protection Courses

The Education and Training of Future Protection Engineers: Challenges, Opportunities and Solutions

C-6 Working Group Members of Power System Relaying Committee

Venkata, De La Ree, Phadke, McLaren, Thorp, Gers, Hunt, Novosel, Levorin, Sachdev, Napikoski, Bolder, Gopalakrishnan, Brahma, Girgis, Thomas Wiedman, Vahid Madani, Mladen Kezunovic

Abstract It is getting increasingly clear that electric power systems are undergoing rapid changes due to deregulation, the penetration of new technologies and the adoption of efficient computation, communications and control mechanisms. The primary goal of this paper is to recognize the importance of education and training of future protection engineers, and secondly to identify course design and content needed to meet this challenge.

Keywords—power system protection, relaying, deregulation, education and training, engineering curriculum

I. INTRODUCTION AND BACKGROUND

A. Background

Electric energy has not only become a basic necessity for human existence but also has become the backbone for our economic development. We have witnessed electric power systems becoming larger and more complex since last sixty years due to the unprecedented population growth and higher standards demanded by society. Unfortunately these complex systems are becoming more vulnerable to natural disturbances and external threats. If one analyzes the de-

traditions of the electric utility company. In order to capture this aspect of freedom of choice, many view relaying as an art. As a matter of fact, the entire power engineering education curriculum is at a crossroads and needs complete rejuvenation [3-5]. Experience to date has shown that students can be attracted to and retained in power programs if they are exposed early to the joys of creation through design, discovery through research and invention through hands-on experimentation.

B. Purpose

The IEEE Power System Relaying Committee (PSRC) recognizes the importance of proper education and training of future protection engineers. With this primary goal, the Committee proposes the development of materials that will enable successful education of engineers at the university level and the subsequent training in the industry. The education and training should be viewed as a seamless process. This is a challenge that the PSRC has undertaken enthusiastically.

C. Organization of Paper

The history of protection is addressed in Section II. The

Teaching Protection Courses

Fields included in teaching of protection courses

The teaching at the universities should consider that engineers working in power system protection may have to work in one of the following aspect of the discipline.

- (1) Define appropriate protection schemes and prepare specifications for acquisition purposes
- (2) Undertake designs of ac and dc schematic drawings as well wiring diagrams for installation purposes
- (3) Carry out power system studies to define settings and parameters of relays to guarantee the required levels of reliability and selectivity.
- (4) Implement parameters, test and commission protection installations
- (5) Analyze relay performance under events and interpret COMTRADE (commercial name? If so delete?) files

Topics for power system protection courses

The following topics, if adequately covered in a power system protection course, should provide adequate knowledge

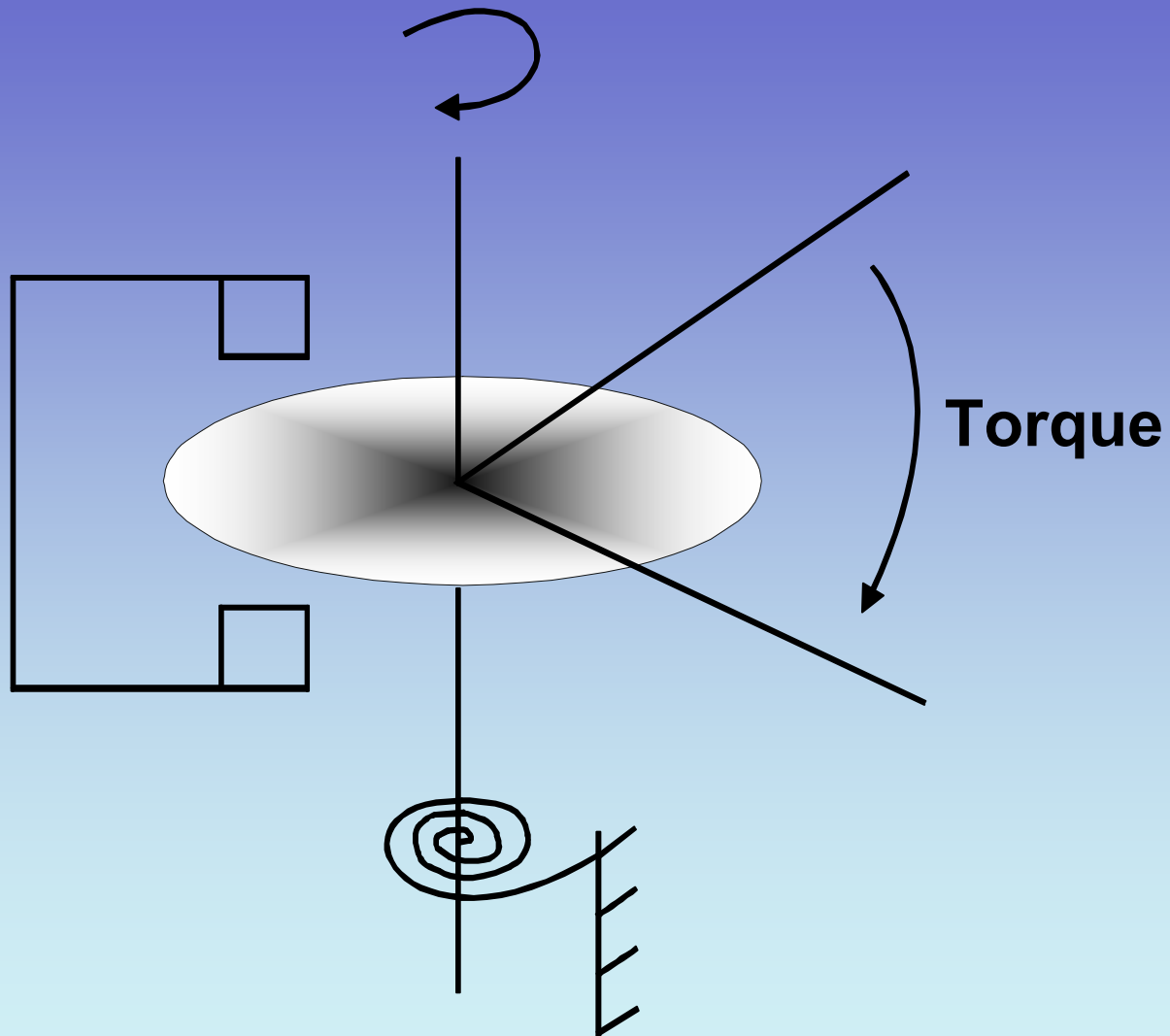
Protection requirements

- Reliability: ability to operate correctly. It has two components:
 - Dependability
 - Security
- Speed: Minimum operating time clear a fault
- Selectivity: maintaining continuity of supply
- Cost: maximum protection at the lowest cost possible

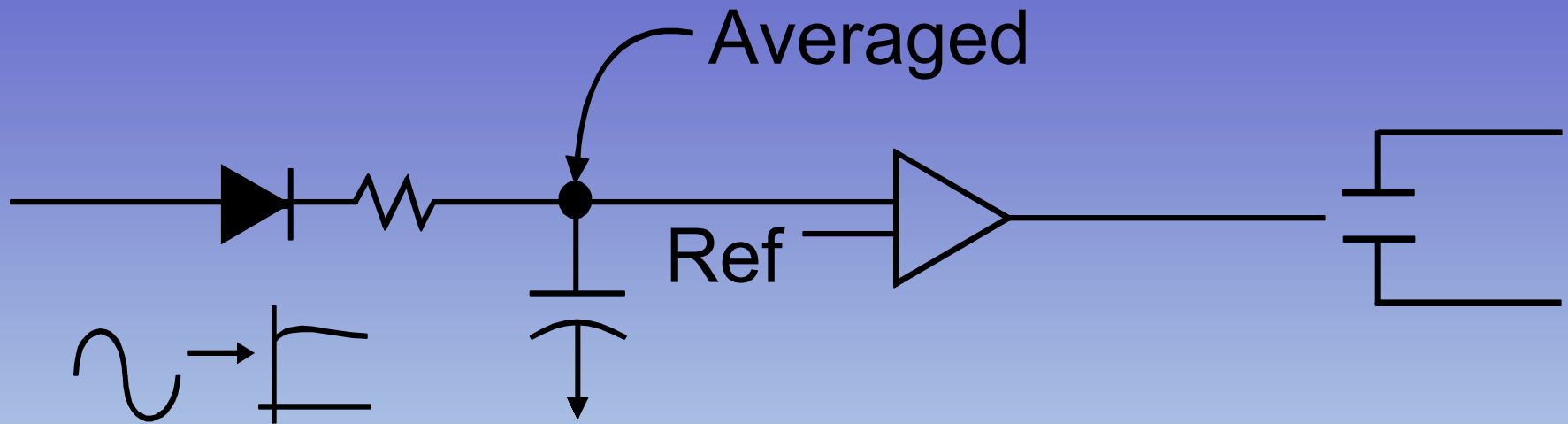
Classification of relays by construction type

- Electromagnetic
- Solid state
- Microprocessor
- Numerical
- Non-electric (thermal, pressure, etc.,)

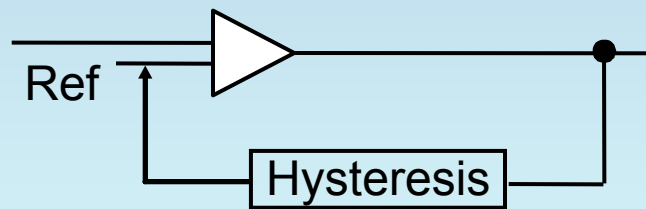
Electromagnetic



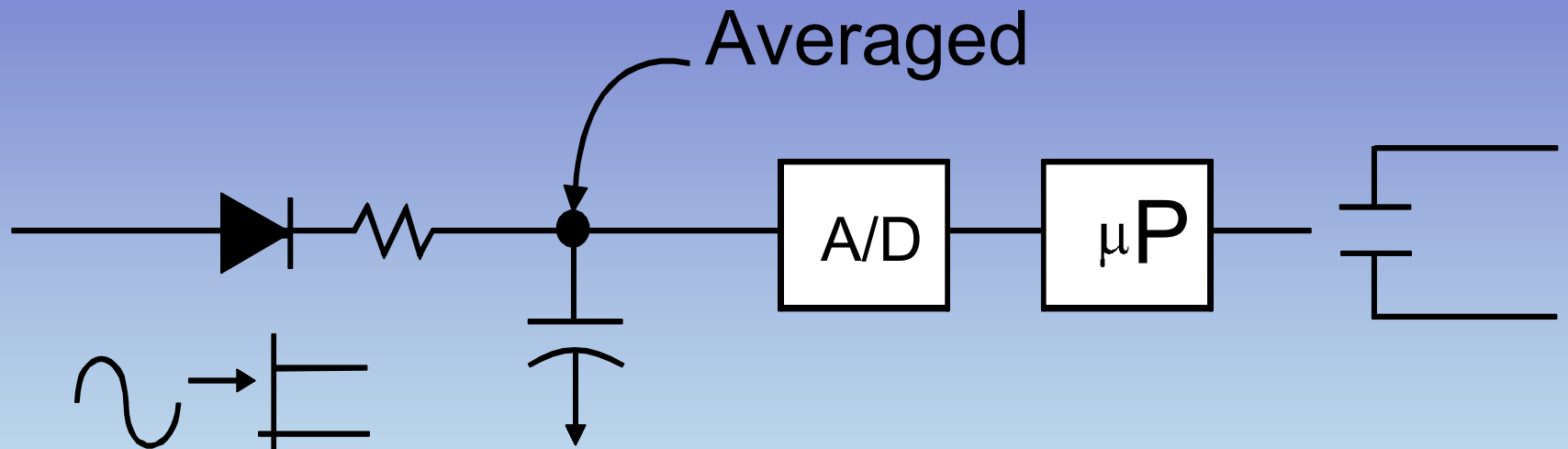
Solid State



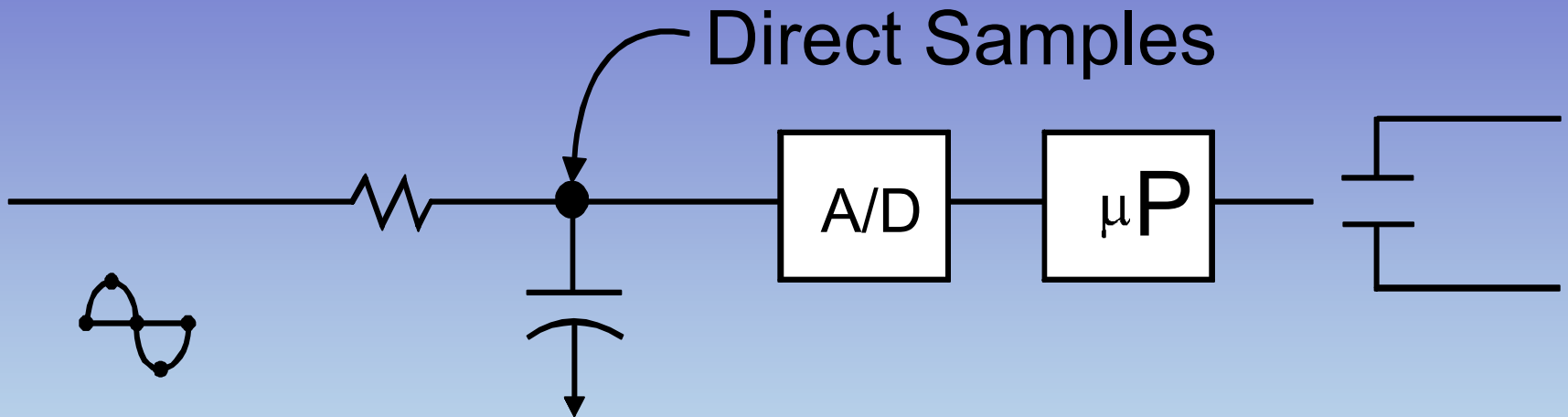
Hysteresis



Microprocessor



Numeric



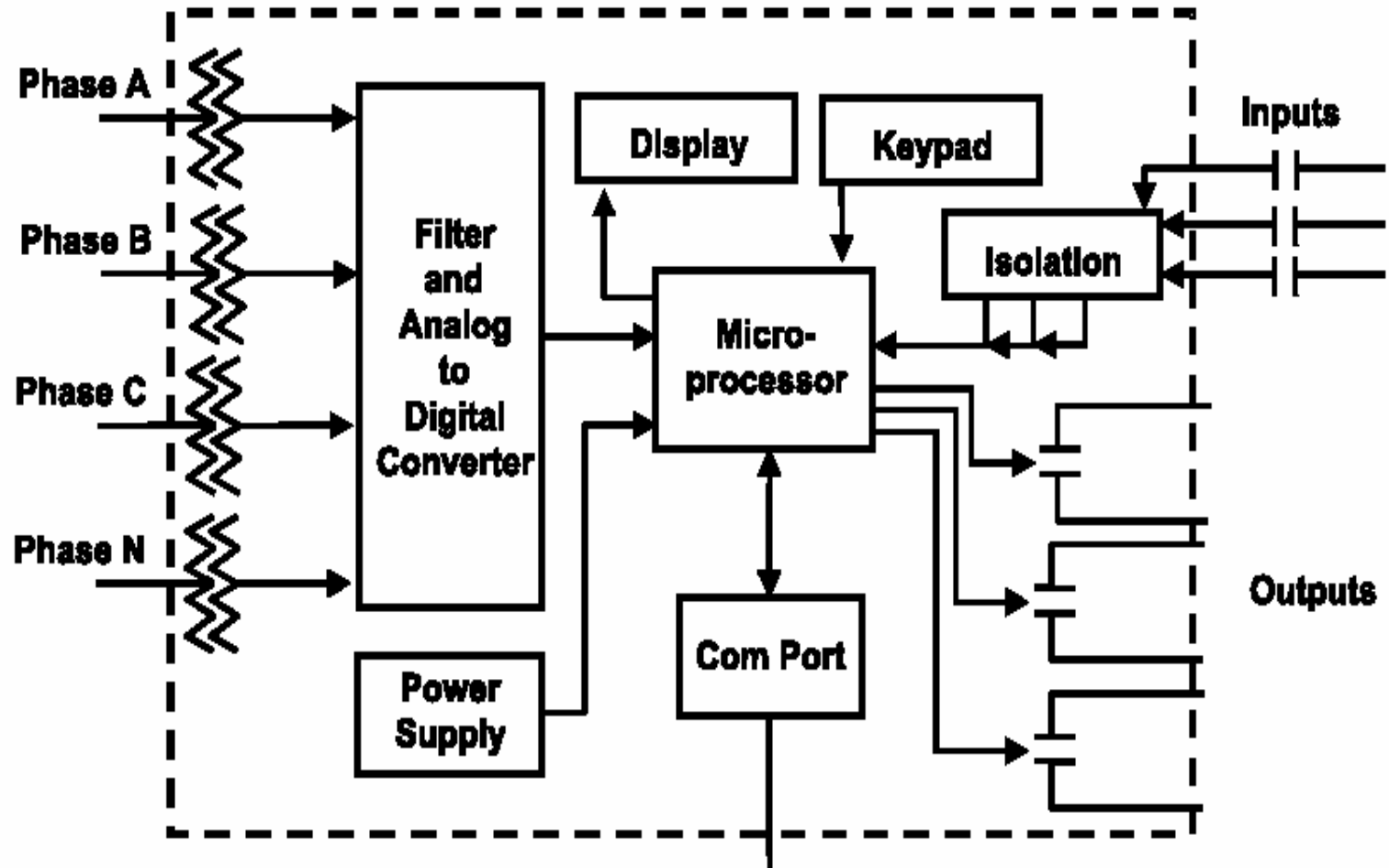
Advantages of numerical relays

- Reliability
- Multifunctionality
- Self-diagnosis
- Event and disturbance records
- Communication capabilities
- Adaptive protection

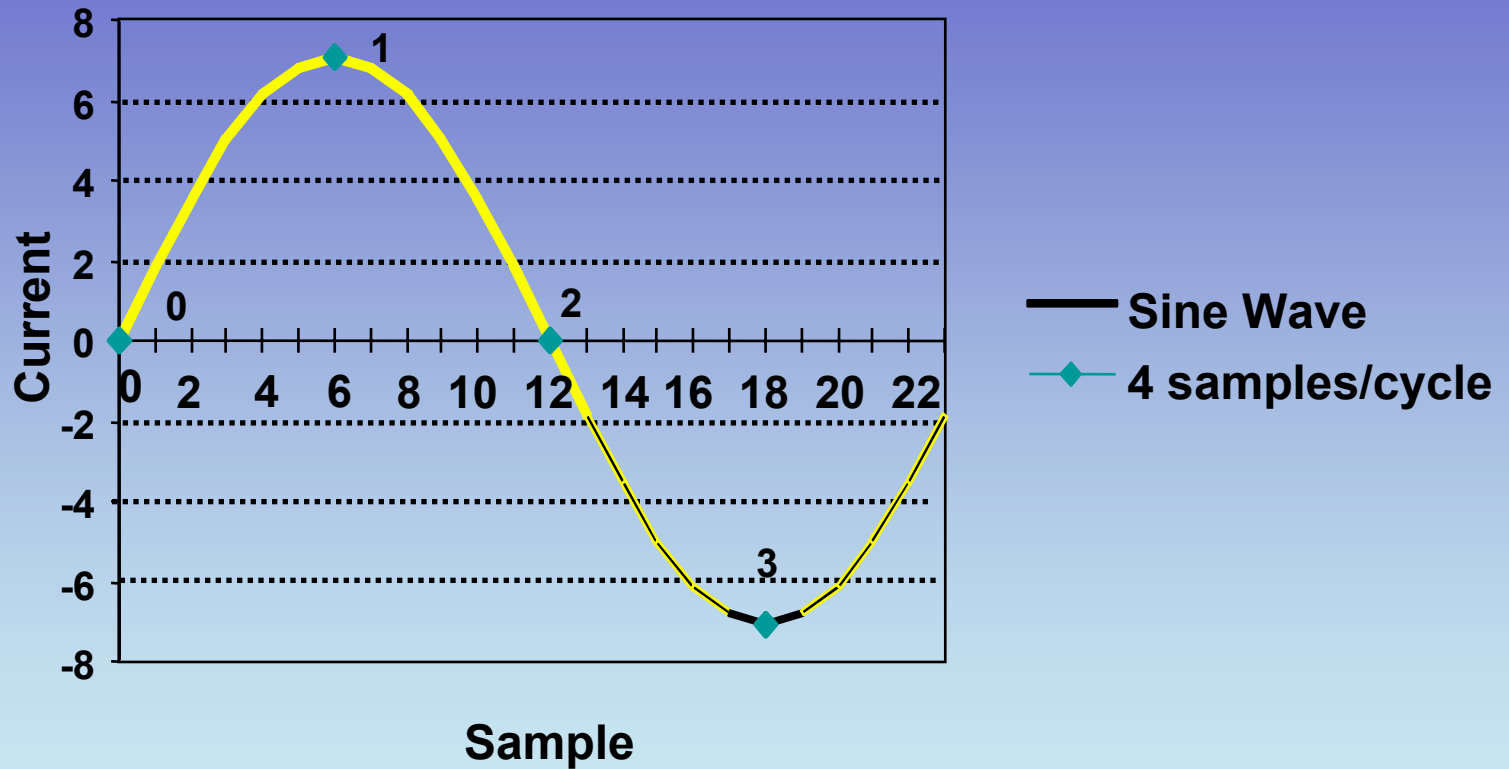
Architecture of numerical relays

- Microprocessor
- Memory module
- Input module
- Output module
- Communication module

Numerical relays



Sampled Waveform



DFT

$$I(n)_{DFT} = \frac{\sqrt{2}}{N} \sum_{k=0}^{N-1} [I_k(\cos(nk \frac{2\pi}{N})) - jI_k(\sin(nk \frac{2\pi}{N}))]$$

N = # samples/cycle fundamental

n = desired harmonic

k = sample index

DFT

For $k = 0$, $n=1$ $\cos(nk\frac{2\pi}{N})=1$ and $\sin(nk\frac{2\pi}{N})=0$

For $k = 1$, $n=1$ $\cos(nk\frac{2\pi}{N})=0$ and $\sin(nk\frac{2\pi}{N})=1$

For $k = 2$, $n=1$ $\cos(nk\frac{2\pi}{N})=-1$ and $\sin(nk\frac{2\pi}{N})=0$

For $k = 3$, $n=1$ $\cos(nk\frac{2\pi}{N})=0$ and $\sin(nk\frac{2\pi}{N})=-1$

$$I_{DFT} = \frac{\sqrt{2}}{N} (I_0 - jI_1 - I_2 + jI_3)$$

ANSI/IEEE device identification

No.	DESCRIPTION
2	Time-delay relay
21	Distance relay
24	Overexcitation / Volts per Hertz
25	Synchronism-check relay
27	Undervoltage relay
27TN	Third-Harmonic Undervoltage relay
30	Annunciator device
32	Reverse power relay
37	Undercurrent or underpower relay
40	Field excitation relay
46	Negative sequence overcurrent relay
47	Negative sequence overvoltage relay
49	Thermal relay
50	Instantaneous AC overcurrent relay
50DT	Split Phase Differential
50/27	Inadvertent Energizing
50BF	Breaker Failure
51	AC Inverse Time Overcurrent relay
52	Circuit breaker
59	Overvoltage relay
59D	Third-Harmonic Voltage Differential Ratio

No.	DESCRIPTION
60	Voltage balance or loss of potential relay
63	Pressure device
64F	Field Ground relay
64B	Brush Lift-Off Detection
64S	100% Stator Ground Protection by Low Frequency Injection
67	AC directional overcurrent relay
68	Power Swing Blocking
69	Permissive relay
74	Alarm relay
76	DC overcurrent relay
78	Out-of-step relay
79	AC reclosing relay
81	Frequency relay
81R	Rate of Change Frequency relay
83	Transfer device
85	Carrier or pilot-wire relay
86	Lock out relay
87	Differential relay
94	Auxiliary tripping relay

Review of Grounding Techniques

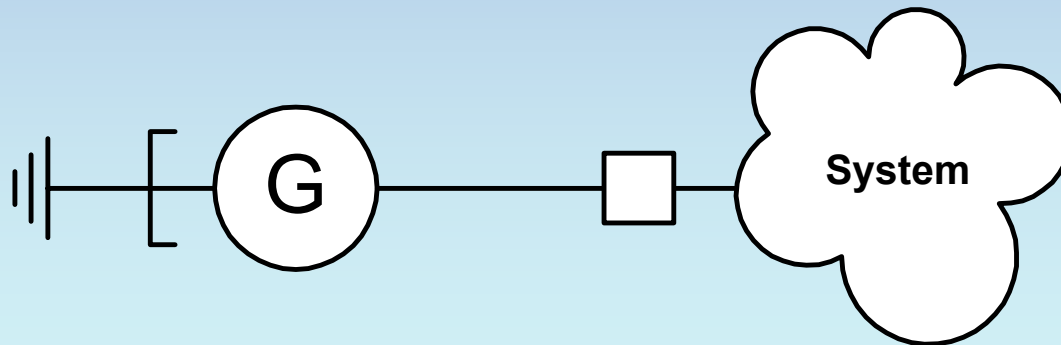
Why Ground?

- Safety
- Ability to detect less harmful (hopefully) phase-to-ground fault before phase-to-phase fault occurs
- Limit damage from ground faults
- Stop transient overvoltages
- Provide ground source for other system protection (other zones)

Types of Generator Grounds

No Impedance

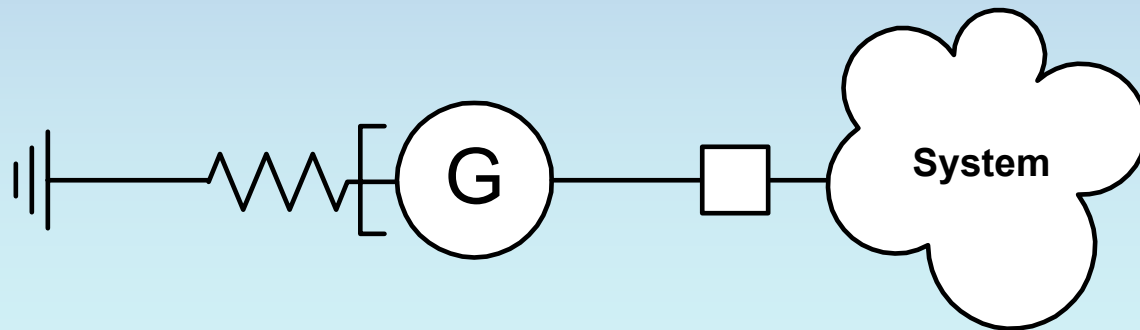
- Cheap
- Usually done only on small generators
- Definitely a good ground source
- Generator likely to get damaged on internal ground fault



Types of Generator Grounds

Low Impedance

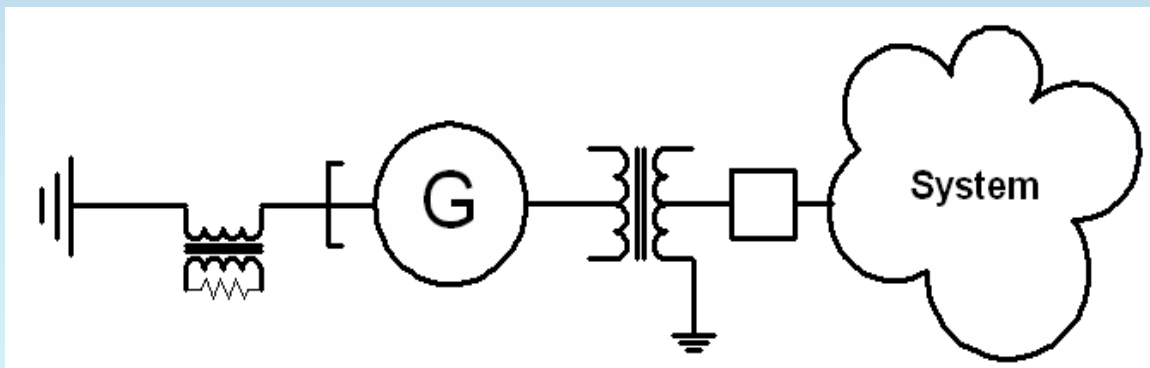
- Can get expensive as resistor size increases
- Usually a good ground source
- Generator still likely to be damaged on internal ground fault
- Ground fault current typically 200-400 A



Types of Generator Grounds

High Impedance

- Moderately expensive
- Used when generators are unit connected
- System ground source obtained from unit xfmr
- Generator damage minimized or mitigated from ground fault
- Ground fault current typically $\leq 10A$



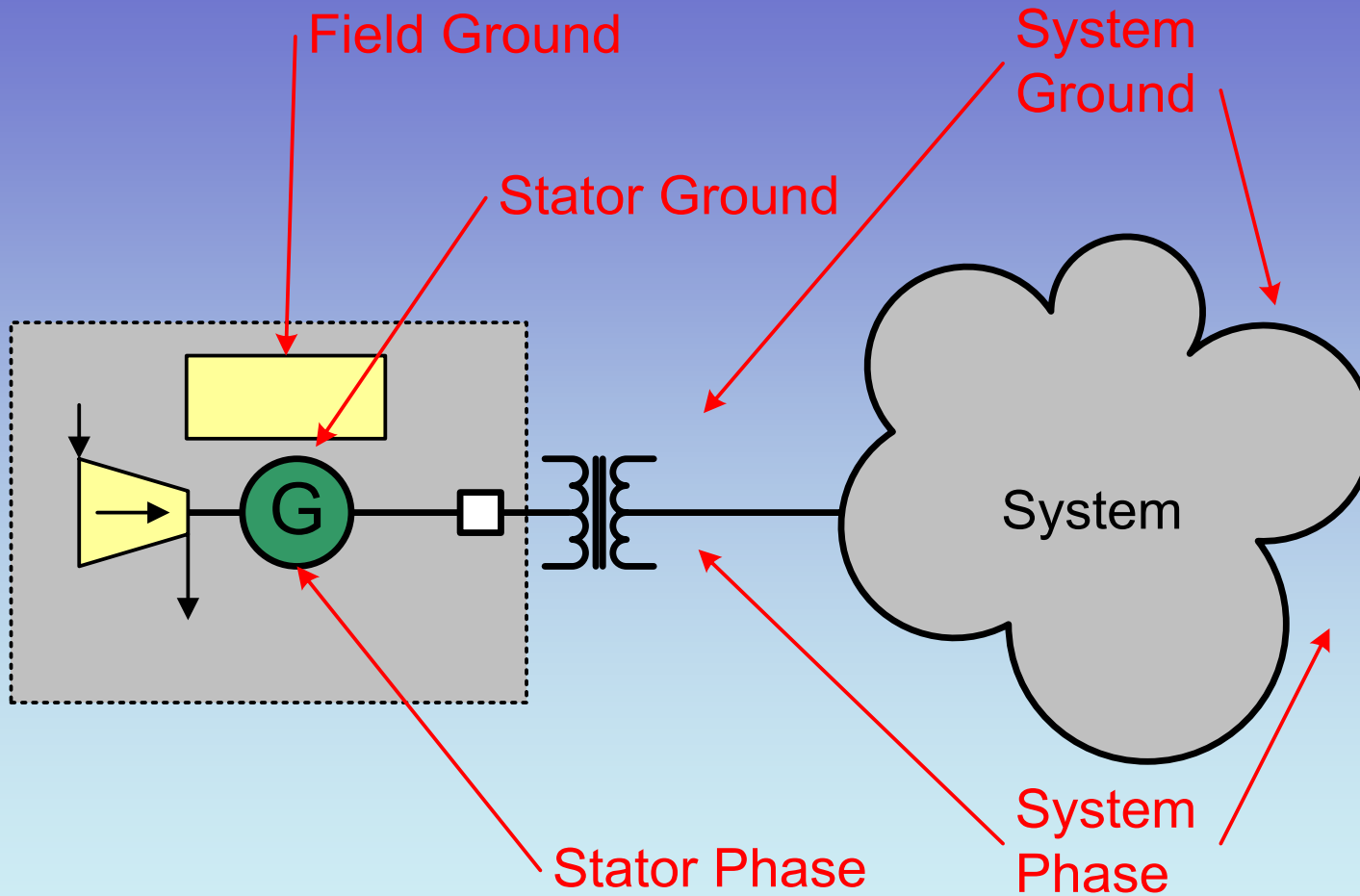
Types of Generator Grounds

Hybrid Impedance

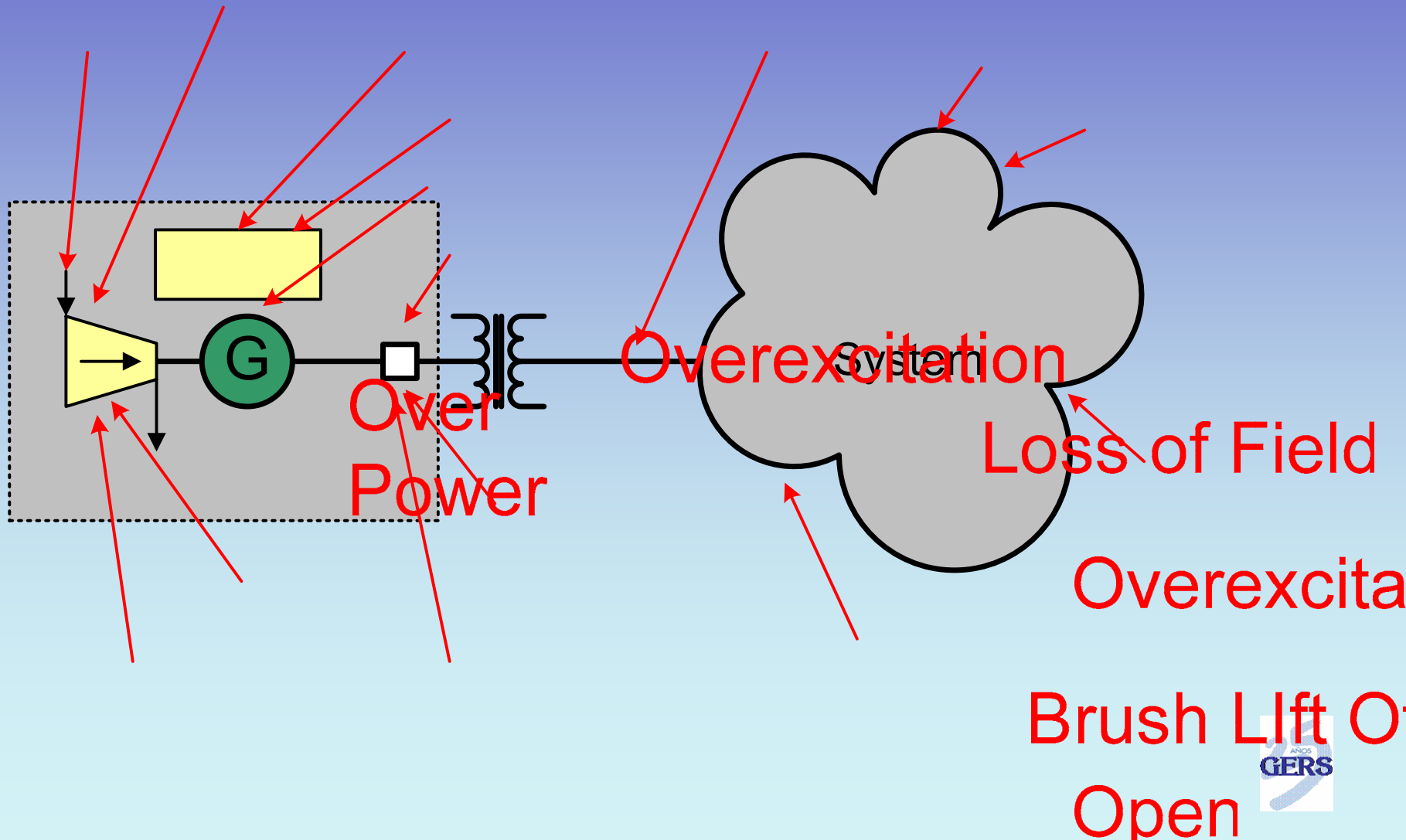
- Combines advantages of Low Z and High Z ground
- Low Z ground provides ground source for normal conditions
- If an internal ground fault (in the generator) is detected by the 87GD element, the Low Z ground path is opened, leaving only the High Z ground path
- The High Z ground path limits fault current to approximately 10A (saves generator!)

The diagram illustrates a power system configuration. On the left, a transformer is connected to a busbar. The busbar has three outgoing lines labeled 52 F1, 52 F2, and 52 F3, each with a 51 N relay. The busbar is also connected to a generator (G) through a 52 G relay. The generator is connected to a load (VS) and a transformer. The load is connected to a 51 G relay. The transformer is connected to a 59 N relay. The diagram shows the interconnection of these components and the flow of signals between them.

Generator Protection: Faults



Generator Protection: Abnormal Conditions



New Std C37.102-2005

IEEE PC37.102/D6 Draft Guide for AC Generator Protection

Prepared by the Revision of AC Generator Protection Guide Working Group of the
Rotating Machinery Protection Subcommittee

Abstract: A review of the generally accepted forms of relay protection for the synchronous generator and its excitation system is presented. This guide is primarily concerned with protection against faults and abnormal operating conditions for large hydraulic, steam, and combustion-turbine generators.

Keywords: ac generator protection, relay protection, synchronous generator.

Copyright © 2004 by the Institute of Electrical and Electronics Engineers, Inc.
Three Park Avenue
New York, NY 10016-5997, USA
All Rights Reserved.

This document is an unapproved draft of a proposed IEEE Standard. As such, this document is subject to change. USE AT YOUR OWN RISK! Because this is an unapproved draft, this document must not be utilized for any conformance/compliance purposes. Permission is hereby granted for IEEE Standards Committee participants to reproduce this document for purposes of IEEE standardization activities only. Prior to submitting this document to another standards development organization for standardization activities, permission must first be obtained from the Manager, Standards Licensing and Contracts, IEEE Standards Activities Department. Other entities seeking permission to reproduce this document, in whole or part, must obtain permission from the Manager, Standards Licensing and Contracts, IEEE Standard Activities Department.

IEEE Standards Department
Copyrights and Permissions
445 Hoes Lane, P.O. Box 1331
Piscataway, NJ 08855-1331, USA



New Std C37.102-2005

Introduction (This introduction is not a part of PC37.102, Guide for AC Generator Protection.)

IEEE Std C37.102, Guide for AC Generator Protection, was initially published in 1987. It was subsequently revised in 1995 and reaffirmed in 2002. The Guide is designed for the protection of typical steam, hydraulic and combustion-turbine generators. Schemes that are judged to be good alternative practice for generator protection are included in the Guide. New schemes that have gained acceptance and usage have been added to the Guide.

In this revision of IEEE C37.102-1995, several areas were improved. Among the most notable:

- A new clause (Clause 6) on Multifunction Generator Protection Systems has been added.
- A new Annex (Annex A) Typical Relay Setting Calculations has been added.

Participants

At the time this of this guide was completed, the Revision of AC Generator Protection Guide Working Group has the following membership:

Murty V.V.S. Yalla, Chair

Kevin Stephan, Vice Chair

Munnu Bajpai
Zeeky Bukhala
Ratan Das
Wayne Hartmann
Prem Kumar
Subhash Patel
Michael Reichard
Joe Uchiyama

Matt Basler
Terry Crawley
Everett Fennell
Patrick Kerrigan
Rai Marttila
Robert Pettigrew
Sam Sambasivan
W. Phil Waudby

Gabriel Benmouyal
Albert Darlington
Jon Gardell
Shoukat Khan
Charles Mozina
R.Vital Rebbapragada
Michael Thompson

What's new in Std C37.102-2005

Section 6 – Multifunction Generator Protection Systems

- Digital technology offers several additional features which could not be obtained in one package with earlier technology
- These features include:
 - Metering of voltages, currents, power and other measurements
 - Oscillography
 - Sequence of events capture with time tagging
 - Remote setting and monitoring through communications
 - User configurability of tripping schemes and other control logic
 - Low burden on the PT's and CT's
 - Continuous self-checking and ease of calibration

What's new in Std C37.102-2005

6.2.1 Protective Functions

- 87G – Generator Phase Differential
- 87GN – Generator Ground Differential
- 59G Stator Ground
- 100% Stator Ground
 - 27TH - Third Harmonic Neutral Undervoltage
 - 59TH – Third Harmonic Voltage Ratio or Differential
 - 64S – Sub-harmonic Voltage Injection
- 46 – Current Unbalance/Negative Sequence

What's new in Std C37.102-2005

- 24 – Overexcitation
- 27 – Undervoltage
- 59 – Overvoltage
- 81U – Underfrequency
- 81O – Overfrequency
- 32 – Reverse Power or Directional Power
- 49 – Thermal Protection
- 51 – Overcurrent
- 51VC/51VR or 21 – System Backup

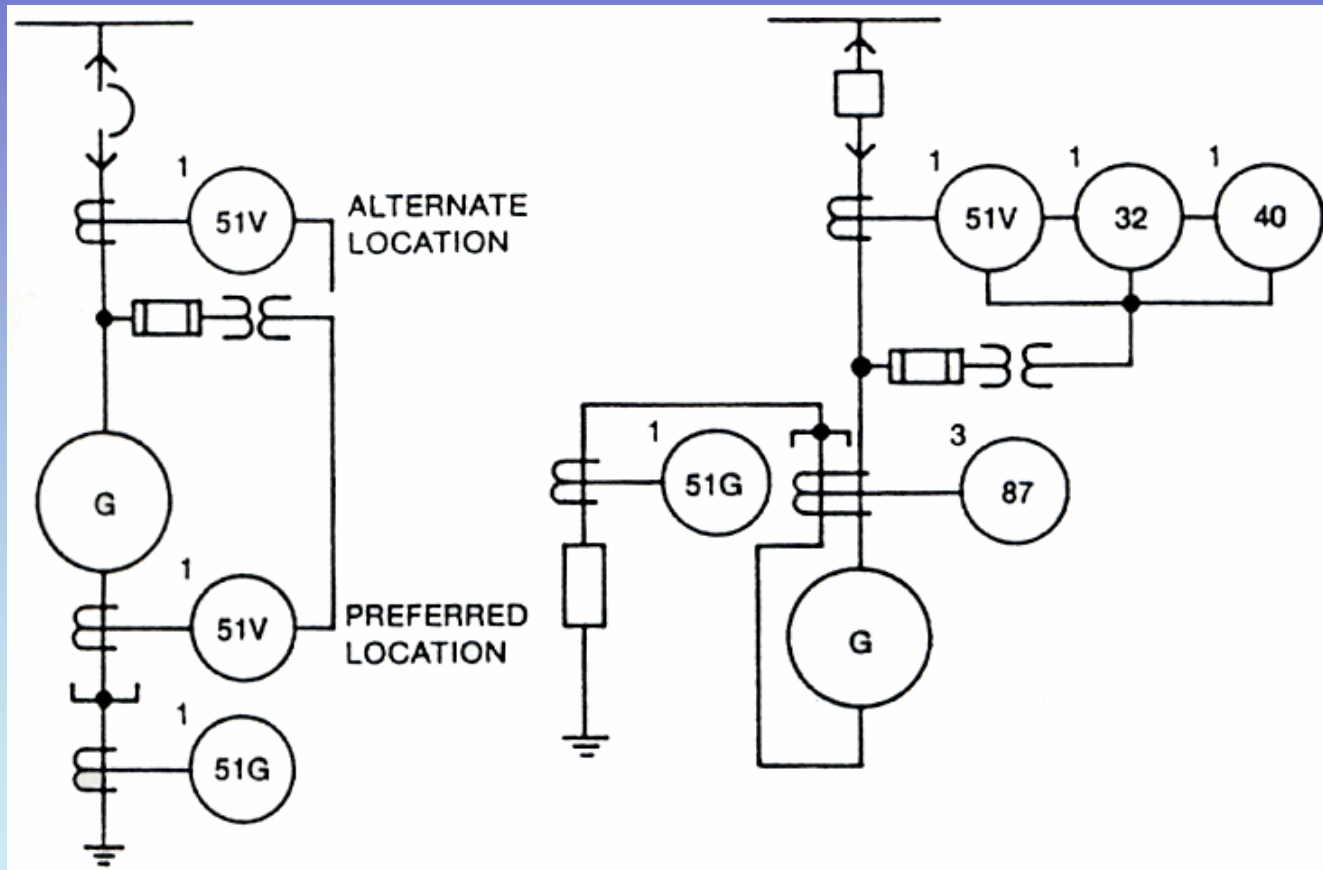
What's new in Std C37.102-2005

- 60 – Loss of Voltage
- 78 – Out-of-Step
- 64F – Field Ground
- Additional functions that may be provided include:
 - Sequential Trip Logic
 - Accidental Energization
 - Open Breaker Detection

What's new in Std C37.102-2005

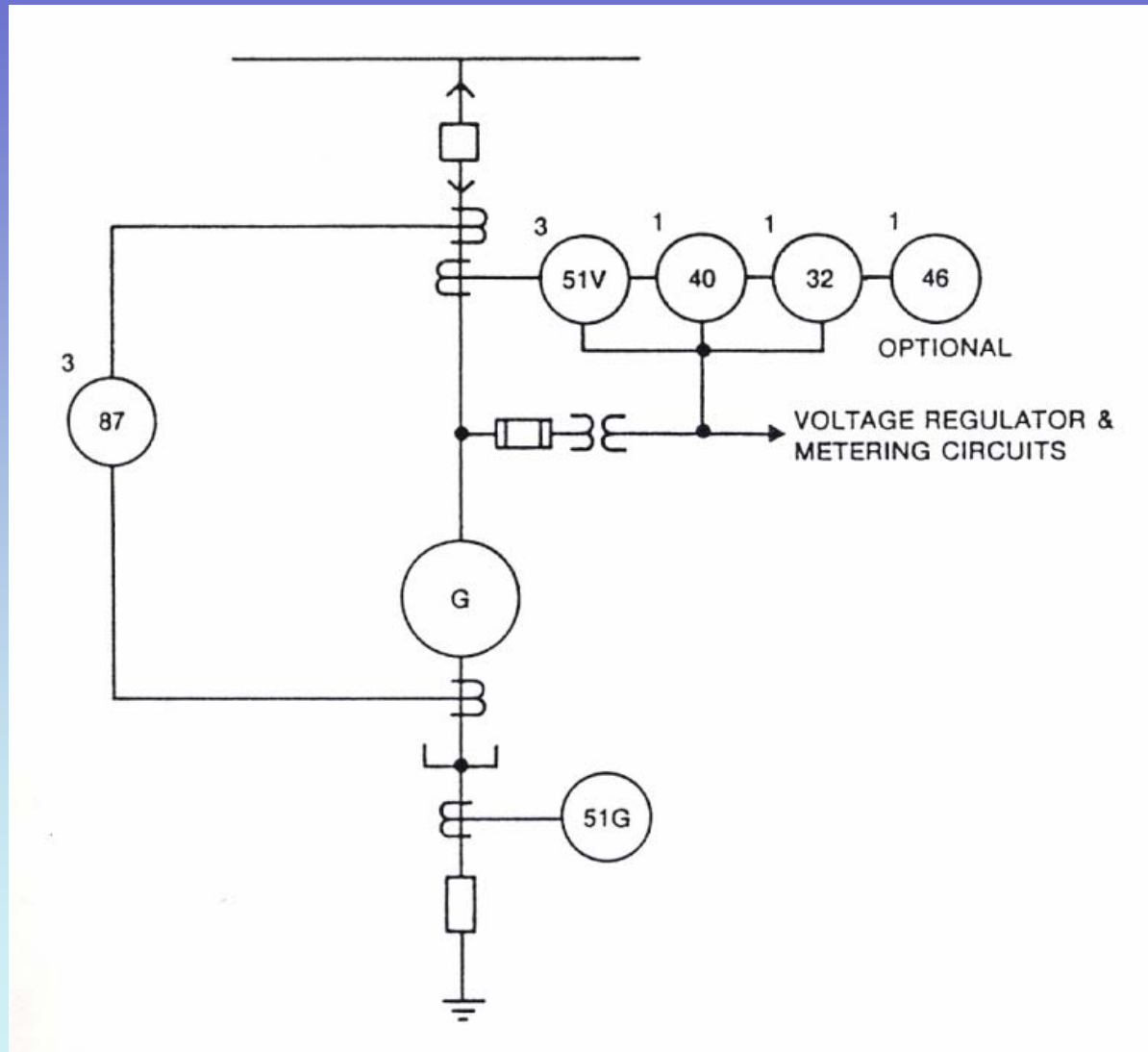
- 60 – Loss of Voltage
- 78 – Out-of-Step
- 64F – Field Ground
- Additional functions that may be provided include:
 - Sequential Trip Logic
 - Accidental Energization
 - Open Breaker Detection

Small Machine Protection IEEE “Buff Book”



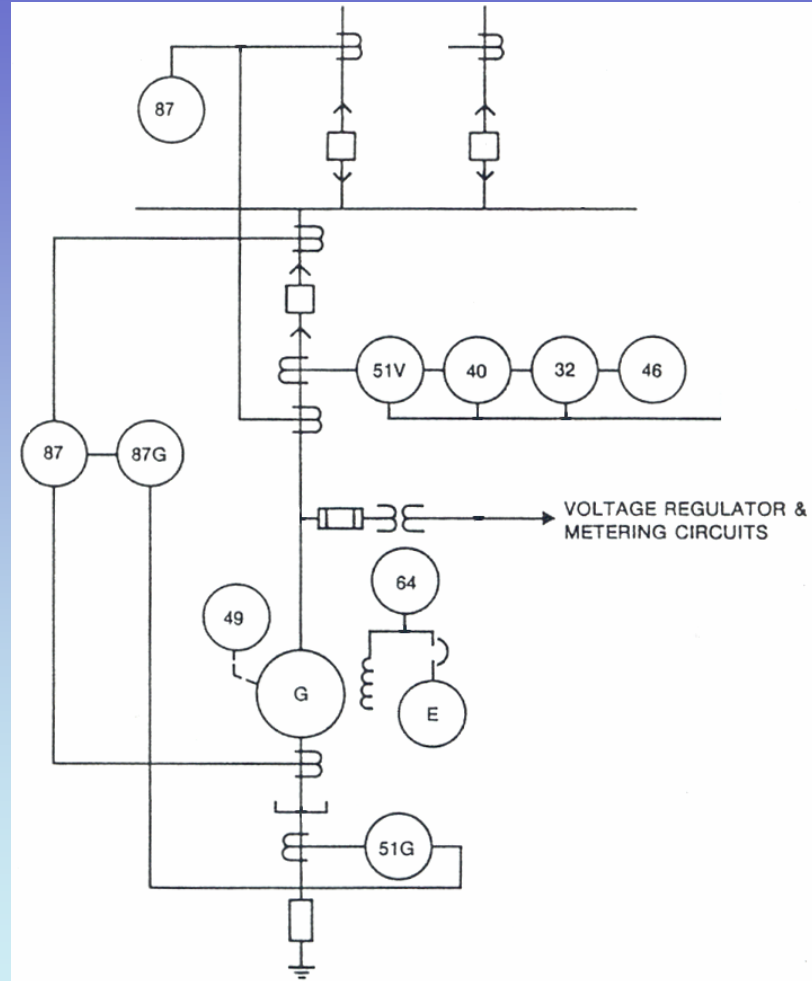
Small – up to 1 MW to 600V, 500 kVA if >600V

Medium Machine Protection IEEE “Buff Book”



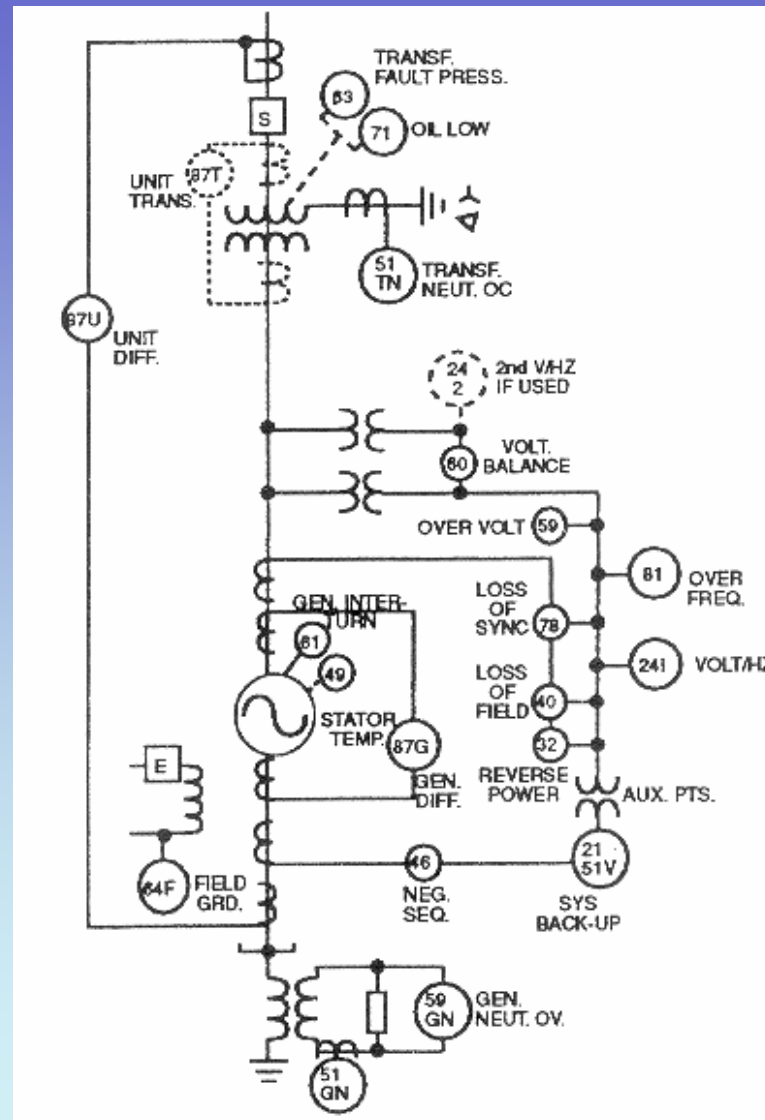
Medium – up to 12.5 MW

Large Machine Protection IEEE “Buff Book”



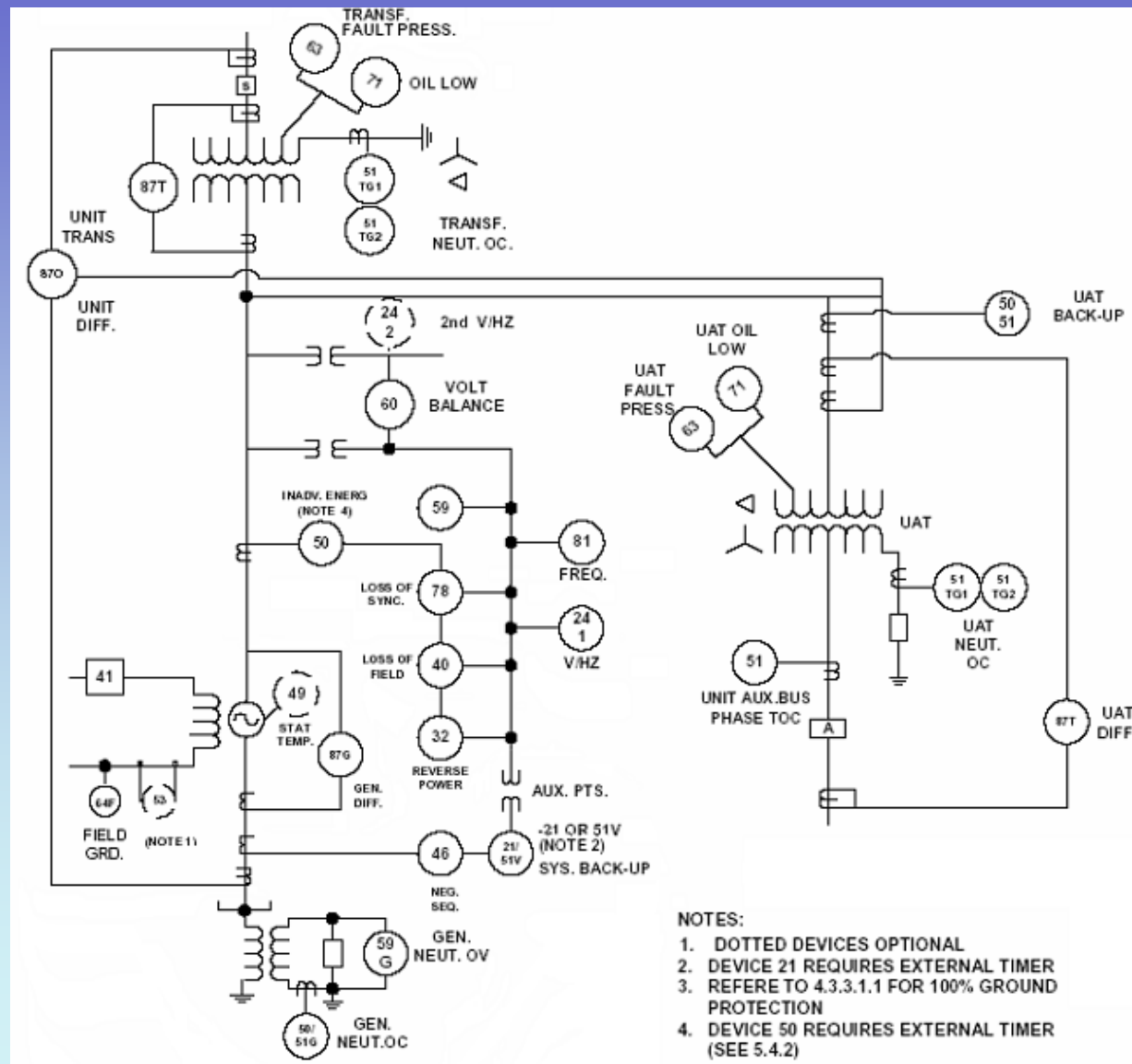
Large – up to 50 MW

Large Machine Protection IEEE C37.102-1995

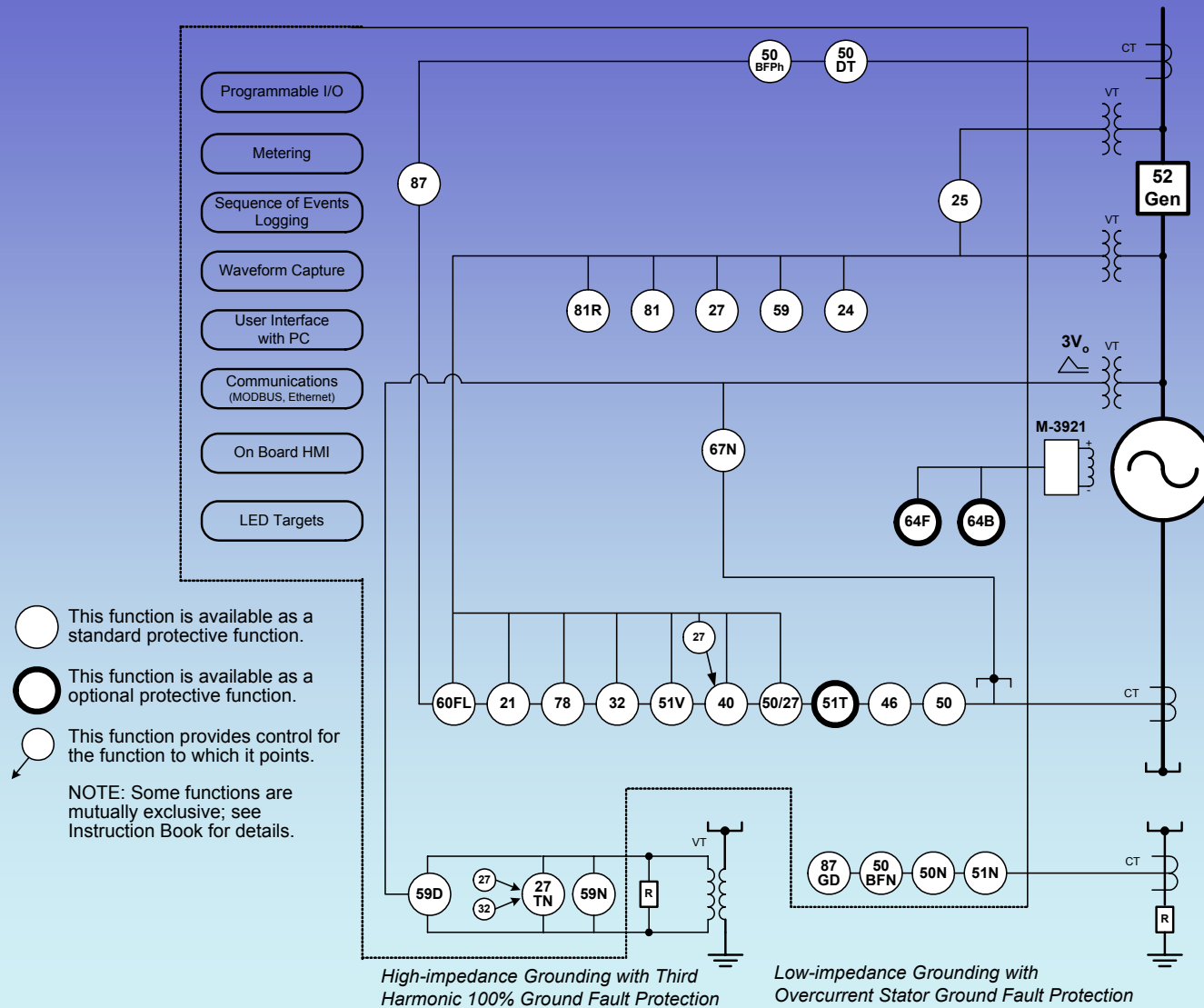


Larger than 50 MW

Large Machine Protection IEEE C37.102-2006



Relay Beckwith M-3425A



IEEE Devices used in Generator Protection

No.	DESCRIPTION
21	Phase Distance protection
24	Overexcitation / Volts per Hertz protection
25	Sync-check
27	Phase Undervoltage protection
27TN	100% Stator Ground Fault protection using 3rd Harmonic Undervoltage Differential
32R	Reverse Power protection
32F, 32LF	Overpower, Low Forward protection
40	Loss of Field protection
46	Negative sequence overcurrent protection

IEEE Devices used in Generator Protection

No.	DESCRIPTION
50	Instantaneous AC Overcurrent protection
50DT	Split Phase Differential protection
50/27	Inadvertent Generator Energizing protection
50BF	Breaker Failure
51	AC Inverse Time Overcurrent protection
51V	Inverse Time Overcurrent protection with Voltage Control/Restraint
59	Overvoltage protection
59D	100% Stator Ground Fault protection using 3rd Harmonic Voltage Comparison
60FL	VT Fuse-loss detection and blocking

IEEE Devices used in Generator Protection

No.	DESCRIPTION
64F	Field Ground protection
64B	Brush Lift-Off Detection
64S	100% Stator Ground Protection by Low Frequency Injection
67N	AC Directional Neutral Overcurrent protection
78	Out-of-step protection
81	Over/Under Frequency protection
81R	Rate of Change Frequency protection
87	Generator Phase Differential protection
87GD	Ground Differential protection

Distance Protection (21)

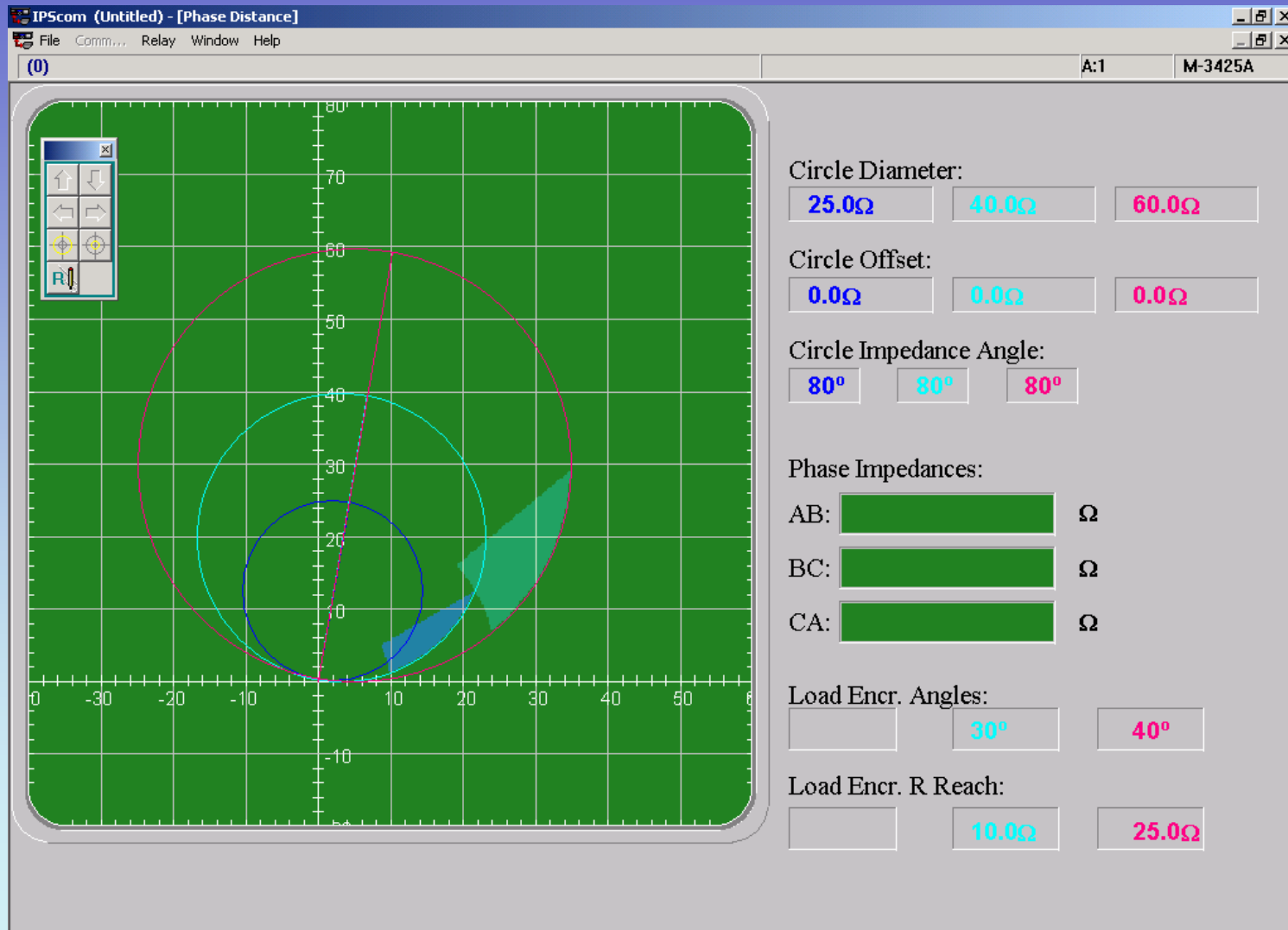
Distance Protection

- Distance relaying with mho characteristics is commonly used for system phase-fault backup.
- These relays are usually connected to receive currents from current transformers in the neutral ends of the generator phase windings and potential from the terminals of the generator.
- If there is a delta grounded-wye step-up transformer between the generator and the system, special care must be taken in selecting the distance relay and in applying the proper currents and potentials so that these relays see correct impedances for system faults.

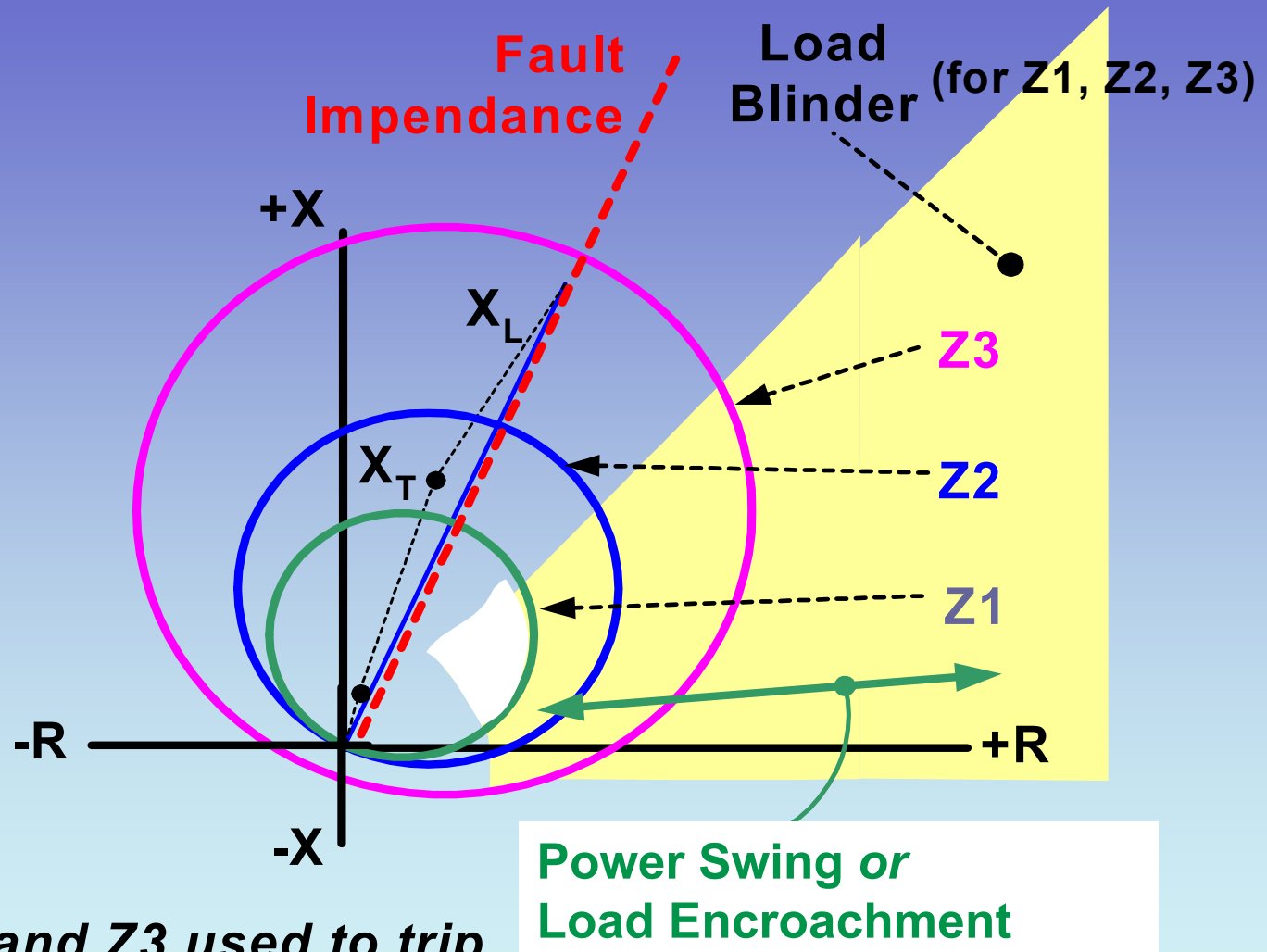
Phase Distance (21)

- **Phase distance backup protection may be prone to tripping on stable swings and load encroachment**
 - Employ three zones
 - Z1 can be set to reach 80% of impedance of GSU for 87G back-up.
 - Z2 can be set to reach 120% of GSU for station bus backup, **or** to overreach remote bus for system fault back up protection. Load encroachment blinder provides security against high loads with long reach settings.
 - Z3 may be used in conjunction with Z2 to form out-of-step blocking logic for security on power swings **or** to overreach remote bus for system fault back up protection. Load encroachment blinder provides security against high loads with long reach settings.
 - Current threshold provides security against loss of potential (machine off line)

3-Zone 21 Function with OSB/Load Encroachment



21 – Distance element

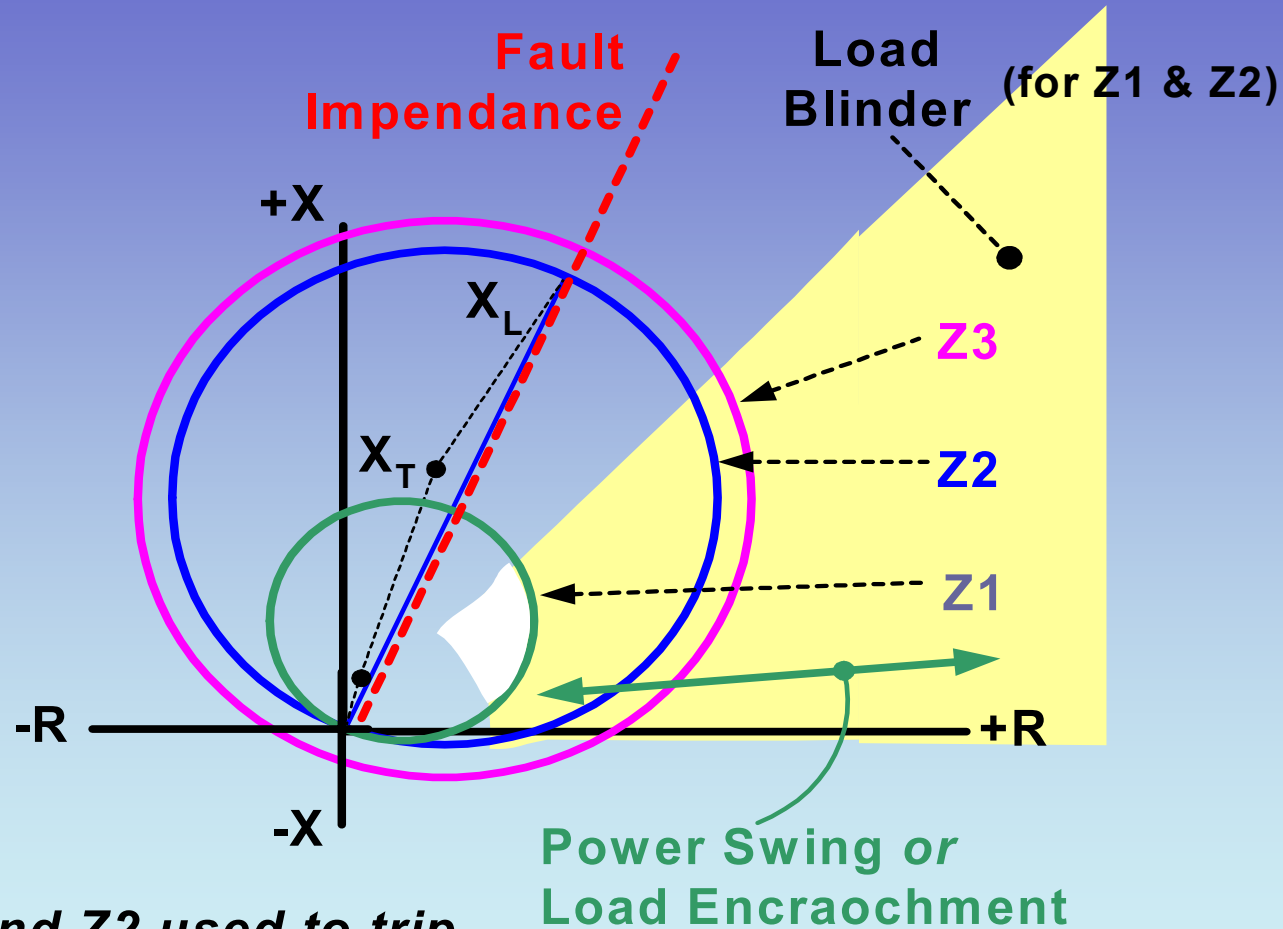


Z1, Z2 and Z3 used to trip

Z1 set to 80% of GSU, Z2 set to 120% of GSU

Z3 set to overreach remote bus

21 – Distance Element



Z1 and Z2 used to trip

Z1 set to 80% of GSU, Z2 set to overreach remote bus

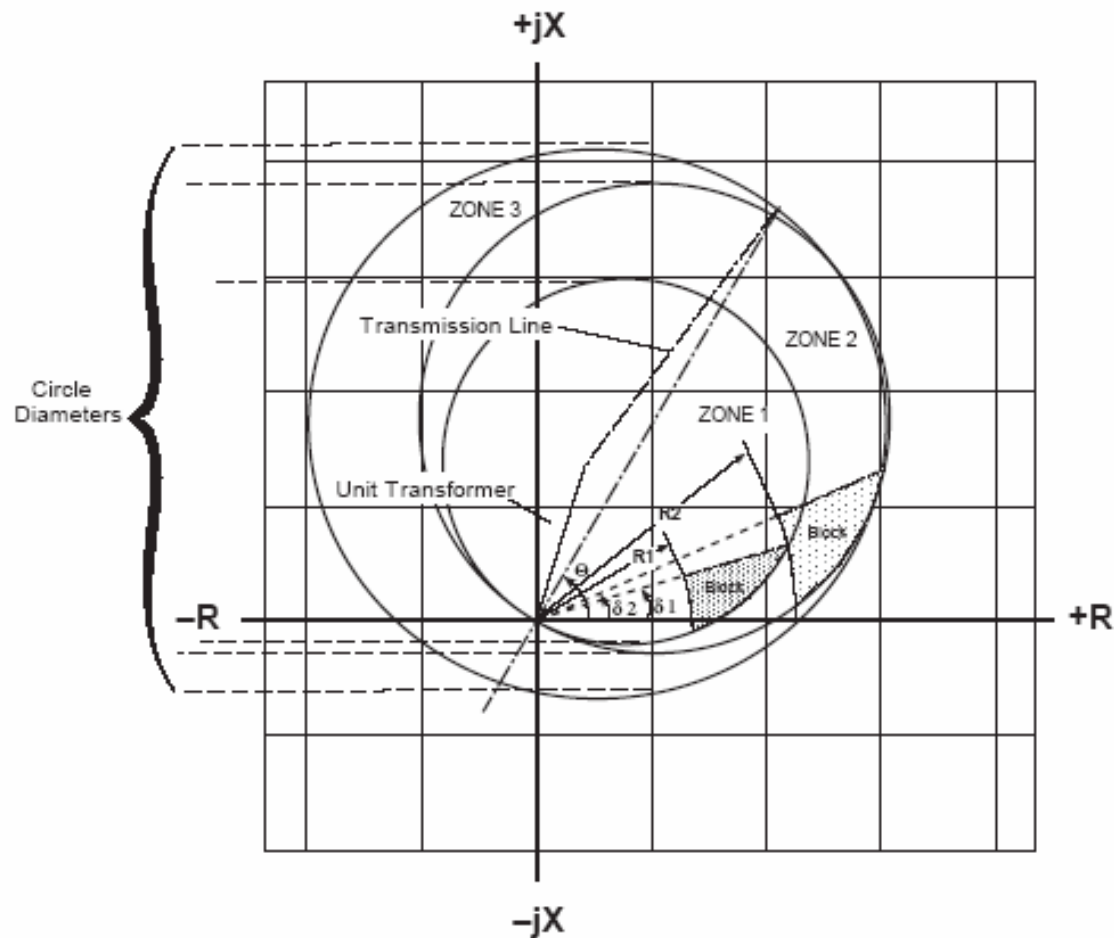
Z3 used for power swing blocking; Z3 blocks Z2

Distance Protection

Settings summary per IEEE C37.102-2005

- Zone-1 = the smaller of the two following criteria:
 1. 120% of unit transformer
 2. 80% of Zone 1 reach setting of the line relay on the shortest line (neglecting in-feed);Time = 0.5 s
- Zone-2 = the smaller of the three following criteria:
 - A. 120% of longest line (with in-feed).
 - B. 50% to 66.7% of load impedance (200% to 150% of the generator capability curve) at the RPF
 - C. 80% to 90% of load impedance (125% to 111% of the generator capability curve) at the maximum torque angle;Zone-2 < 2Z maxload @ RPF
Time > 60 cycles

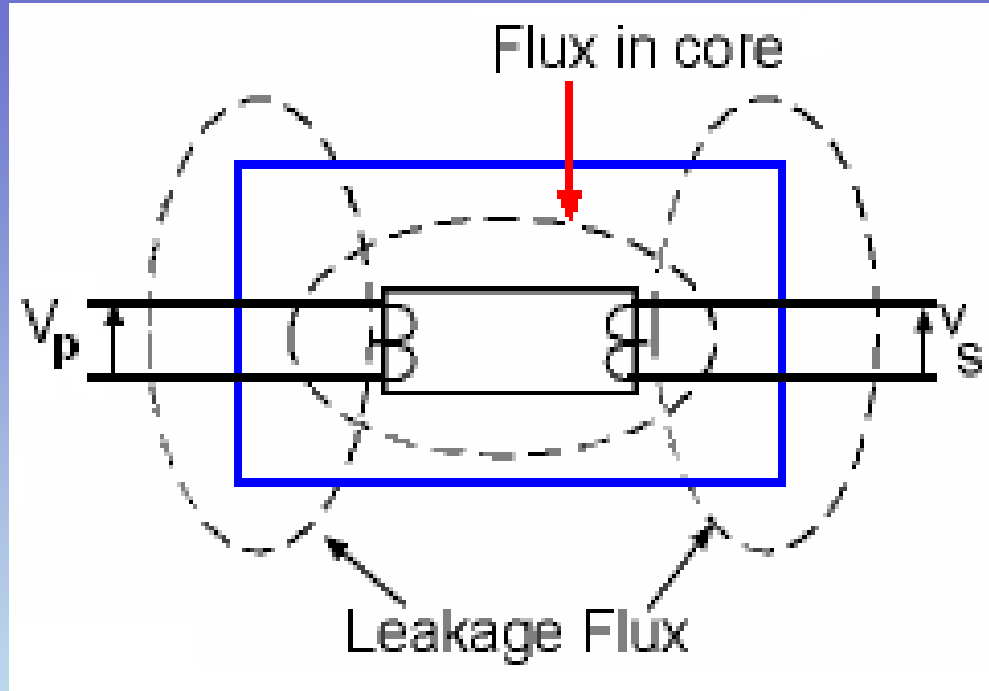
Distance Protection



- R1 Zone 1 Load Encroachment Blinder R Reach
- R2 Zone 2 Load Encroachment Blinder R Reach
- δ_1 Zone 1 Load Encroachment Blinder Angle
- δ_2 Zone 2 Load Encroachment Blinder Angle
- θ Impedance Angle Setting

Overexcitation/ Volts per Hertz (24)

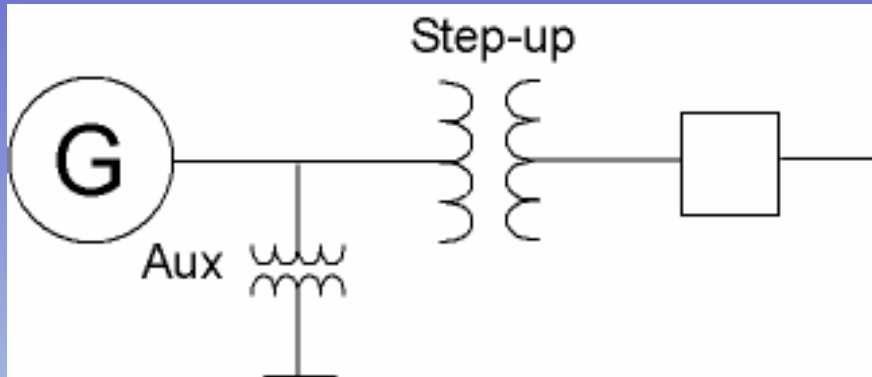
Overexcitation/Volts per Hertz



PHYSICAL INSIGHTS

- As voltage rises above rating leakage flux increases
- Leakage flux induces current in transformer support structure causing rapid localized heating.

Overexcitation/ Volts per Hertz



$$\text{GENERATOR TRANSFORMER EXCITATION} \approx \frac{\text{Voltage}}{\text{Freq.}} \quad \frac{\text{V}}{\text{Hz}}$$

GENERATOR LIMITS (ANSI C 50.13)

Full Load $\text{V/Hz} = 1.05 \text{ pu}$

No Load $\text{V/Hz} = 1.05 \text{ pu}$

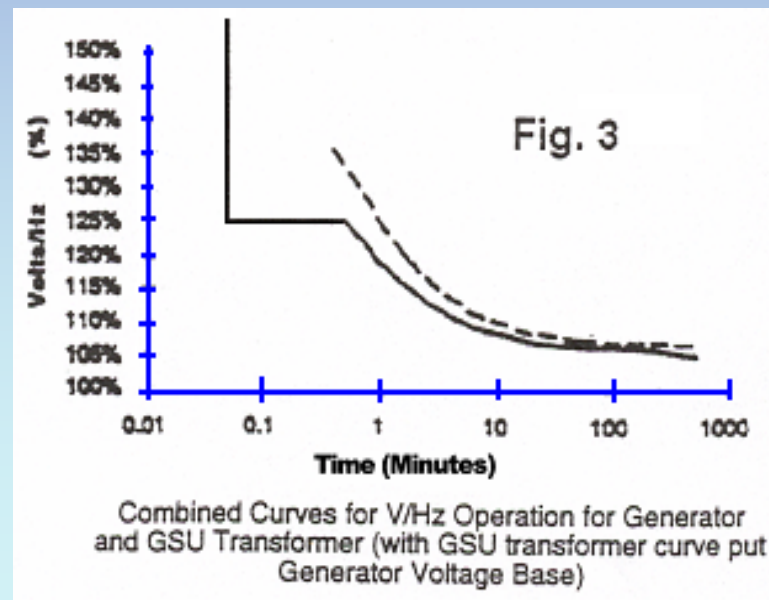
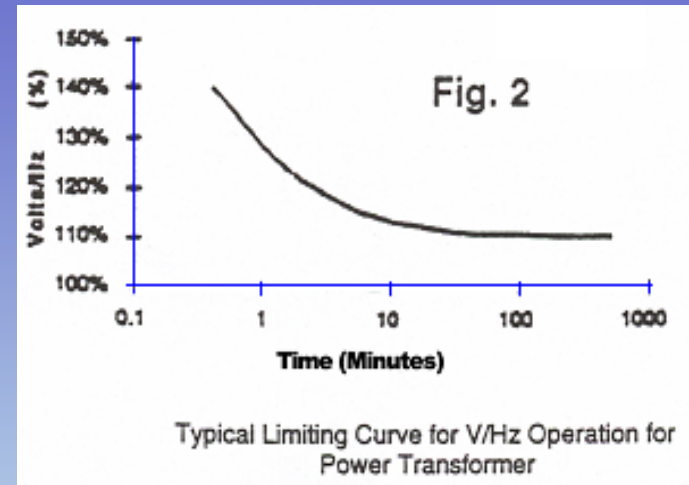
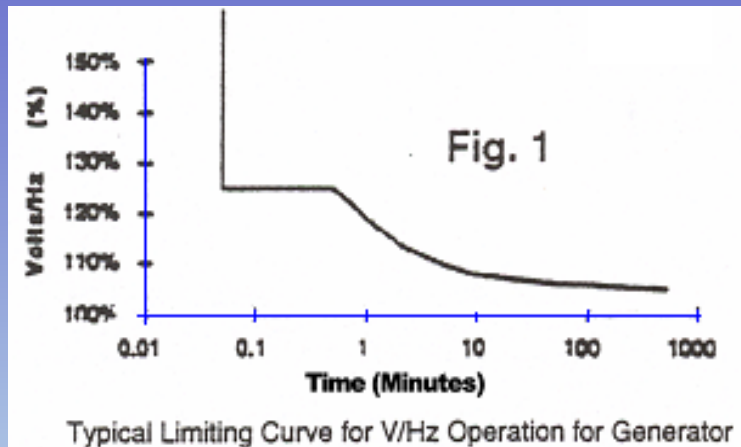
TRANSFORMER LIMITS

Full Load $\text{V/Hz} = 1.05 \text{ pu (HV Terminals)}$

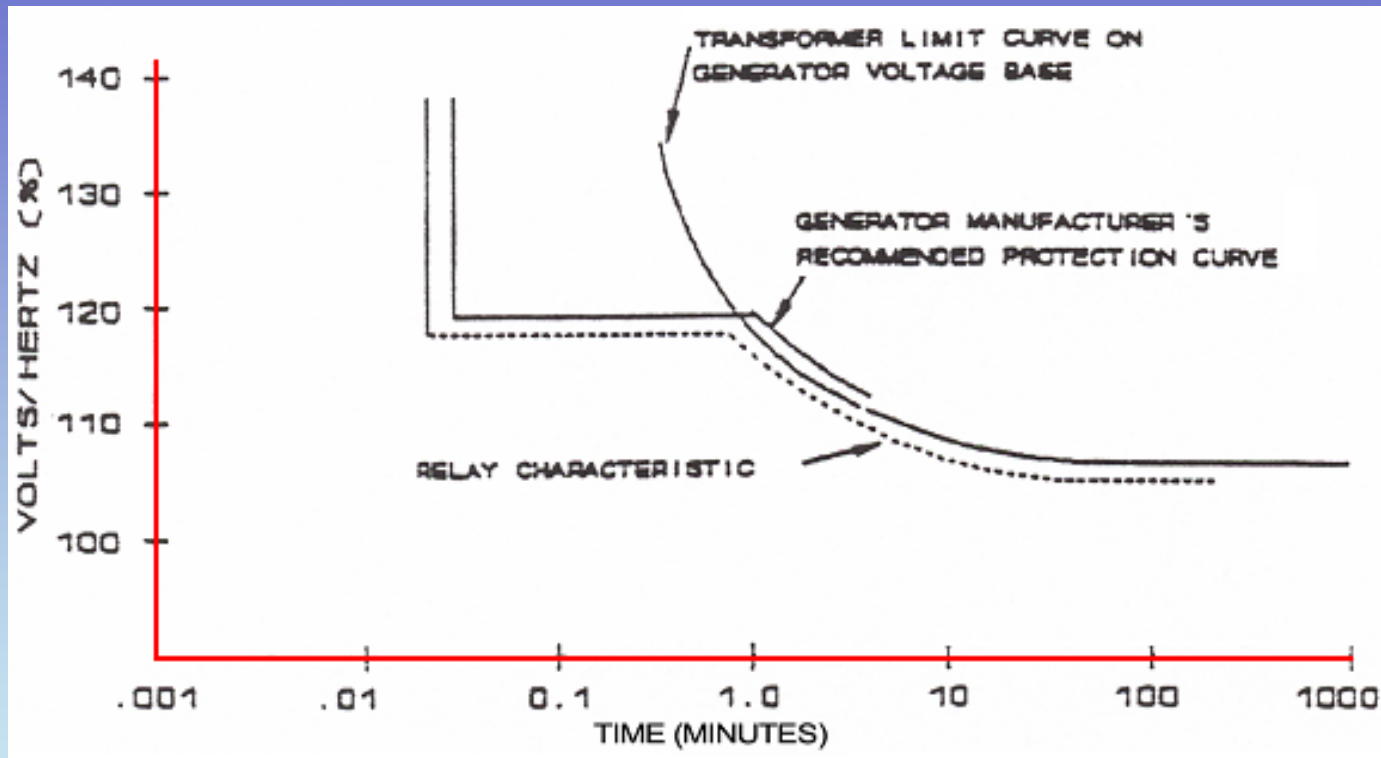
No Load $\text{V/Hz} = 1.10 \text{ pu (HV Terminals)}$

Overexcitation/Volts per Hertz

Typical Curves



Overexcitation/Volts per Hertz



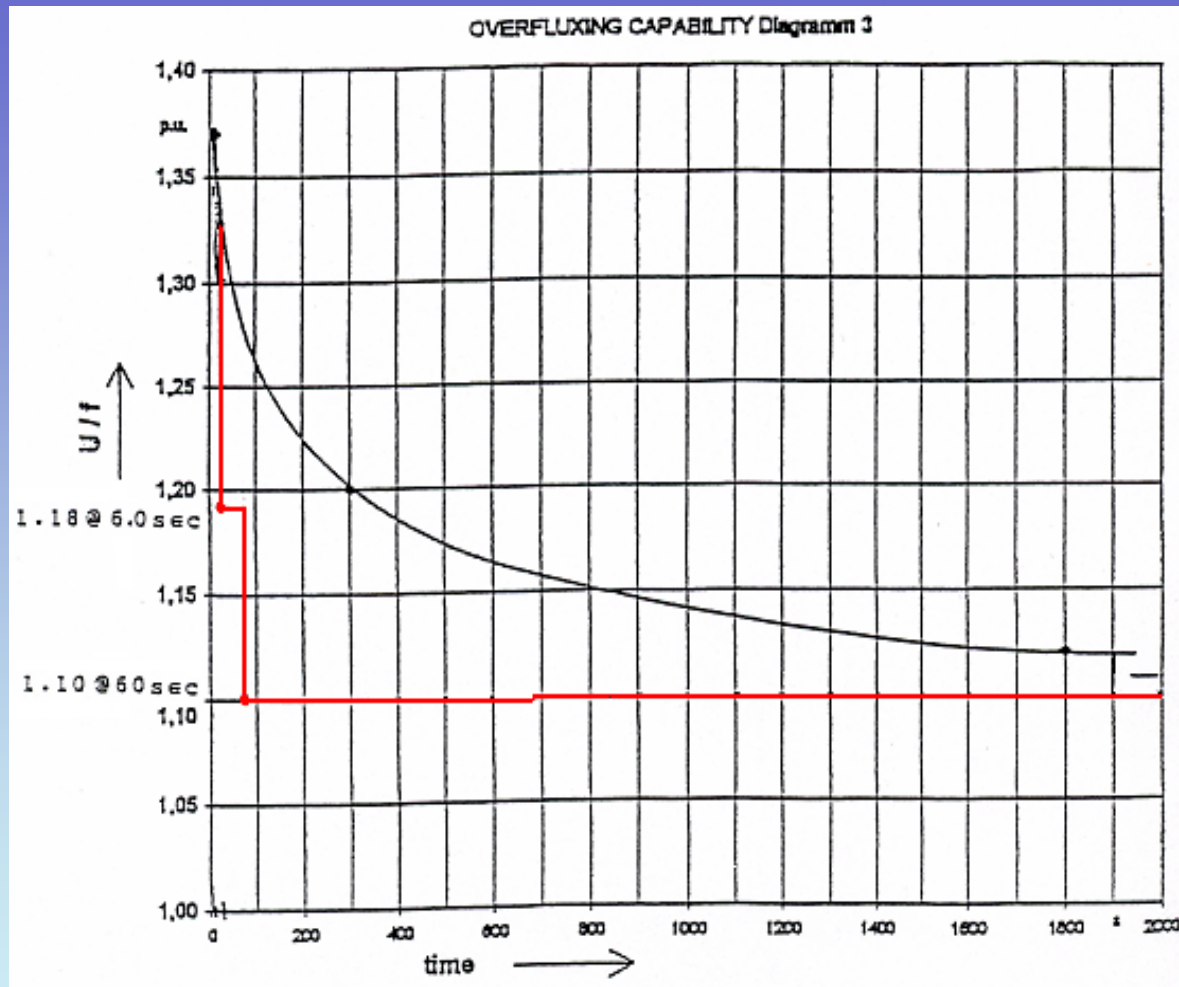
Example of inverse volts/hertz setting

Overexcitation/ Volts per Hertz

Settings summary per IEEE C37.102

- Single relay: $PU = 110\%$ p.u. time = 6 s
- Two stages relay: alarm pu = 110%; $45 < t < 60$ s
trip pu = 118% - 120%, $2 < t < 6$ s

Overexcitation/Volts per Hertz



Overfluxing Capability, Diagram 3

Siemens V84.3 165 MW Generator 12/1/94 MET-ED, FPC

Synchronizing (25)

Synchronizing

- Improper synchronizing of a generator to a system may result in damage to the generator step-up transformer and any type of generating unit.
- The damage incurred may be slipped couplings, increased shaft vibration, a change in bearing alignment, loosened stator windings, loosened stator laminations and fatigue damage to shafts and other mechanical parts.
- In order to avoid damaging a generator during synchronizing, the generator manufacturer will generally provide synchronizing limits in terms of breaker closing angle and voltage matching.

Synchronizing

Settings summary per IEEE C37.102

- Breaker closing angle: within ± 10 elect. degrees
- Voltage matching: 0 to +5%
- Frequency difference < 0.067 Hz

Undervoltage (27)

Undervoltage

- Generators are usually designed to operate continuously at a minimum voltage of 95% of its rated voltage, while delivering rated power at rated frequency.
- Operating generator with terminal voltage lower than 95% of its rated voltage may result in undesirable effects such as reduction in stability limit, import of excessive reactive power from the grid to which it is connected, and malfunctioning of voltage sensitive devices and equipment.

Undervoltage

Settings summary per IEEE C37.102

Relays with inverse time characteristic and instantaneous

- PU : 90%Vn; $t = 9.0$ s at 90% of PU setting
- Inst : 80% Vn

Relays with definite time characteristic and two stages

- Alarm PU : 90%Vn; $10 < t < 15$ s
- Trip PU : 80% Vn; time: 2s

Reverse Power

(32)

Reverse Power

- Prevents generator from motoring on loss of prime mover
- From a system standpoint, motoring is defined as the flow of real power into the generator acting as a motor.
- With current in the field winding, the generator will remain in synchronism with the system and act as a synchronous motor.
- If the field breaker is opened, the generator will act as an induction motor.
- A power relay set to look into the machine is therefore used on most units.
- The sensitivity and setting of the relay is dependent upon the type of prime mover involved.

Reverse Power

Settings summary per IEEE C37.102

Pickup setting should be below the following motoring limits:

- Gas : 50% rated power; time < 60 s
- Diesel : 25% rated power; time < 60 s
- Hydro turbines : 0.2% - 2% rated power; time < 60 s
- Steam turbines : 0.5% - 3% rated power; time < 30 s

Sequential Tripping

- ❖ Used on steam turbine generators to prevent overspeed
- ❖ Recommended by manufacturers of steam turbine generators as a result of field experience
- ❖ This trip mode used only for boiler/reactor or turbine mechanical problems
- ❖ Electrical protection should not trip through this mode

Sequential Tripping

STEP 1

- ❖ Abnormal turbine/boiler/reactor condition is detected

STEP 2

- ❖ Turbine valves are closed; generator allowed to briefly “motor” (I.e., take in power)

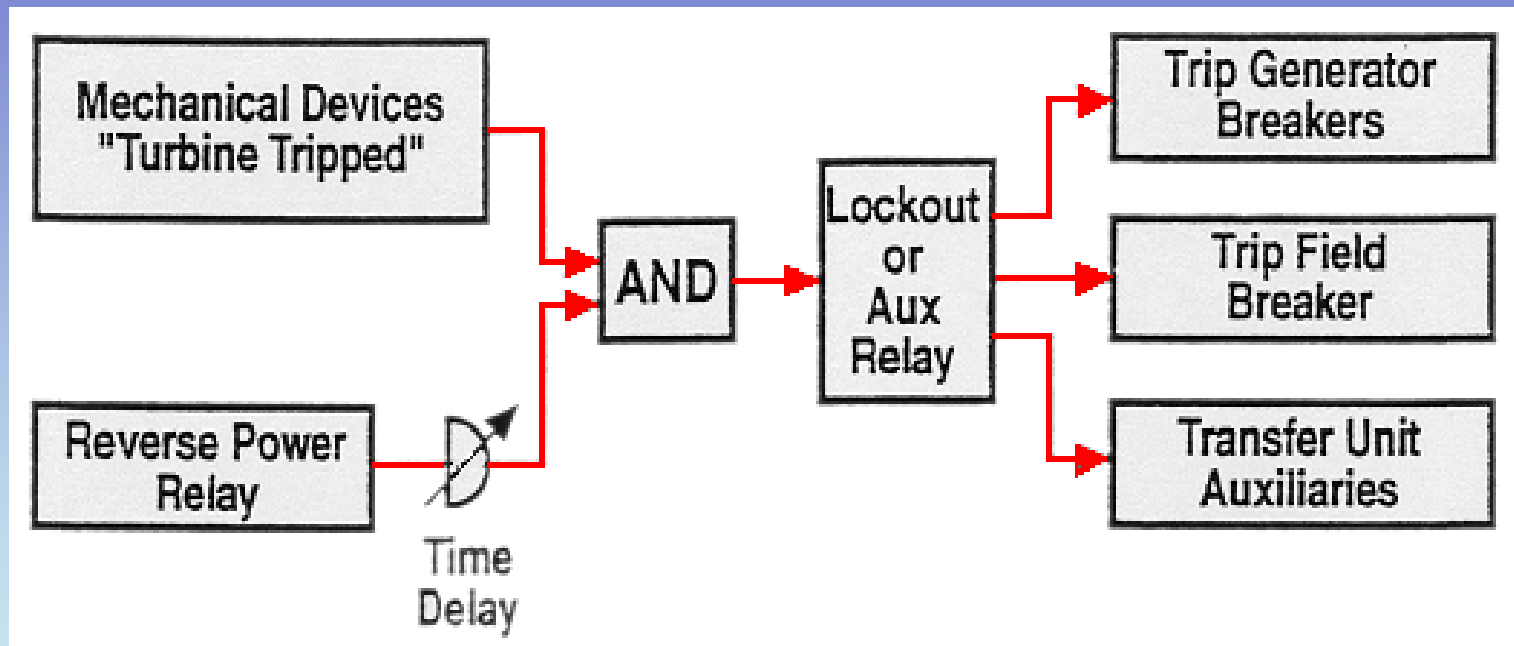
STEP 3

- ❖ A reverse power (32) relay in series with turbine valves position switches confirms all valves have closed

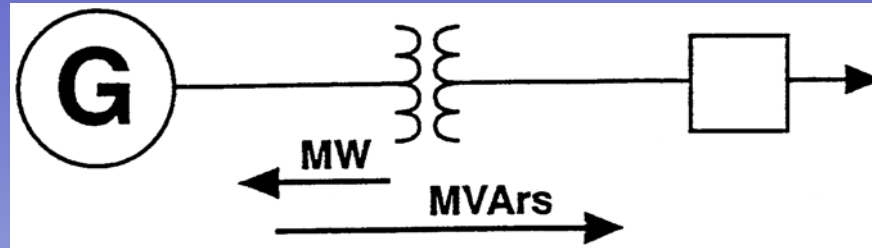
STEP 4

- ❖ Generator is separated from power system

Sequential Tripping Logic

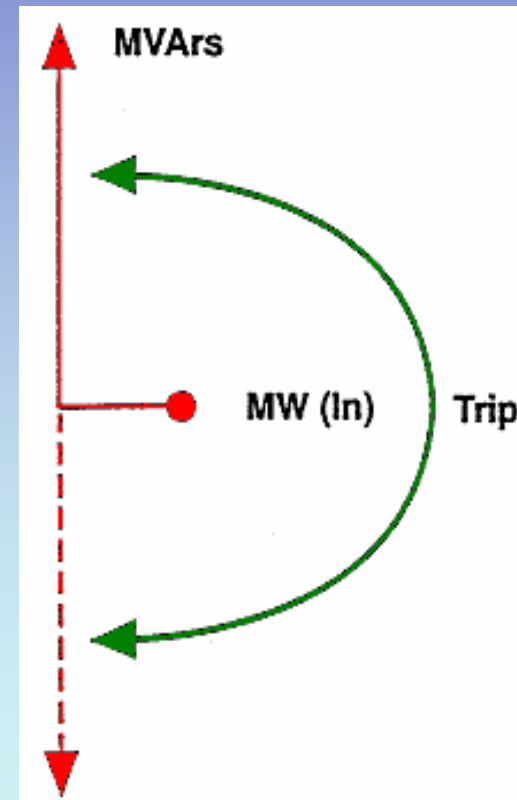


Sequential Tripping Problem



CONSIDER

- ❖ High MVARs (out)
- ❖ Low MW (in)
- ❖ E-M relay can be fooled



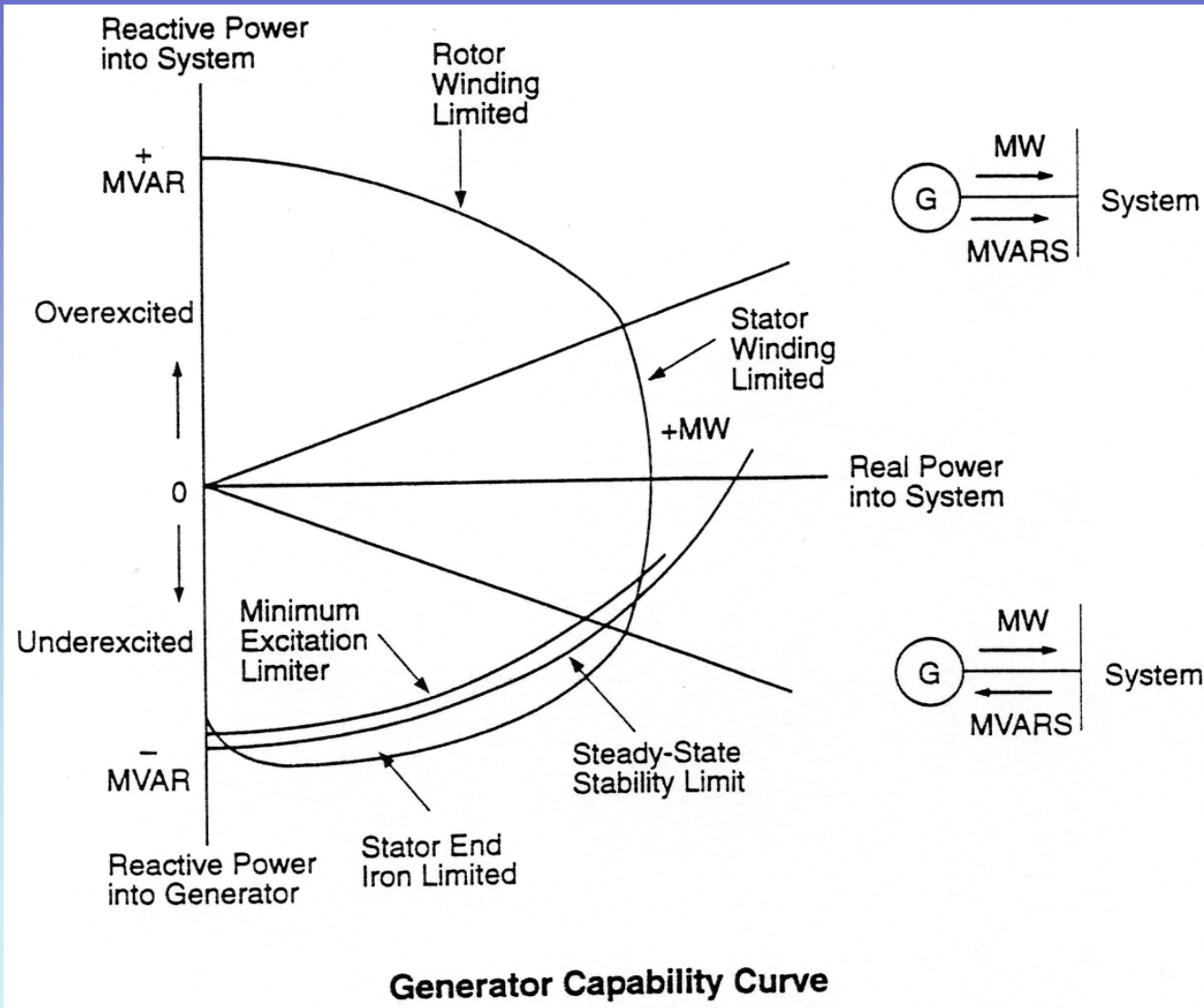
Loss-of-Field (40)

Loss of Field

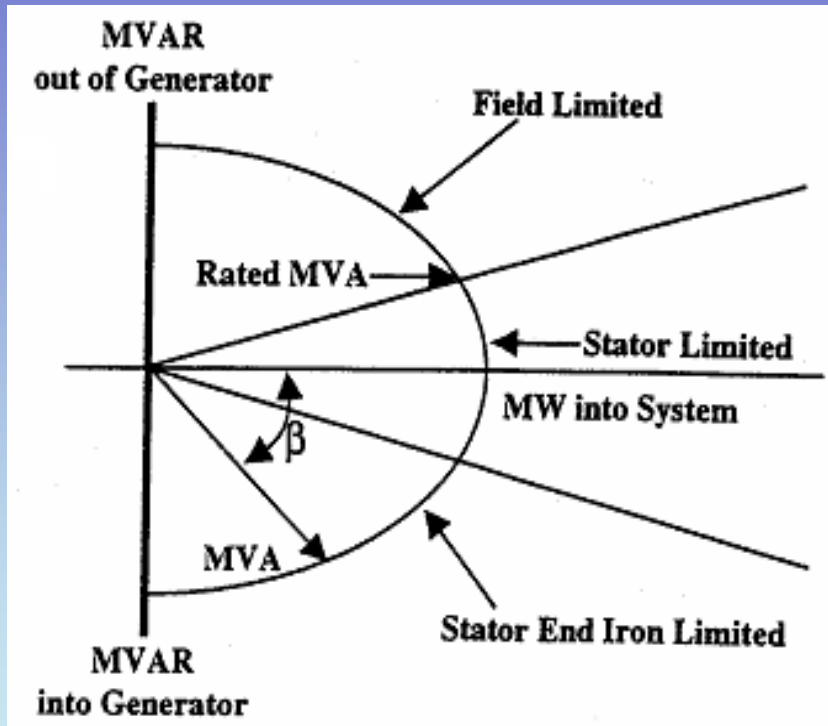
CAUSES

- Field open circuit
- Field short circuit
- Accidental tripping of field breaker
- Regulator control failure
- Loss of main exciter

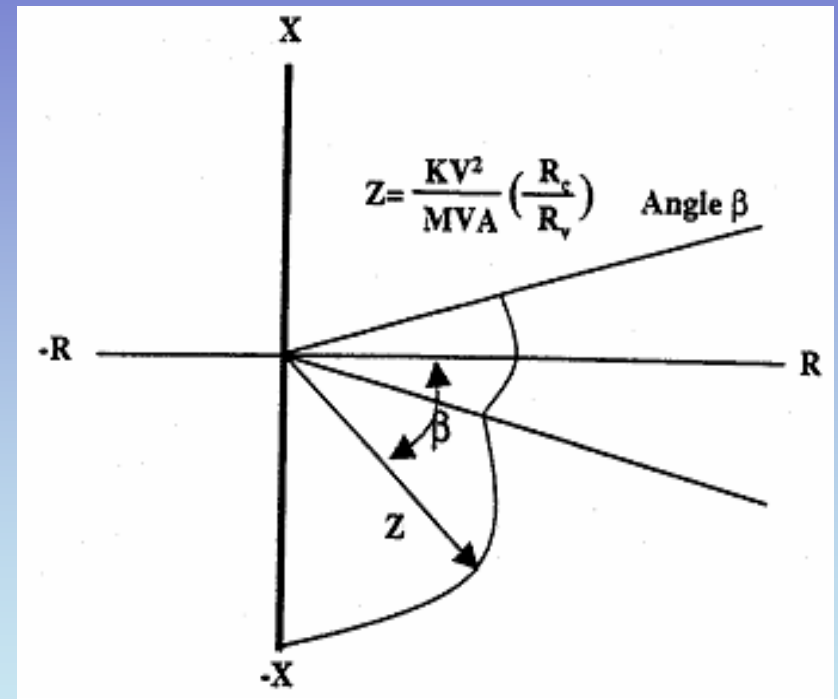
Loss of Field



Transformation from KW-KVAR plot to R-X Plot

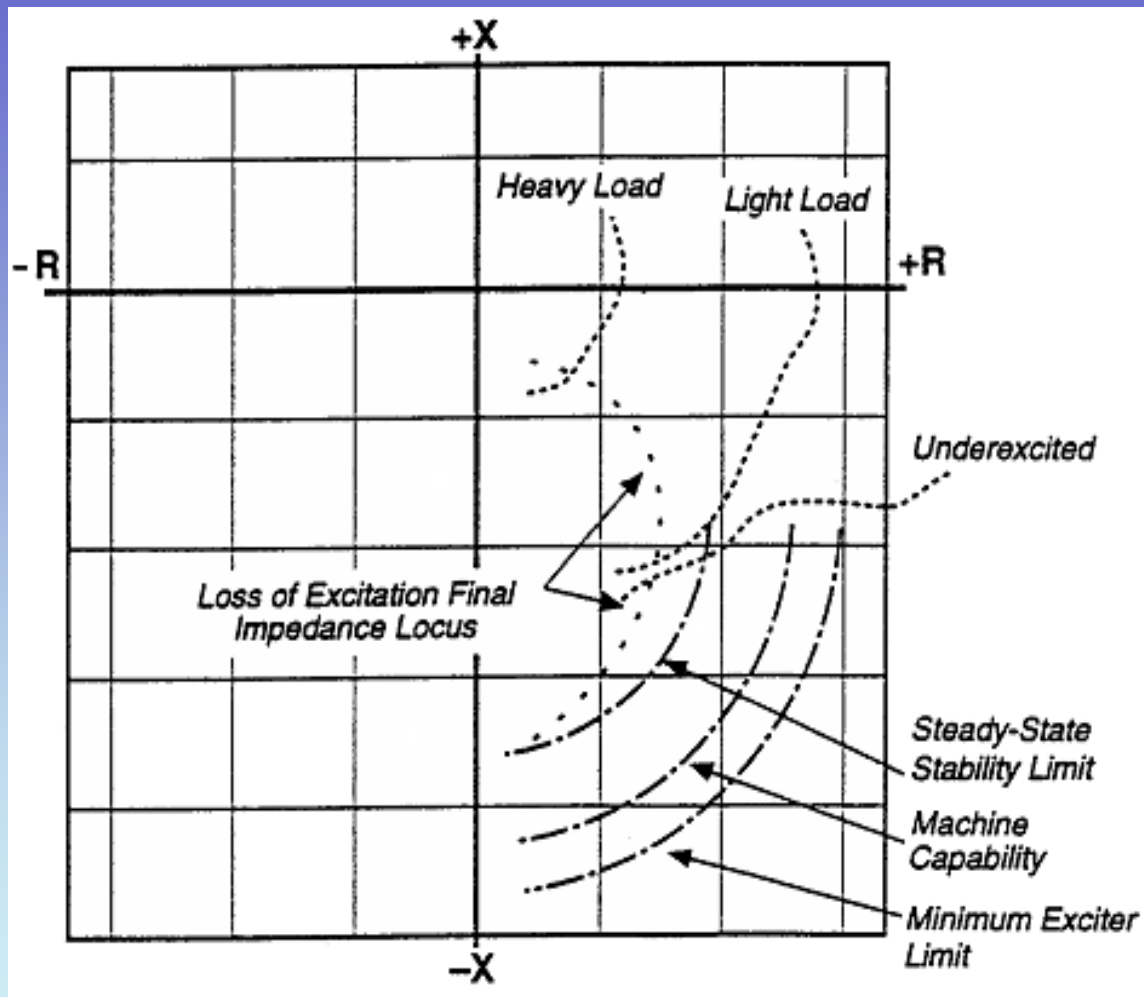


Machine Capability Curve



R-X Plot

Loss of Field



Loss of Field Impedance Characteristics

Loss of Field

Settings summary per IEEE C37.102

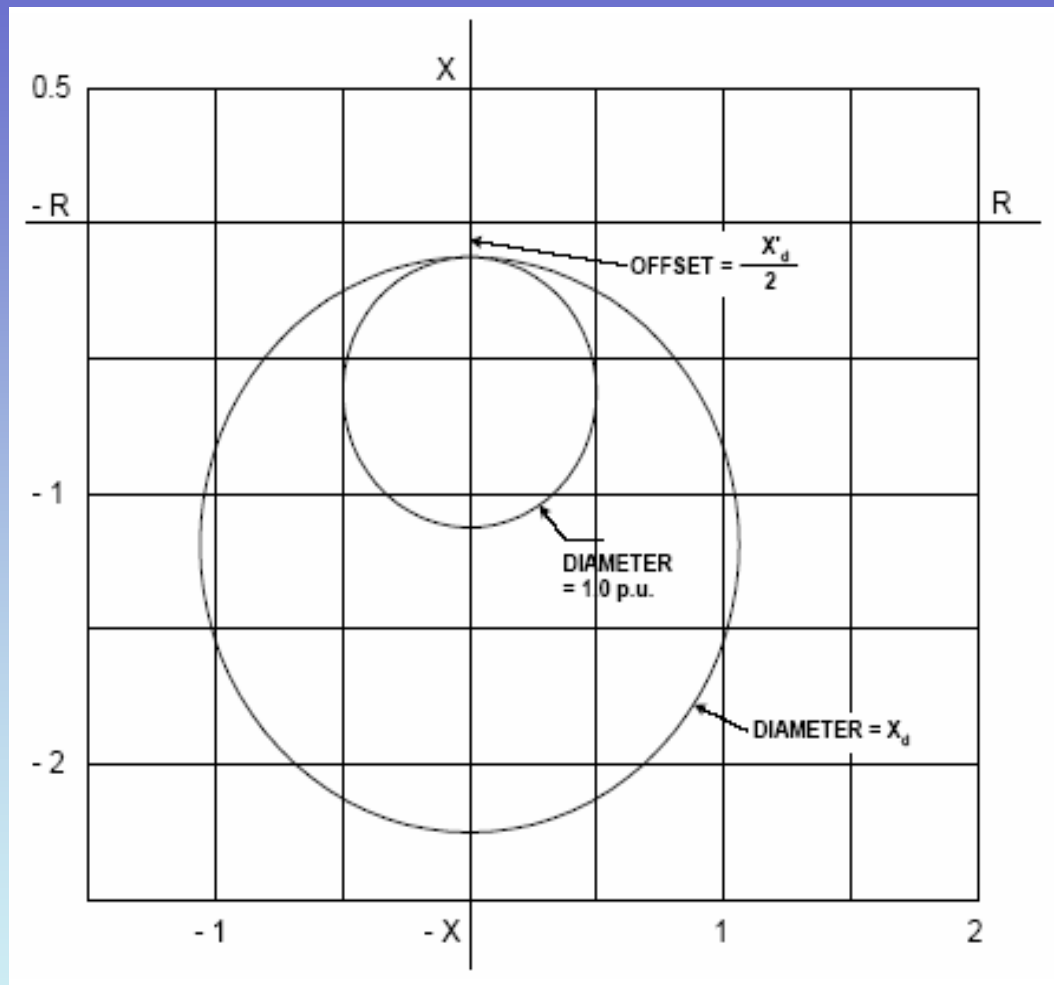
➤ UNIT 1

Offset: $X'd/2$; Diameter: 1.0 pu; time: 0.1 s

➤ UNIT 2

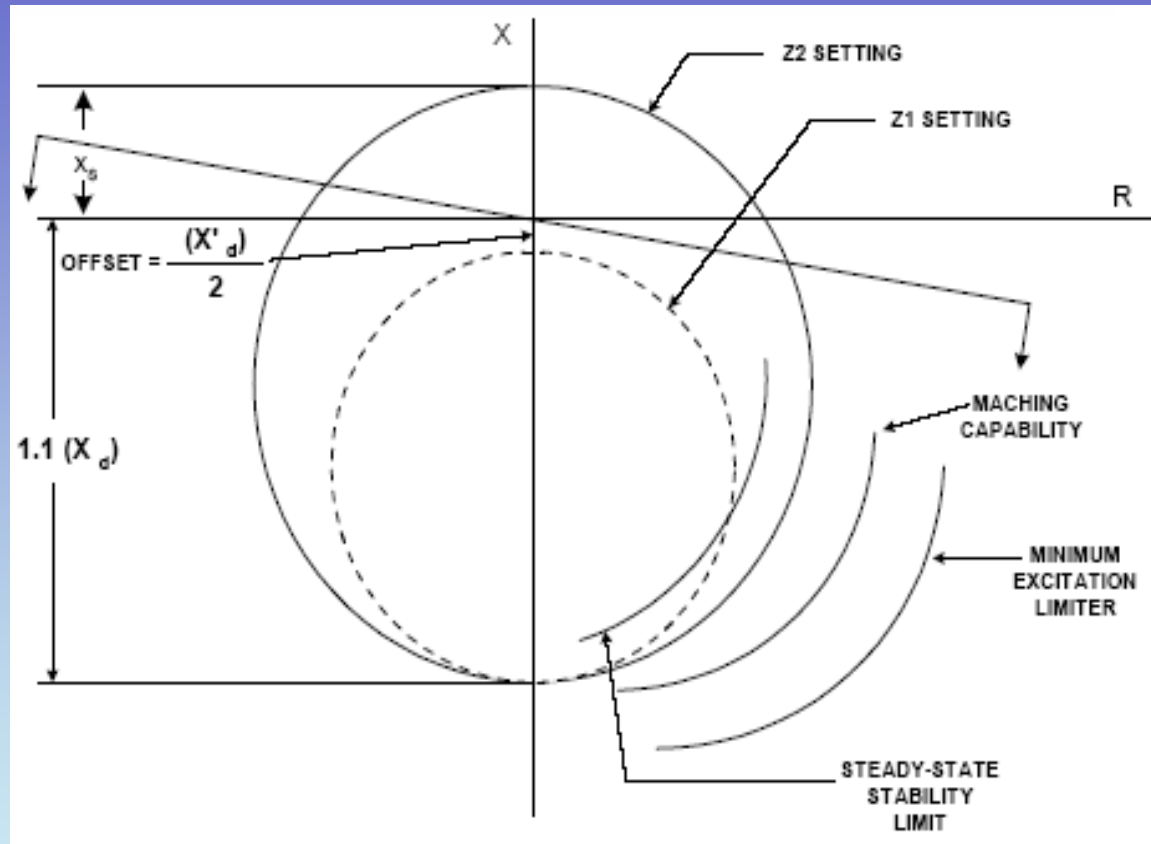
Offset: $X'd/2$; Diameter: X_d ; time: 0.5 to 0.6 s

Loss of Field



Protective Approach # 1

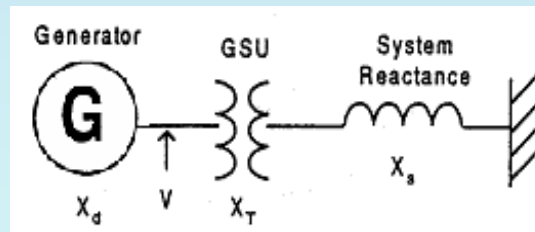
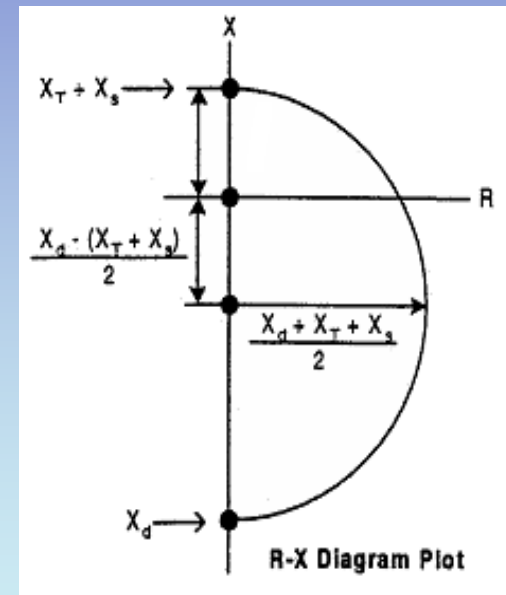
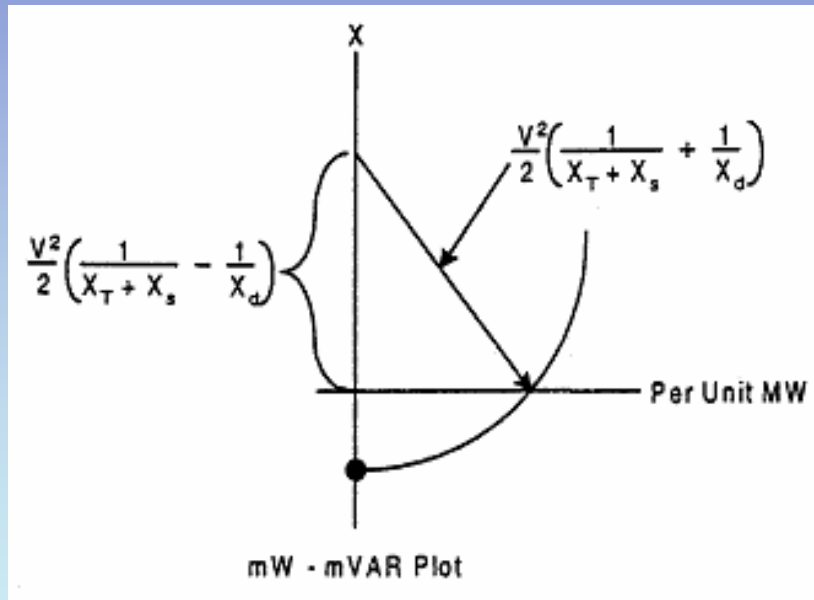
Loss of Field



Protective Approach # 2

Graphical Method For Steady-state Stability

The Steady-State Stability limit can be a significant limit that should be related to both the machine capability curve (MW-MVAR Plot) and the loss-of-field (40) relay operating characteristics (R-X Diagram Plot). In the figures below, V is the per-unit terminal generator voltage, X_T and X_s the per-unit Generator Step Up (GSU) transformer and system impedances respectively as viewed from the generator terminals. X_d is the per-unit unsaturated synchronous reactance of the generator. All reactances should be placed on the generator MVA base.



Negative Sequence (46)

Negative Sequence

- Unbalanced phase currents create negative sequence current in generator stator

$$I_2 = 1/3 (I_a + a^2 I_b + a I_c)$$

where

$$a = 1 \angle 120$$
$$a^2 = 1 \angle 240$$

I_a, I_b, I_c = phase currents

- Negative sequence current interacts with normal positive sequence current to induce a double frequency current (120 Hz)
- Current (120 Hz) is induced into rotor causing surface heating
- Generator has established short-time rating,

$$I_2^2 t = K$$

where

K=Manufacturer Factor (the larger the generator the smaller the K value)

Negative Sequence

Settings summary per IEEE C37.102

TYPE OF GENERATOR	PERMISSIBLE I_2 PERCENT OF STATOR RATING
Salient Pole	
With connected amortisseur windings	10
With non-connected amortisseur windings	5
Cylindrical Rotor	
Indirectly cooled	10
Directly cooled to 960 MVA	8
961 to 1200 MVA	6
1200 to 1500 MVA	5

†These values also express the negative-phase –sequence current capability at reduced generator KVA capabilities.

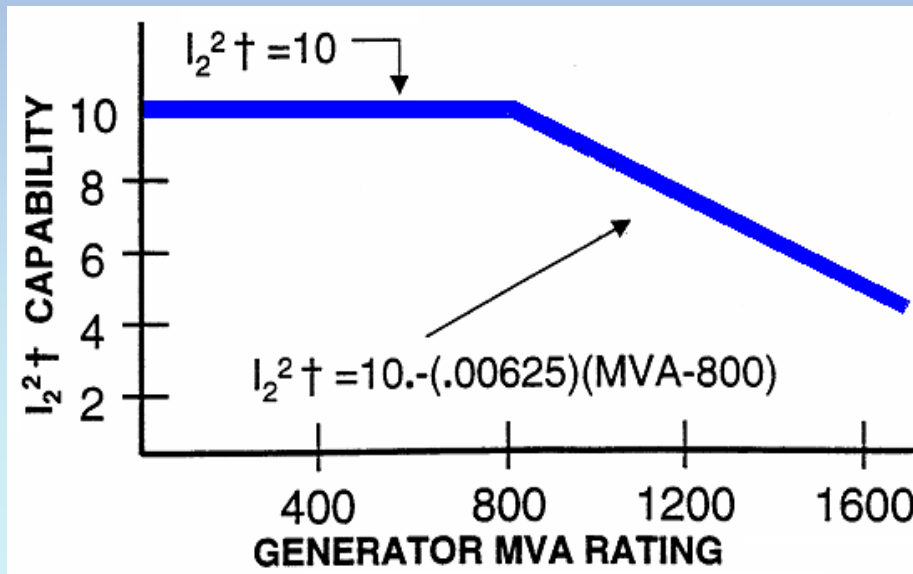
‡ The short time (unbalanced fault) negative sequence capability of a generator is also defined in ANSI C50.13.

Negative Sequence

Type of Generator

Permissible I_2^{2t}

Salient pole generator	40
Synchronous condenser	30
Cylindrical rotor generators	
Indirectly cooled	30
Directly cooled (0-800 MVA)	10
Directly cooled (801-1600 MVA)	see curve below



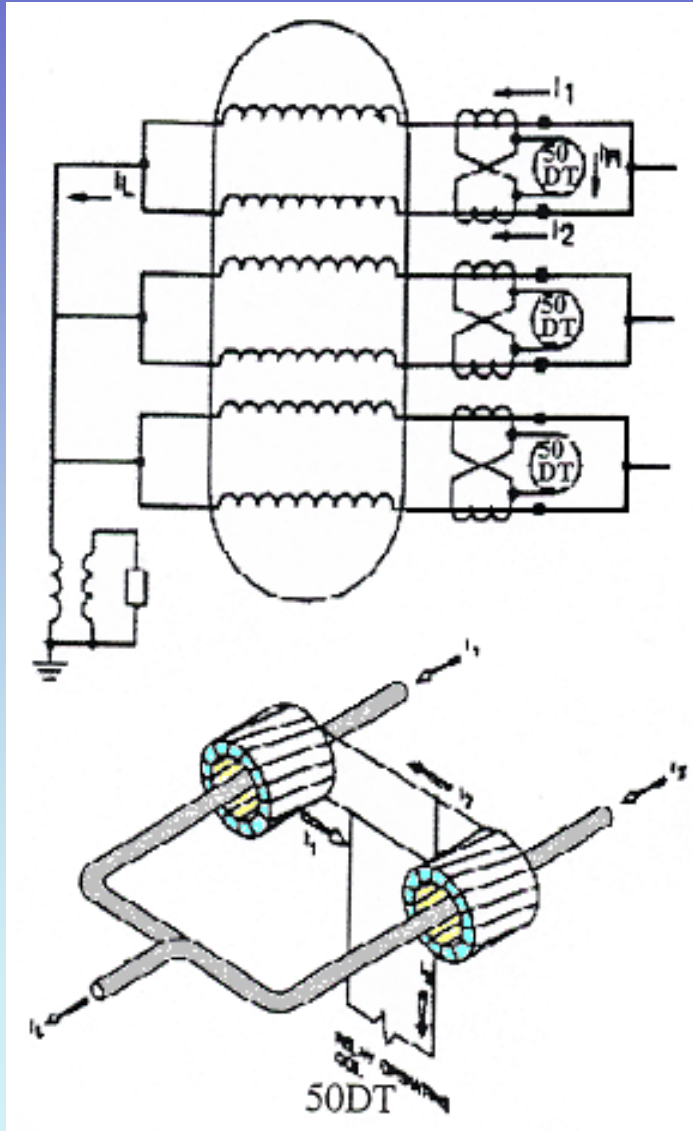
(VALUES TAKEN
FROM ANSI C50.13-1989)

Split Phase Differential (50DT)

Split-Phase Differential

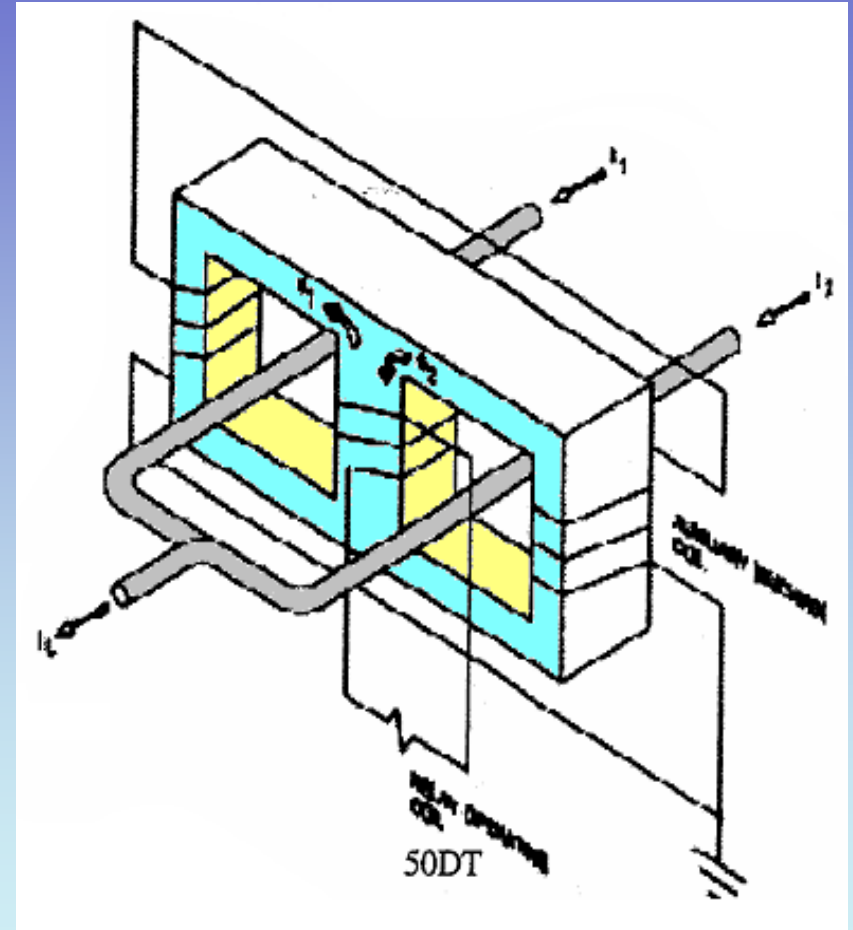
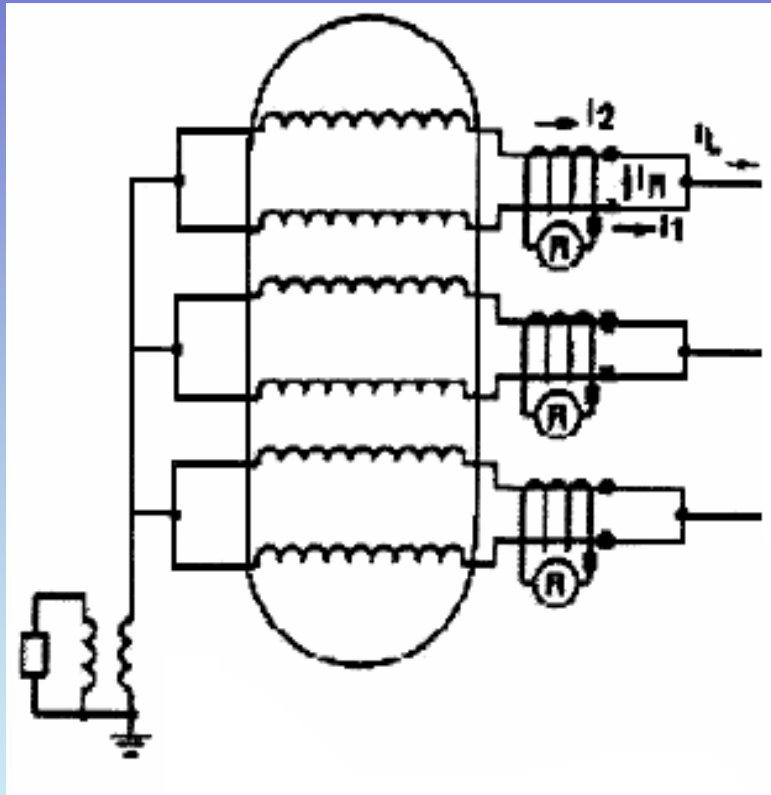
- Most turbine generators have single turn stator windings. If a generator has stator windings with multiturn coils and with two or more circuits per phase, the split-phase relaying scheme may be used to provide turn fault protection.
- In this scheme, the circuits in each phase of the stator winding are split into two equal groups and the currents of each group are compared.
- A difference in these currents indicates an unbalance caused by a single turn fault.

Split-Phase Differential

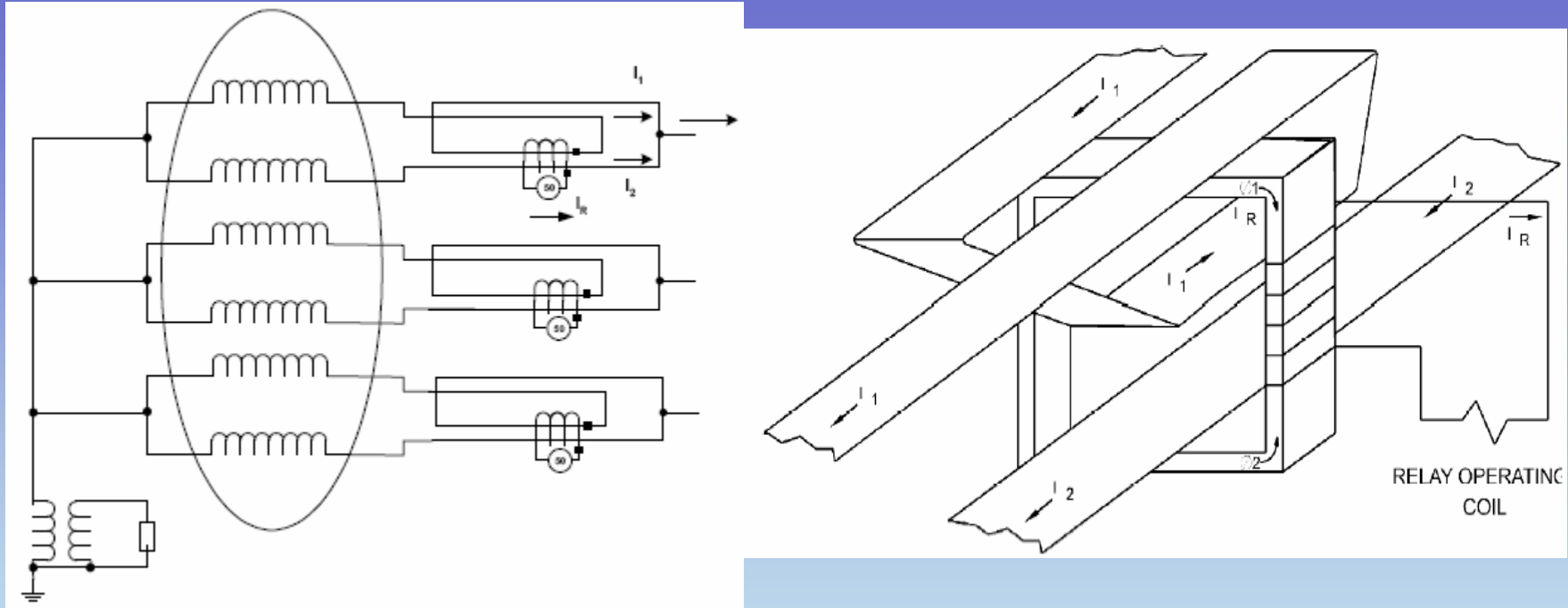


- Scheme detects turn to turn fault not involving ground.
- Generator must have two or more windings per phase to apply scheme.
- Used widely on salient-pole hydro generators. Used on some steam generators.
- Difference between current on each phase indicates a turn to turn fault.
- Need to have separate pick-up levels on each phase to accommodate practice of removal of shorted terms.

Typical Split-Phase Differential Using Window CT's



Split-phase protection using a single window current transformer



Settings summary per IEEE C37.102

The pickup of the instantaneous unit must be set above CT error currents that may occur during external faults.

Inadvertent Off-Line Generator Protection (50/27)

Why Inadvertent Energizing Occurs

- Operating errors
- Breaker head flashover
- Control circuit malfunctions
- Combination of above

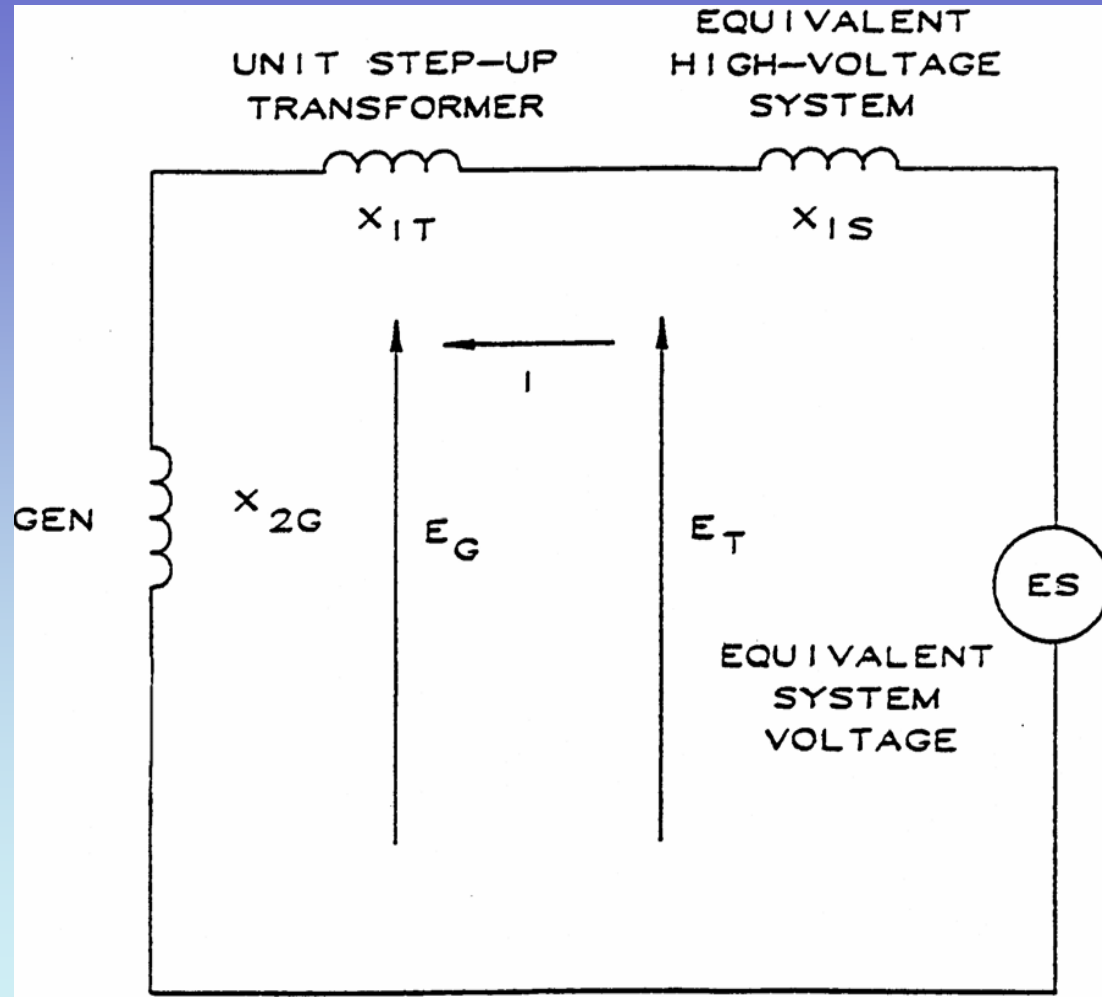
Inadvertent Energizing Protection

- ❖ Inadvertent energizing is a serious industry problem
- ❖ Damage occurs within seconds
- ❖ Conventional generator protection will not provide protection
 - marginal in detecting the event
 - disabled when machine is inadvertently energized
 - operates too slowly to prevent damage
- ❖ Need to install dedicated protection scheme

Generator Response and Damage to Three-Phase Energizing

- ❖ Generator behaves as an induction motor
- ❖ Rotating flux induced into the generator rotor
- ❖ Resulting rotor current is forced into negative sequence path in rotor body
- ❖ Machine impedance during initial energizing is equivalent to its negative sequence impedance
- ❖ Rapid rotor heating occurs $I^2t = K$

Inadvertent Energizing Equivalent Circuit



Response of Conventional Generator Protection to Inadvertent Energizing

Some relays may detect inadvertent generator energizing but can:

- ❖ Be marginal in their ability to detect the condition
- ❖ Operate too slowly to prevent damage

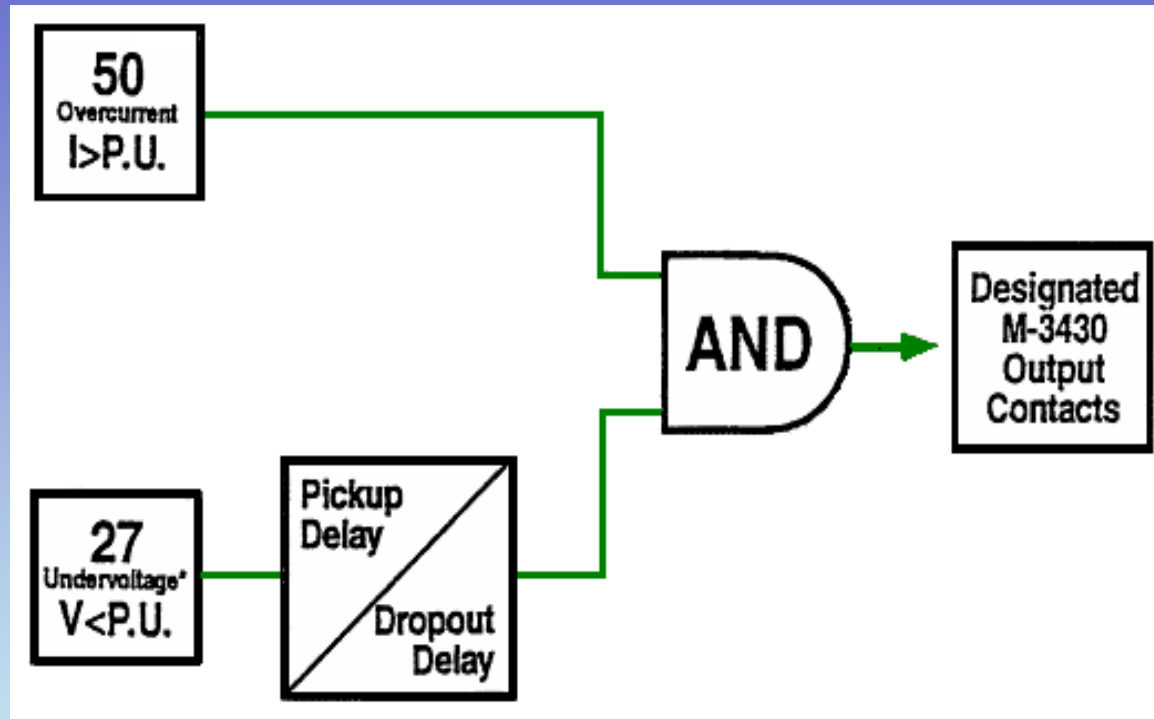
Many times conventional protection is disabled when the unit is off-line

- ❖ Removal of AC potential transformer fuses or links
- ❖ Removal of D.C. control power
- ❖ Auxiliary contact (52a) of breaker or switches can disable tripping

Dedicated Protection Schemes to Detect Inadvertent Energizing

- ❖ Frequency supervised overcurrent scheme
- ❖ Voltage supervised overcurrent scheme
- ❖ Directional overcurrent scheme
- ❖ Impedance relays scheme
- ❖ Auxiliary contact enabled overcurrent scheme

Inadvertent Energizing Protection



*Positive Sequence Voltage

Inadvertent Energizing Protection

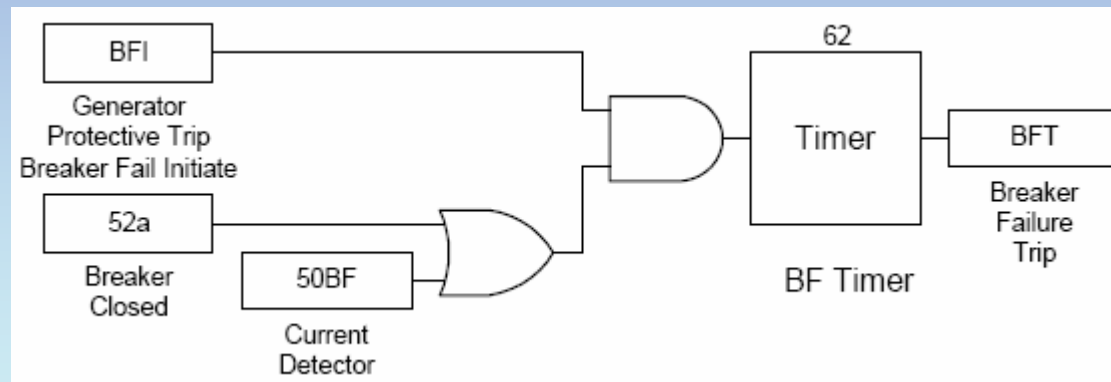
Settings summary per IEEE C37.102

- 50: $P.U \leq 50\%$ of the worst-case current value and should be $< 125\%$ generator rated current.
- 27: $70\% V_n$, time: 1.5 s

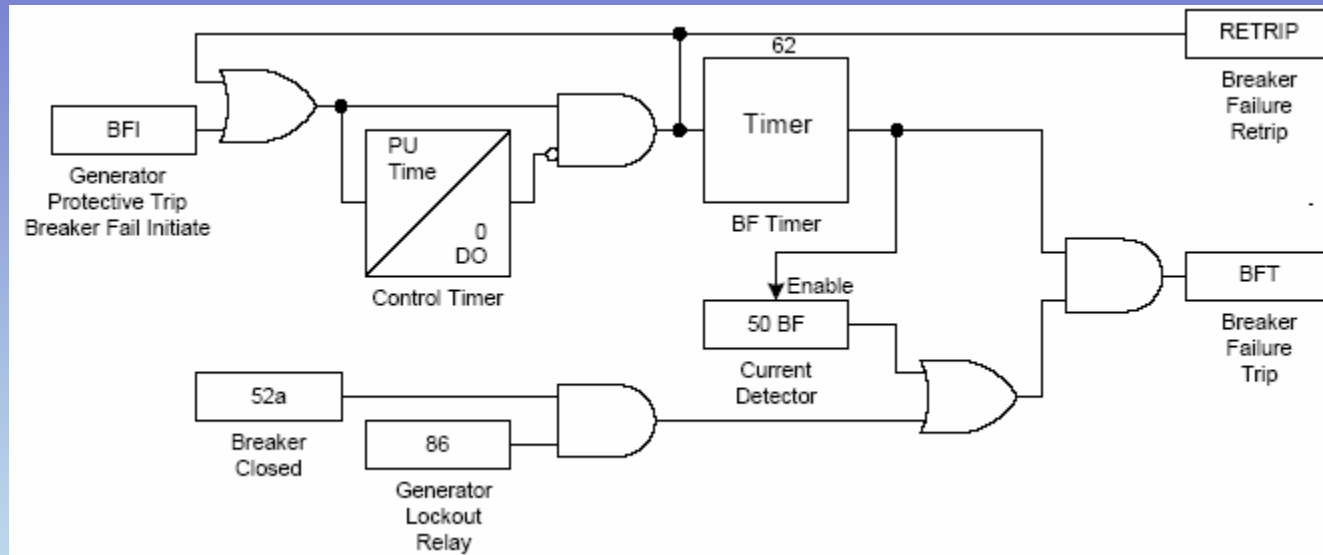
Generator Circuit Breaker Failure (50BF)

Generator Circuit Breaker Failure

- If a breaker does not clear the fault or abnormal condition in a specified time, the timer will trip the necessary breakers to remove the generator from the system.
- To initiate the breaker-failure timer, a protective relay must operate and a current detector or a breaker "a" switch must indicate that the breaker has failed to open, as shown in the Figure.



Generator Circuit Breaker Failure



Functional diagram of alternate generator breaker failure scheme

Generator Circuit Breaker Failure

Settings summary per IEEE C37.102

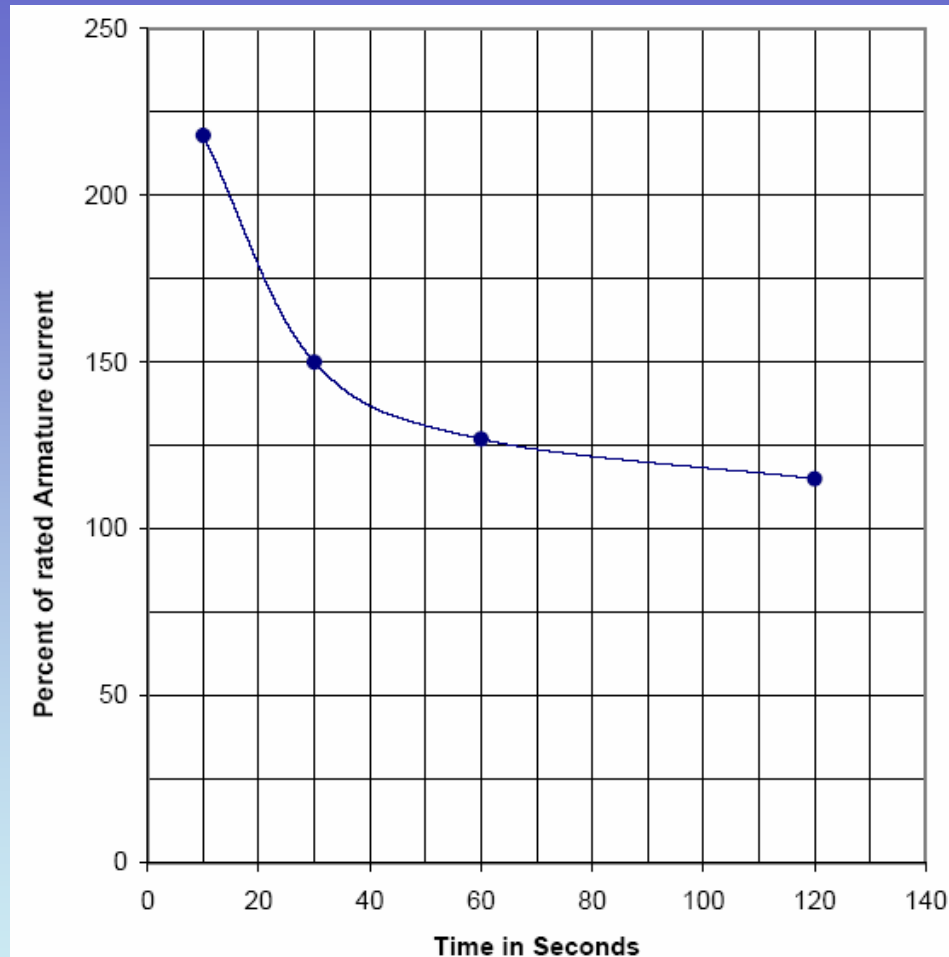
- Current detector PU: should be more sensitive than the lowest current present during fault involving currents.
- Timer: $> \text{Gen breaker interrupting time} + \text{Current detector dropout time} + \text{safety margin}$

Overcurrent Protection (50/51)

Overcurrent Protection

- In some instances, generator overload protection may be provided through the use of a torque controlled overcurrent relay that is coordinated with the ANSI C50.13-2004 short-time capability curve
- This relay consists of an instantaneous overcurrent unit and a time overcurrent unit having an extremely inverse characteristic.
- An overload alarm may be desirable to give the operator an opportunity to reduce load in an orderly manner.
- This alarm should not give nuisance alarms for external faults and should coordinate with the generator overload protection if this protection is provided.

Overcurrent Protection



Turbine-generator short-time thermal capability for balanced 3-phase loading (From ANSI C50.13-2004)

Overcurrent Protection

Settings summary per IEEE C37.102

- 51PU: 75-100% FLC, time: 7 s at 226% FLC.
Where FLC: full load current.
- 50PU: 115% FLC, time: instantaneous
Dropout: 95% of 50PU or higher

Voltage Controlled or Voltage Restrained Time Overcurrent (51 V)

Voltage Controlled or Voltage Restrained Time Overcurrent

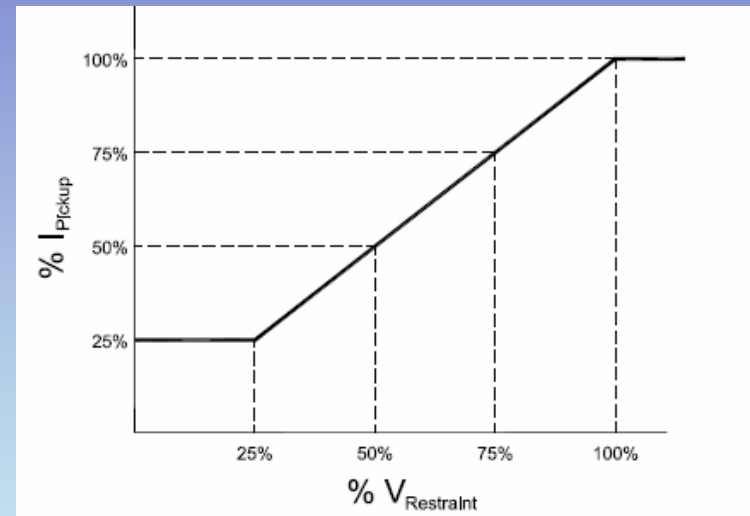
- Faults close to generator terminals may result in voltage drop and fault current reduction, especially if the generators are isolated and the faults are severe.
- Therefore, in generation protection it is important to have voltage control on the overcurrent time-delay units to ensure proper operation and co-ordination.
- These devices are used to improve the reliability of the relay by ensuring that it operates before the generator current becomes too low.
- There are two types of overcurrent relays with this feature – voltage-controlled and voltage-restrained, which are generally referred to as type 51V relays.

Voltage Controlled or Voltage Restrained Time Overcurrent

- The voltage-controlled (51/27C) feature allows the relays to be set below rated current, and operation is blocked until the voltage falls well below normal voltage.
- The voltage-controlled approach typically inhibits operation until the voltage drops below a pre-set value.
- It should be set to function below about 80% of rated voltage with a current pick-up of about 50% of generator rated current.

Voltage Controlled or Voltage Restrained Time Overcurrent

- The voltage-restrained (51/27R) feature causes the pick-up to decrease with reducing voltage, as shown in Figure.
- For example, the relay can be set for 175% of generator rated current with rated voltage applied. At 25% voltage the relay picks up at 25% of the relay setting ($1.75 \times 0.25 = 0.44$ times rated).
- The varying pick-up level makes it more difficult to co-ordinate the relay with other fixed pick-up overcurrent relays.



Voltage Controlled or Voltage Restrained Time Overcurrent

Settings summary per IEEE C37.102

Voltage Controlled:

- Overcurrent PU: 50% FLC
- Control voltage: 75%VNOM.
- Inverse time curve and dial settings should be set to coordinate with system line relays for close-in faults on the transmission lines at the plant.

Voltage Restrained:

- Overcurrent PU: 150% FLC at rated voltage
- Inverse time curve and dial settings should be set to coordinate with system line relays for close-in faults on the transmission lines at the plant.

Overvoltage (59)

Overvoltage

- Generator overvoltage may occur without necessarily exceeding the V/Hz limits of the machine.
- Protection for generator overvoltage is provided with a frequency-compensated (or frequency insensitive) overvoltage relay.
- The relay should have both an instantaneous unit and a time delay unit with an inverse time characteristic.
- Two definite time delay relays can also be applied.

Overvoltage

Settings summary per IEEE C37.102

Relays with inverse time characteristic and instantaneous

- PU : $110\%V_n$; $t = 2.5$ s at 140% of PU setting
- Inst : 130 - 150% V_n

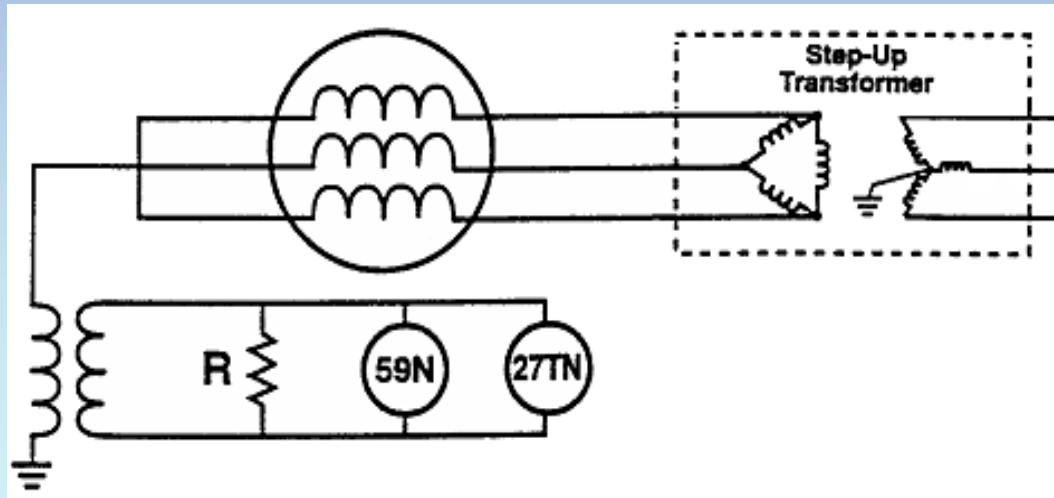
Relays with definite time characteristic and two stages

- Alarm PU : $110\%V_n$; $10 < t < 15$ s
- Trip PU : 150% V_n ; time: 2s

100% Stator Ground (59N/27TH)

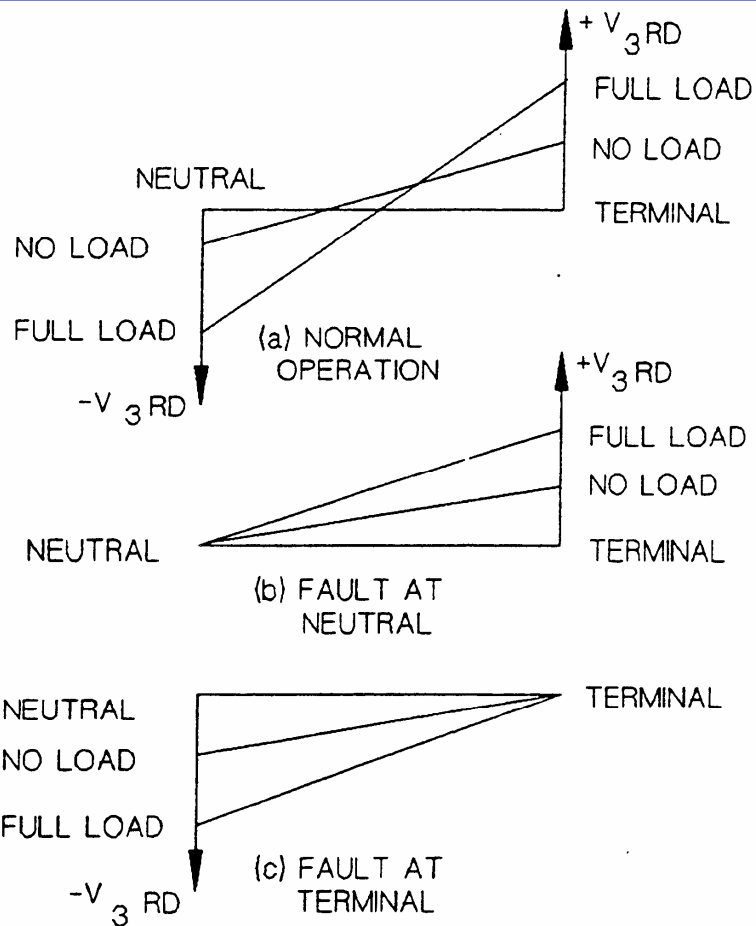
Stator Ground Protection

- ✓ Provides protection for stator ground fault on generators which are high impedance grounded
- ✓ Used on unit connected generators
- ✓ Ground current limited to about 10A primary
- ✓ Provides 100% stator ground protection (entire winding)



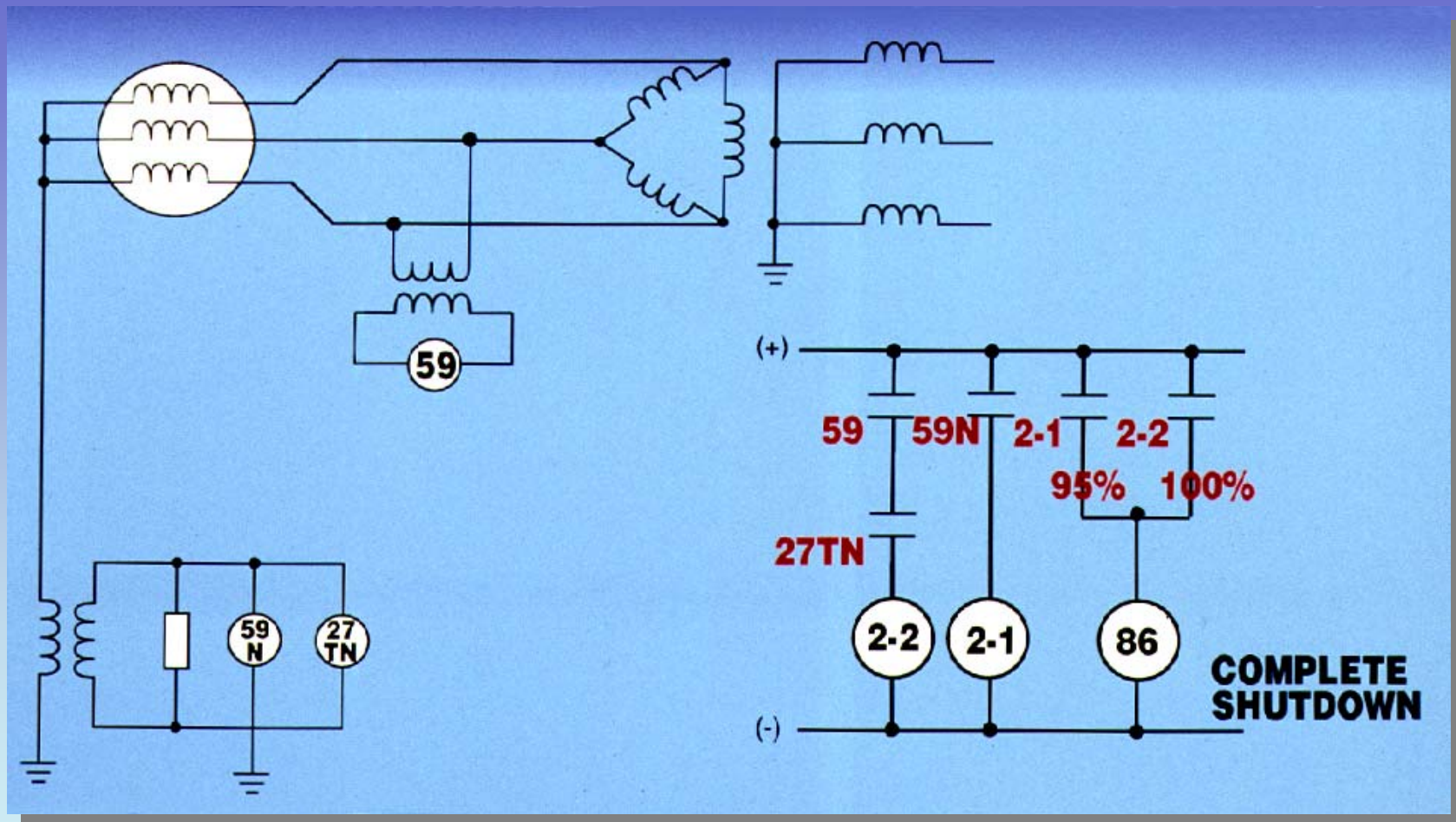
High Impedance Grounding

3rd Harmonic Comparator for 100% Stator Ground Fault Protection



- 3rd harmonic levels change with position of ground fault and loading
- Using a comparator technique of 3rd harmonic voltages at line and neutral ends allows an overvoltage element to be applied

100% Stator Ground Fault (59N/27TN)



Third-Harmonic Undervoltage Ground-Fault Protection Scheme

Stator Ground

Settings summary per IEEE C37.102

- 59G element: Pickup = 5 V; t = 5 s

Note: Time setting must be selected to provide coordination with other system protective devices.

- 27TH element: Pickup = 50% of minimum normal generator 3rd harmonic. t = 5 s

Field Ground (64F)

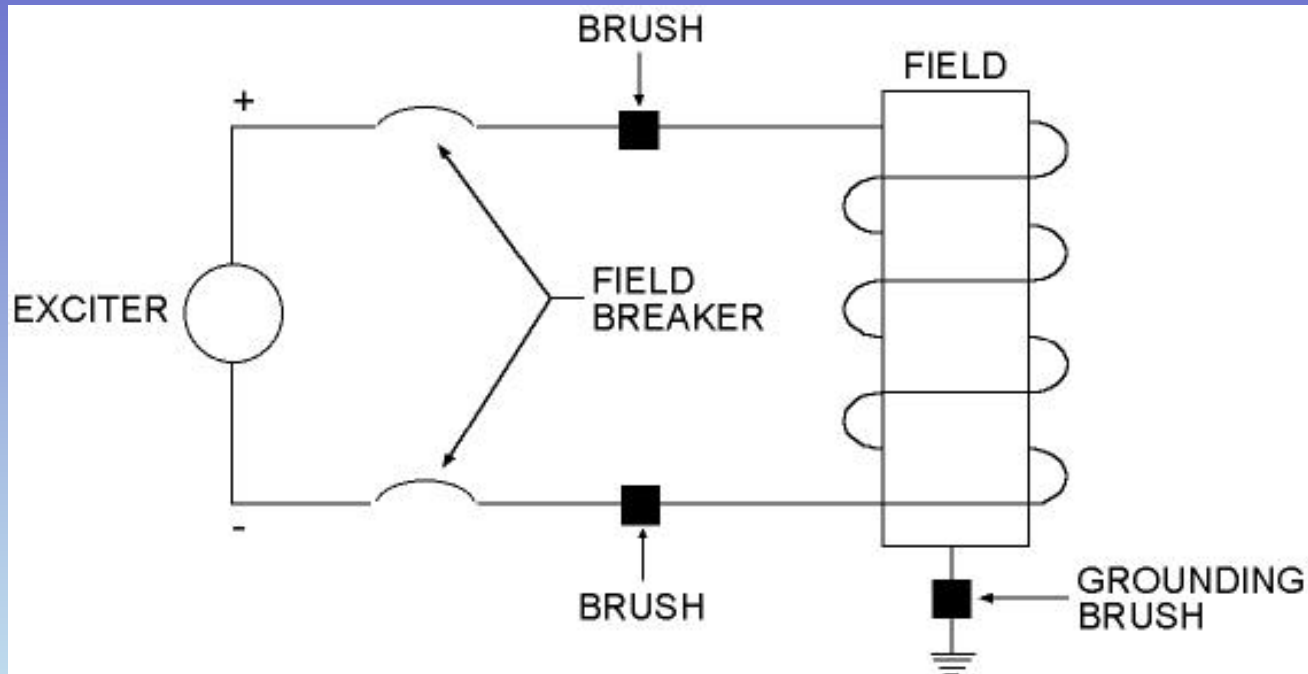
Field (Rotor) Ground Fault Protection

- The field circuit of a generator is an ungrounded system. As such, a single ground fault will not generally affect the operation of a generator.
- However, if a second ground fault occurs, a portion of the field winding will be short circuited, thereby producing unbalanced air gap fluxes in the machine.
- These unbalanced fluxes may cause rotor vibration that may quickly damage the machine; also, unbalanced rotor winding and rotor body temperatures caused by uneven rotor winding currents may cause similar damaging vibrations.

Field (Rotor) Ground Fault Protection

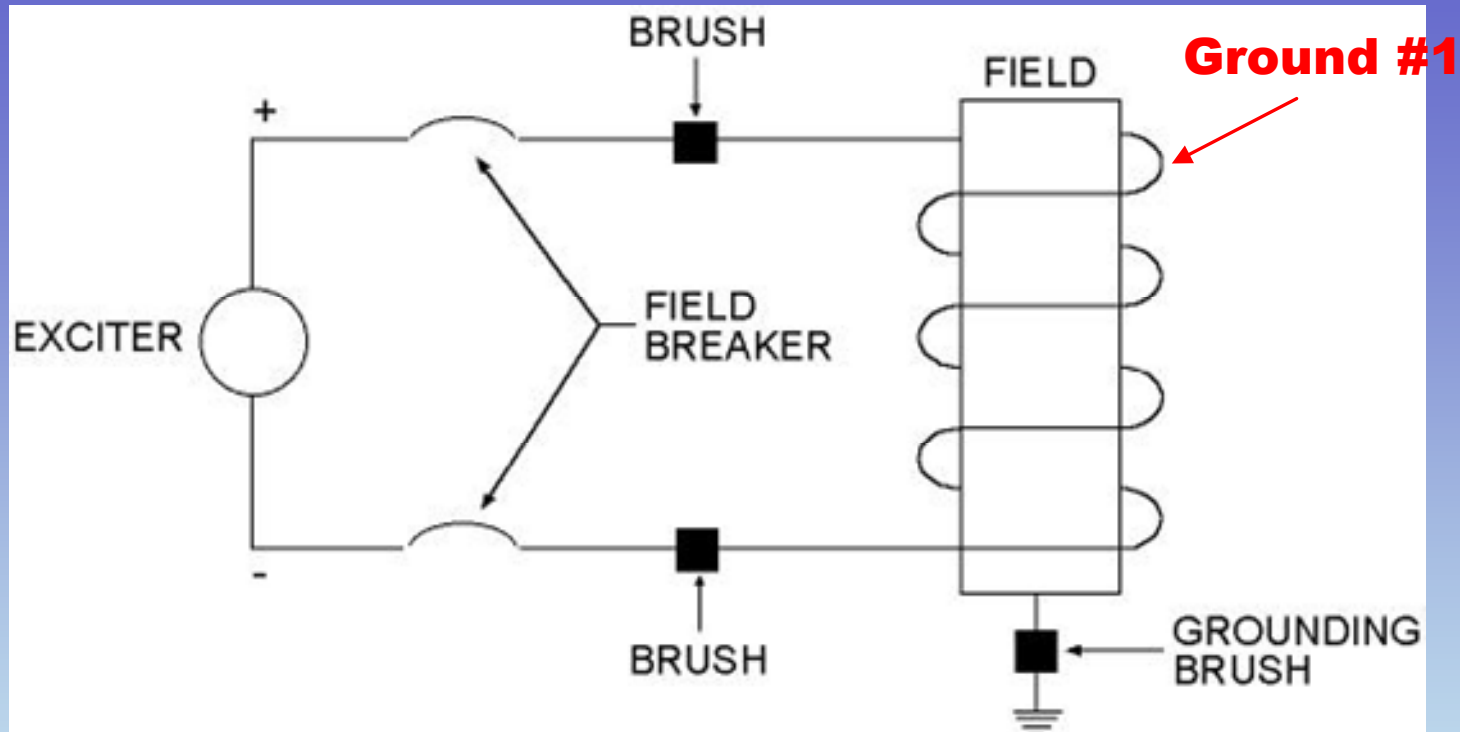
- The probability of the second ground occurring is greater than the first, since the first ground establishes a ground reference for voltages induced in the field by stator transients, thereby increasing the stress to ground at other points on the field winding.
- On a brushless excitation system continuous monitoring for field ground is not possible with conventional field ground relays since the generator field connections are contained in the rotating element.
- Insurance companies consider this is the most frequent internal generator fault
- Review existing 64F voltage protection methods

Typical Generator Field Circuit



- A single field ground fault will not:
 - affect the operation of a generator
 - produce any immediate damaging effects

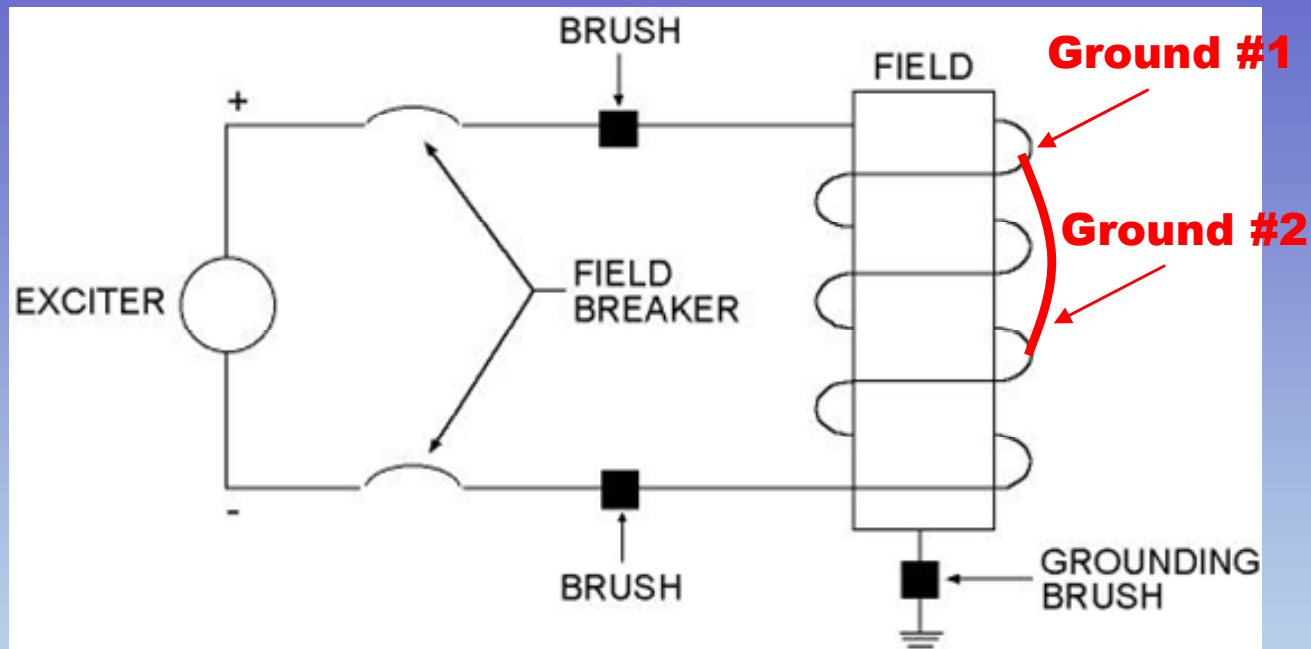
Typical Generator Field Circuit



■ The first ground fault will:

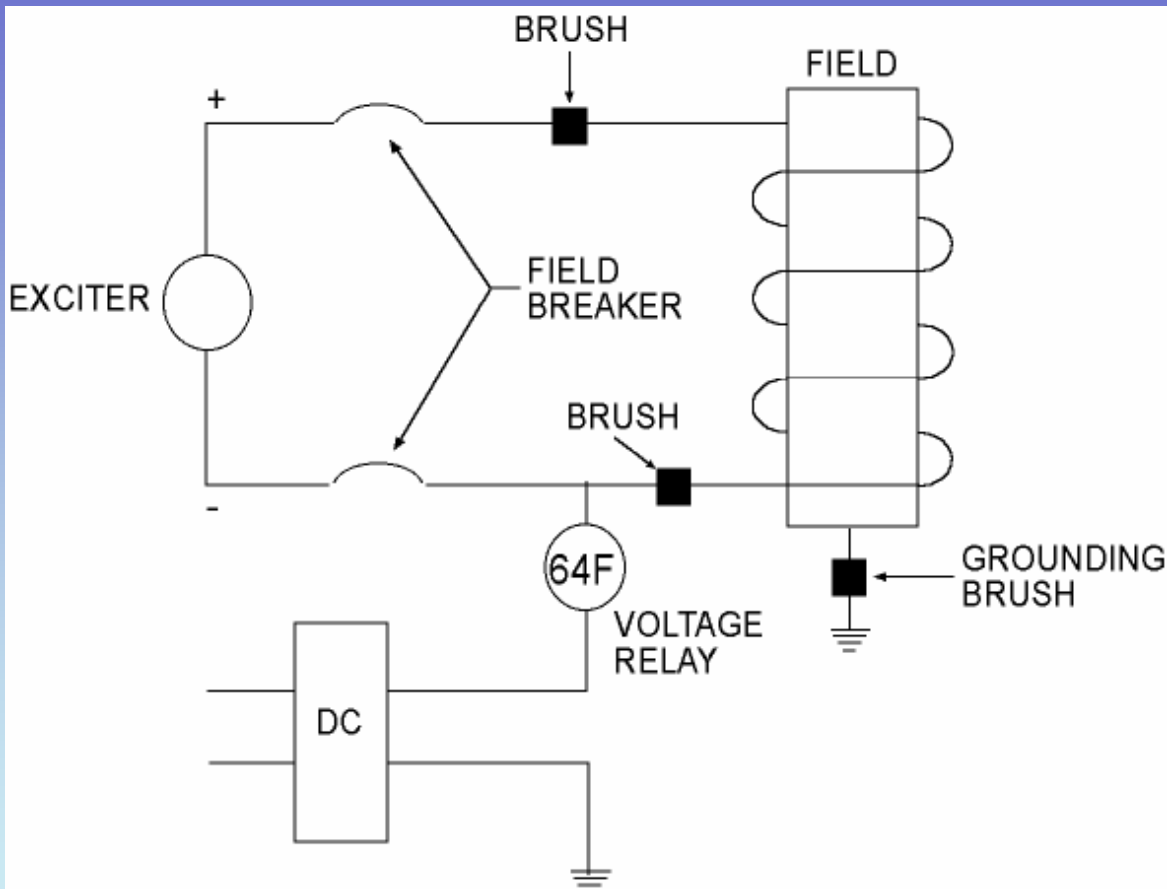
- establish a ground reference making a second ground fault more likely
- increase stress to ground at other points in field winding

Typical Generator Field Circuit



- The second ground fault will:
 - short out part of field winding causing unit vibrations
 - cause rotor heating from unbalanced currents
 - cause arc damage at the points of fault

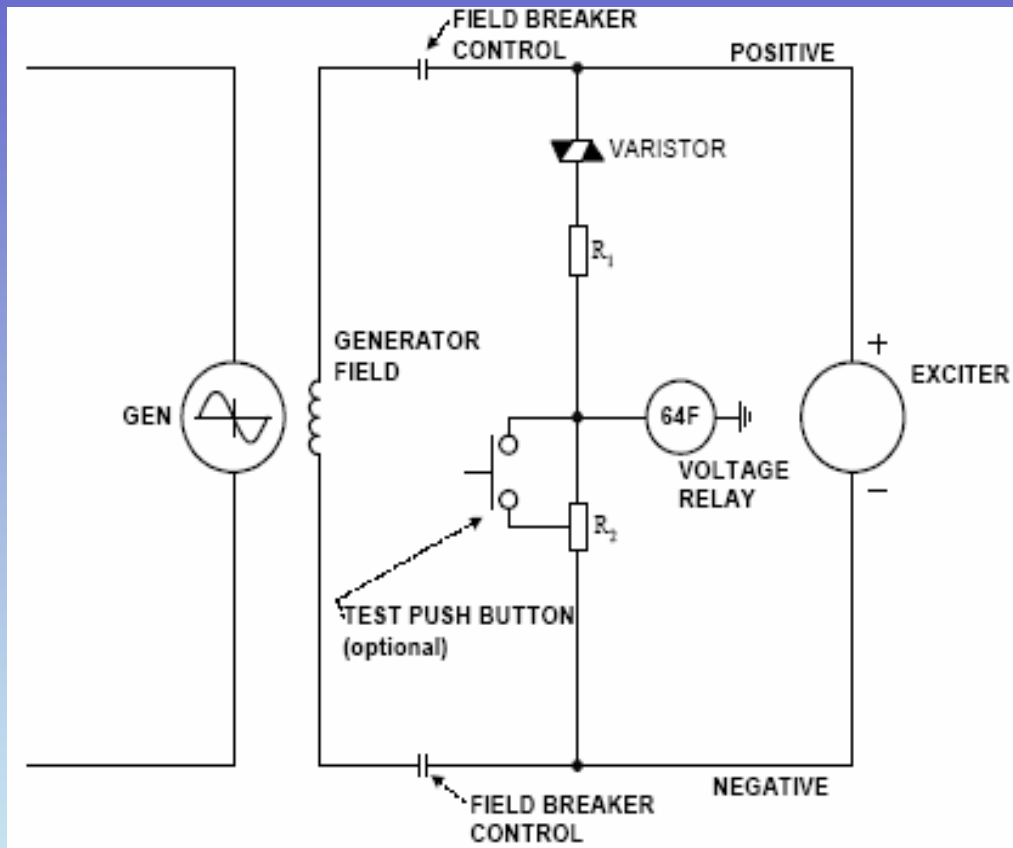
Detection Using a DC Source



➤ A dc voltage source in series with an overvoltage relay coil is connected between the negative side of the generator field winding and ground.

➤ A ground anywhere in the field will cause the relay to operate.

Detection Using a Voltage Divider



➤ This method uses a voltage divider and a sensitive overvoltage relay between the divider midpoint and ground.

➤ A maximum voltage is impressed on the relay by a ground on either the positive or negative side of the field circuit.

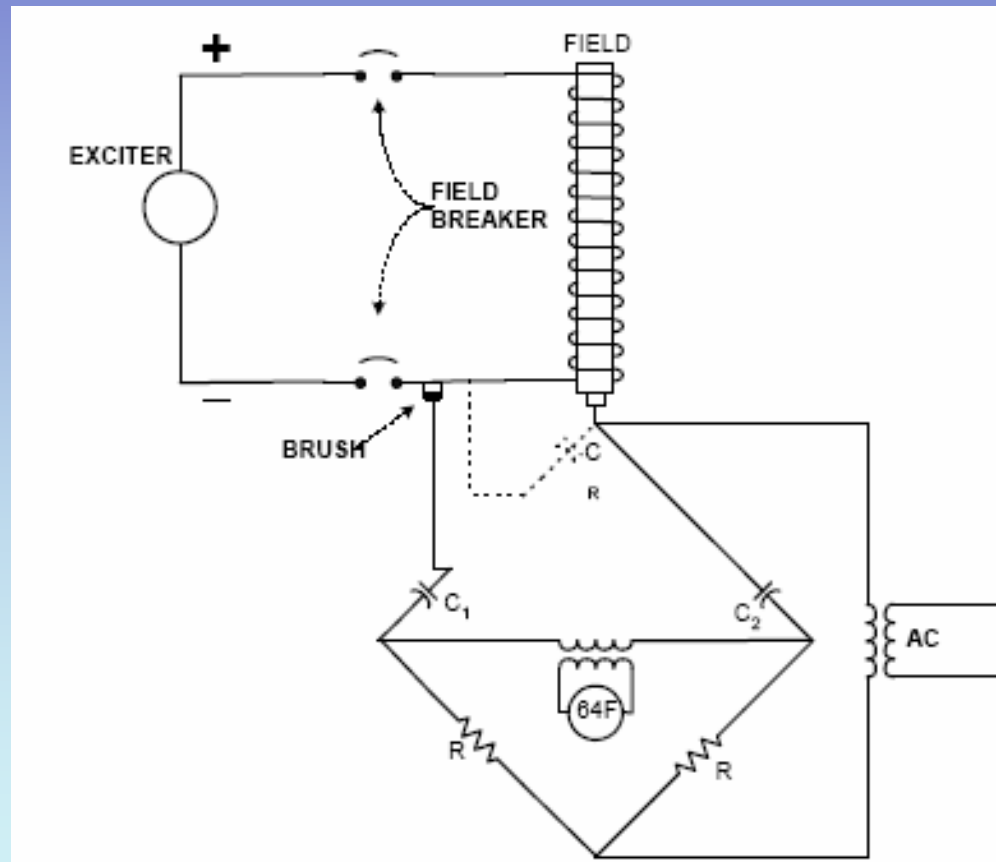
This generator field ground relay is designed to overcome the null problem by using a nonlinear resistor (varistor) in series with one of the two linear resistors in the voltage divider.

Detection Using Pilot Brushes

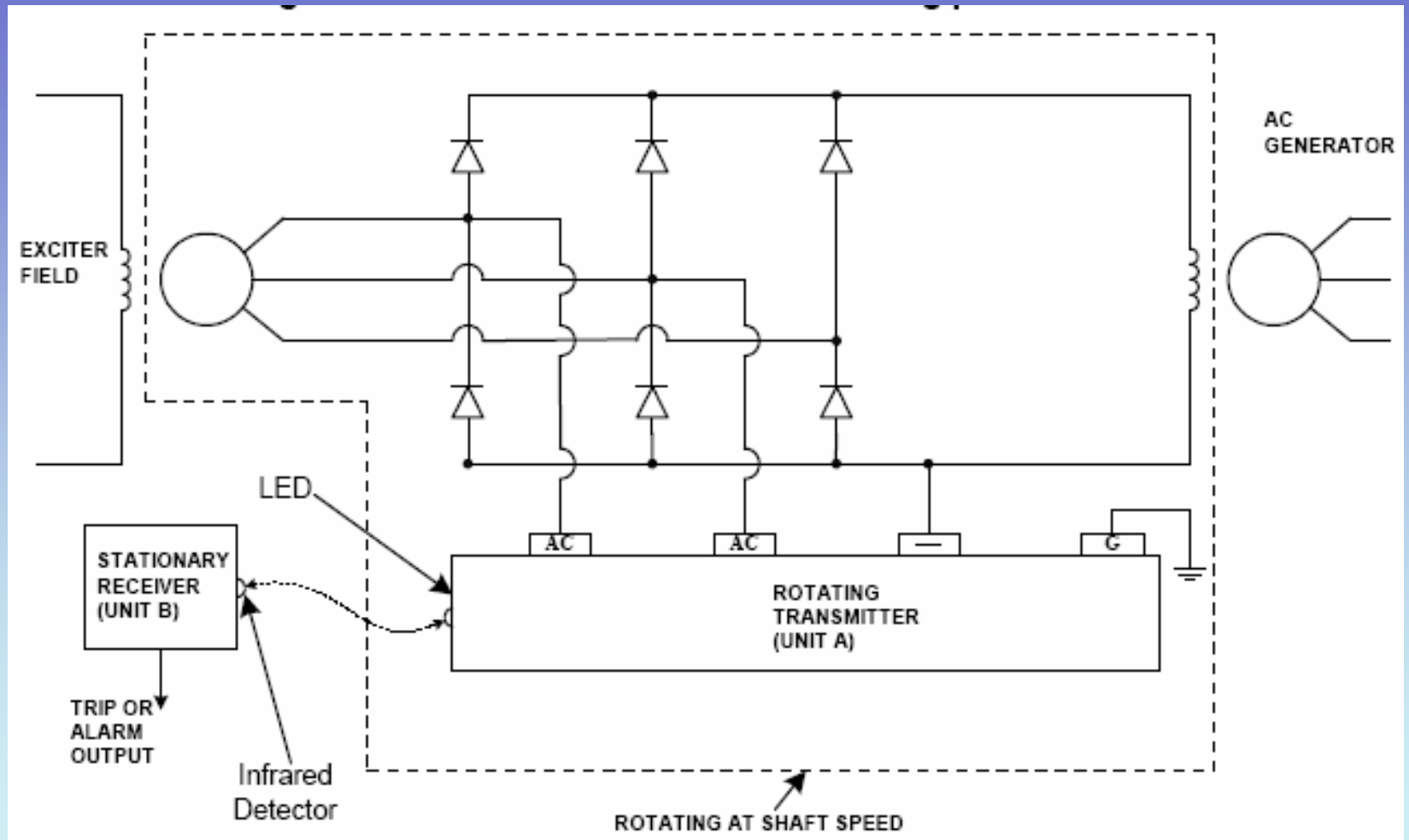
- The addition of a pilot brush or brushes is to gain access to the rotating field parts.
- Normally this is not done since eliminating the brushes is one of the advantages of a brushless system.
- A ground fault shorts out the field winding to rotor capacitance, CR, which unbalances the bridge circuit.
- If a voltage is read across the 64F relay, then a ground exists
- Detection systems may be used to detect field grounds if a collector ring is provided on the rotating shaft along with a pilot brush that may be periodically dropped to monitor the system.

Detection Using Pilot Brushes

- The brushes used in this scheme are not suitable for continuous contact with the collector rings.



Field Ground Detection for Brushless Machines LED Communications



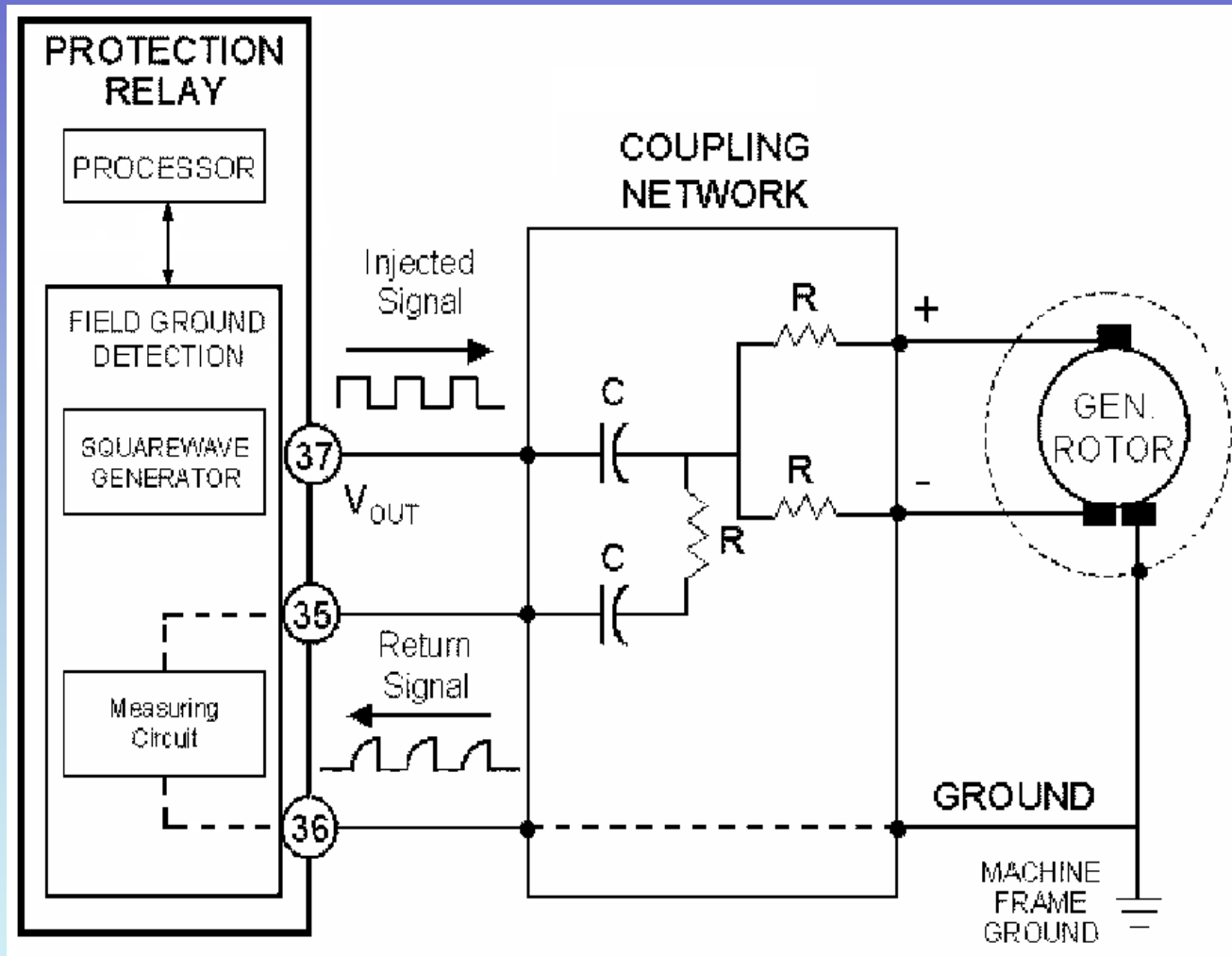
Field Ground Detection for Brushless Machines with Infrared LED Communications

- The relay's transmitter is mounted on the generator field diode wheel.
- Its source of power is the ac brushless exciter system. Two leads are connected to the diode bridge circuit of the rotating rectifier to provide this power.
- Ground detection is obtained by connecting one lead of the transmitter to the negative bus of the field rectifier and the ground lead to the rotor shaft.
- Sensing current is determined by the field ground resistance and the location of a fault with respect to the positive and negative bus.

Field Ground Detection for Brushless Machines with Infrared LED Communications

- The transmitter Light Emitting Diodes (LEDs) emit light for normal conditions.
- The receiver's infrared detectors sense the light signal from the LED across the air gap.
- Upon detection of a fault, the LED's are turned off. Loss of LED light to the receiver will actuate the ground relay and initiate a trip or alarm

Using Injection Voltage Signal



Using Injection Voltage Signal

- In addition, digital relays may provide real-time monitoring of actual insulation resistance so deterioration with time may be monitored.
- The passive coupling network is used to isolate high dc field voltages from the relay.
- Backup protection for the above described schemes usually consists of vibration detecting equipment.
- Contacts are provided to trip the main and field breakers if vibration is above that associated with normal short circuit transients for faults external to the unit.

Field (Rotor) Ground Fault Protection

Settings summary per IEEE C37.102

- Field ground detection using DC a source: $1 < t < 3 \text{ s}$
- Field ground detection for Brushless Machines with infrared LED communications: time up to 10 s
- Field ground detection using low frequency square wave voltage injection: ALARM = $20 \text{ k}\Omega$
TRIP = $5 \text{ k}\Omega$

Generator Out-Of-Step Protection (OSP) (78)

When is OSP needed?

1. When critical switching times are short enough to warrant concern that backup clearing of a system fault could exceed critical switching time.
2. This swing locus passes through the generator or GSU
3. Credible loss of transmission lines could result in high transfer reactance between the generator and the power system

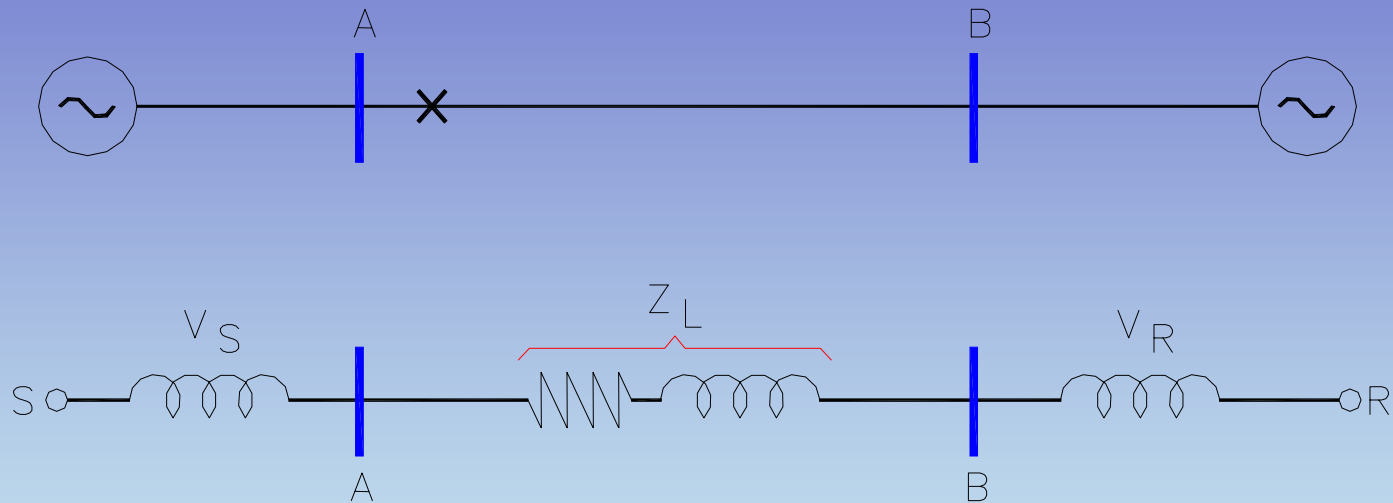
Background

- Power system stability enables the synchronous machines of a system to respond to a disturbance such as transmission system faults, sudden load changes, loss of generating units or line switching.
- Loss of synchronism is produced when the angle of the EMF of a machine increases to a level that does not allow any recovery of the plant when the machine is said to have reached a slip.
- Transient stability studies allow to determine if the system will remain in synchronism following major disturbance

OST & PSB Functions

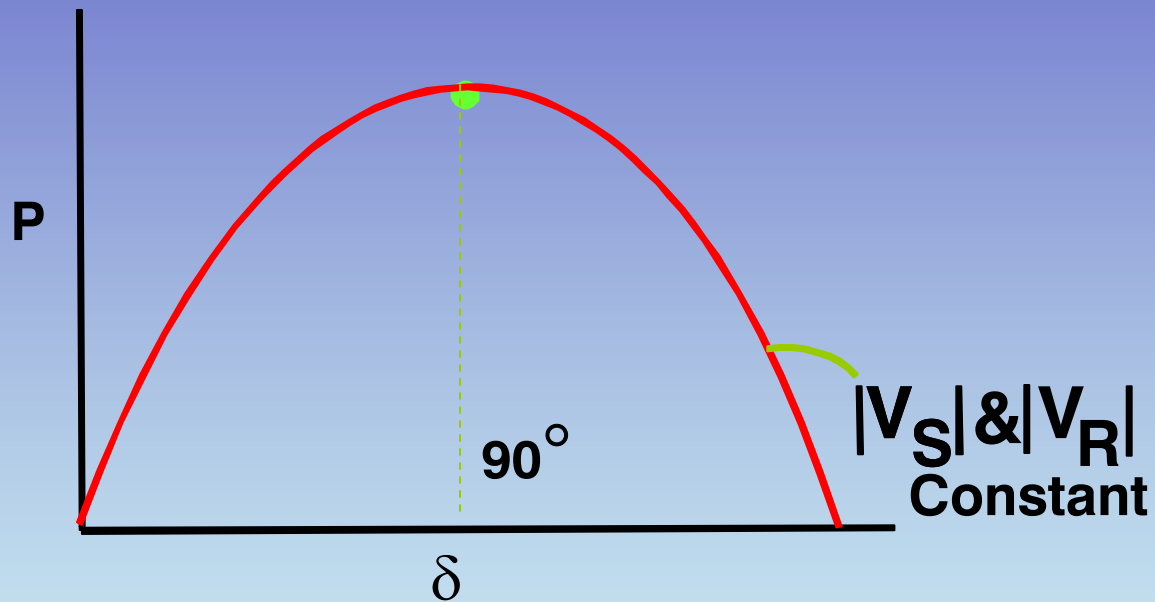
- During power system disturbances, the voltage and current which feed the relays vary with time and, as a result, the relays will also see an impedance that is varying with time.
- Certain power system disturbances may cause loss of synchronism between a generator and the rest of the utility system, or between neighboring utility interconnected power systems.
- If such a loss of synchronism occurs, it is imperative that the generator or system areas operating asynchronously are separated immediately through controlled islanding of the power system using out-of-step protection systems-OST.
- OST systems must be complemented with Power Swing Blocking (PSB) of distance relay elements prone to operate during unstable power swings. PSB prevents system separation from occurring at any locations other than the pre-selected ones.

Power Transfer Equation



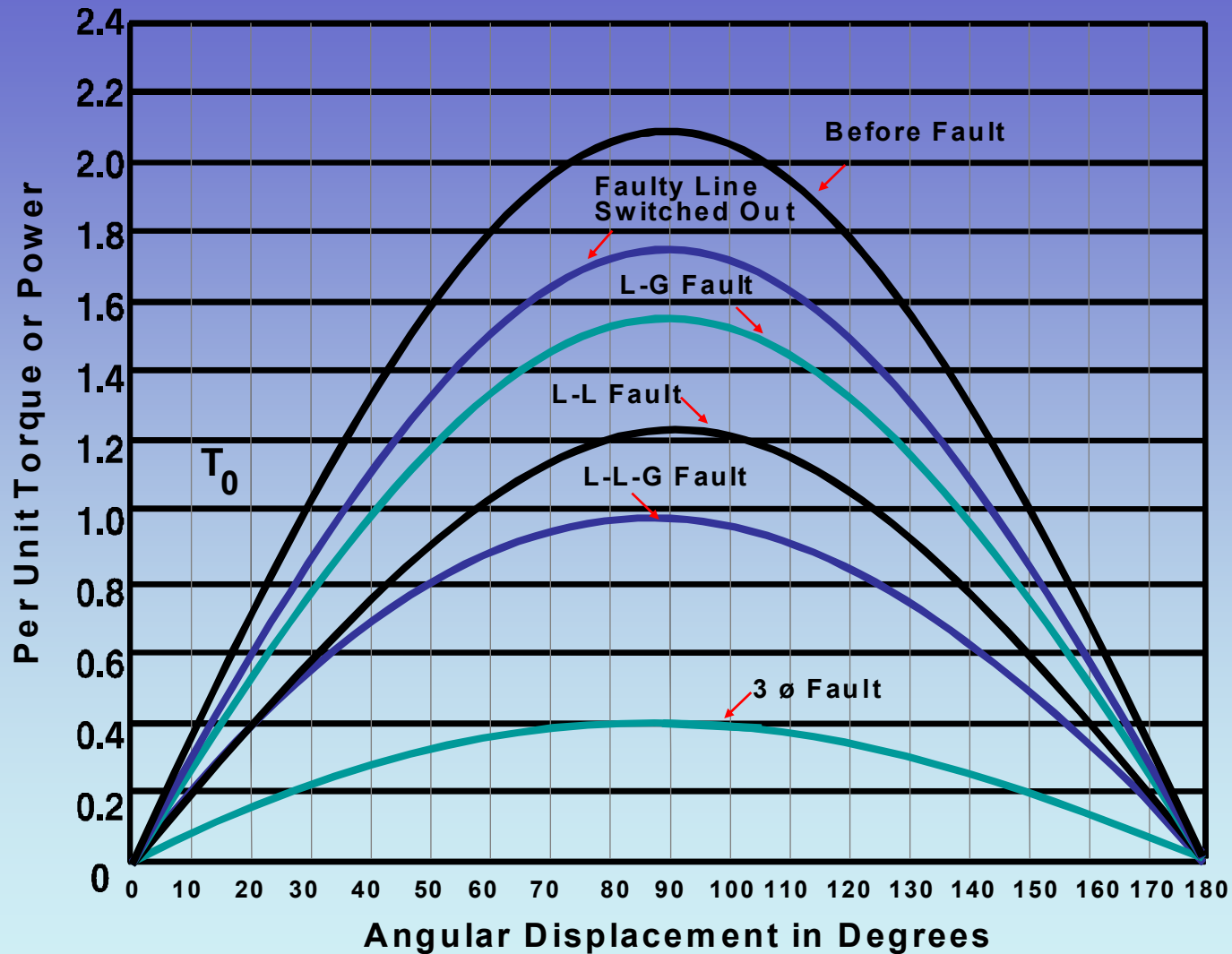
$$P = \frac{|V_S| \times |V_R|}{X} \sin \delta$$

Two-Machine System

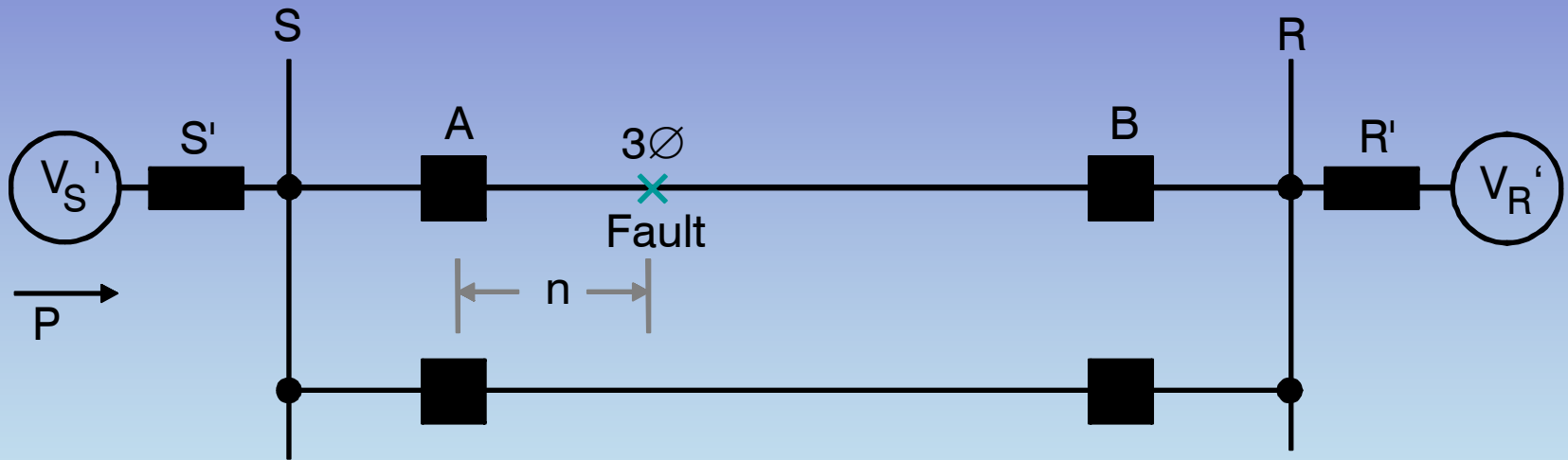


$$P = \frac{|V_S| \times |V_R|}{X} \sin \delta$$

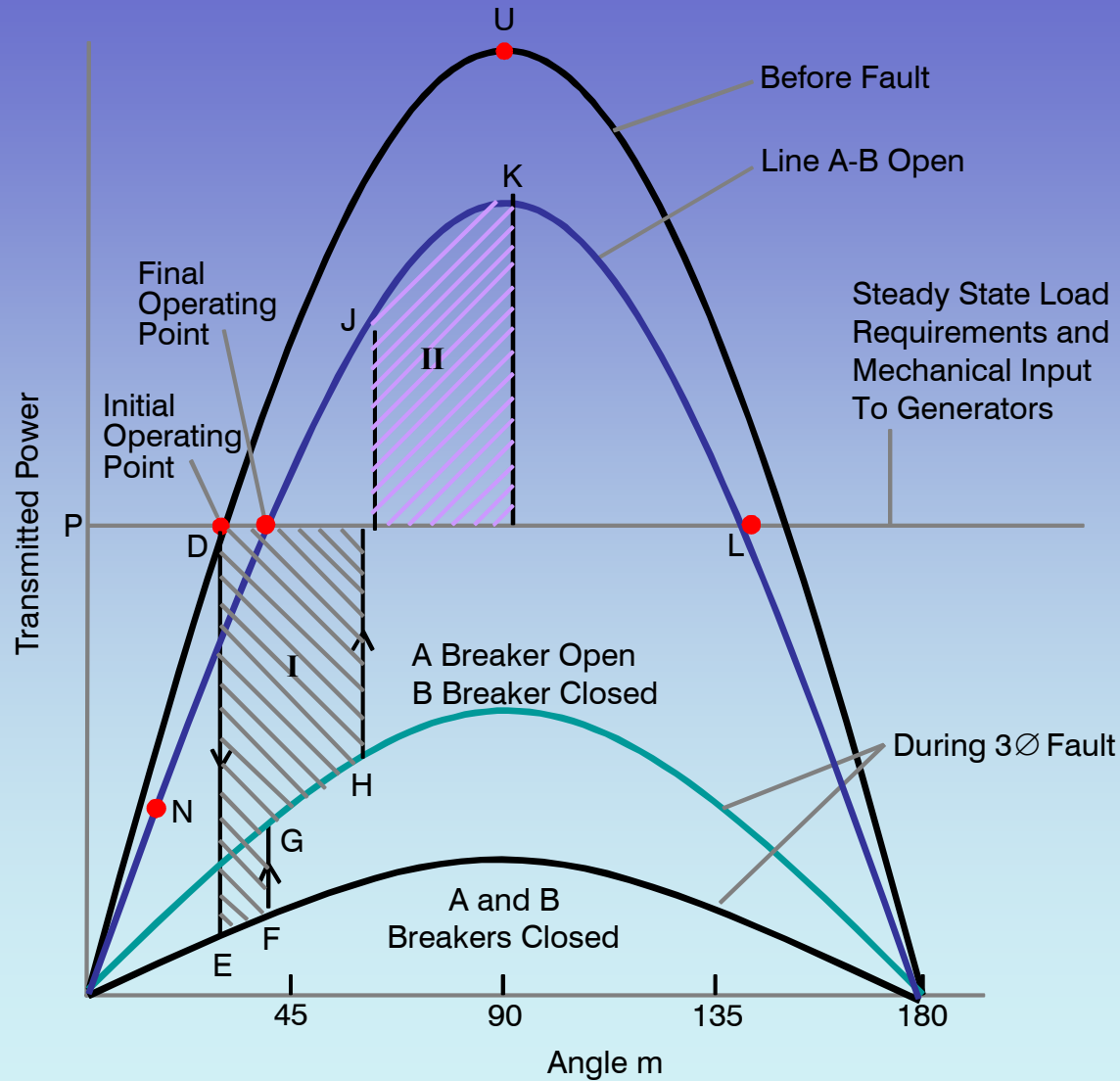
Effect of Faults on Power Transfer



Network with Three Phase Fault



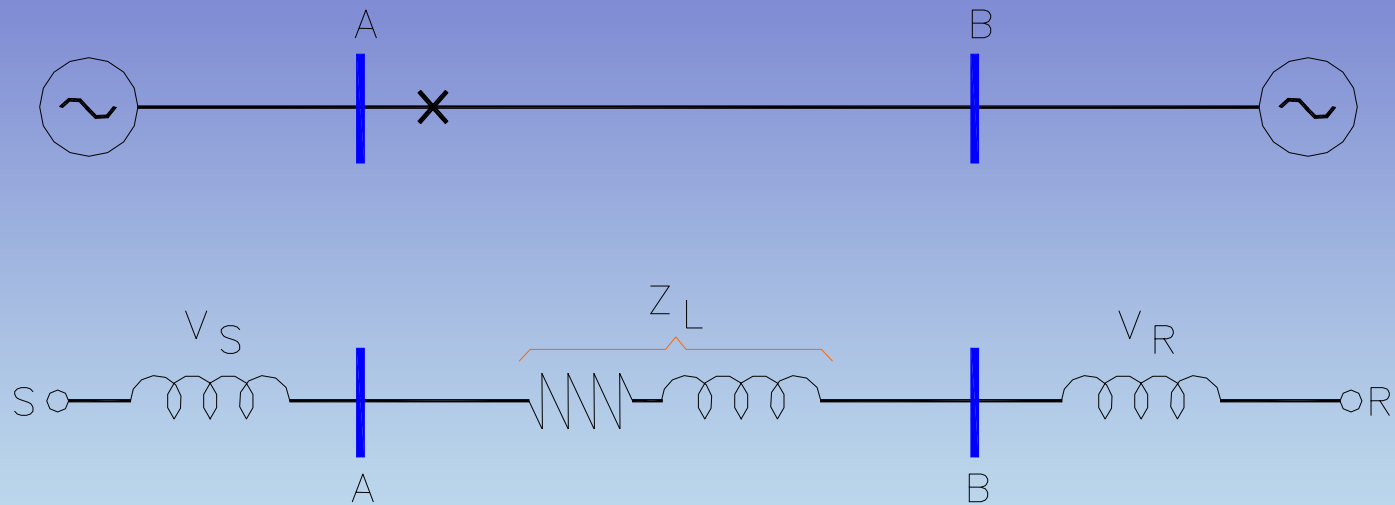
Power Transfer Curve



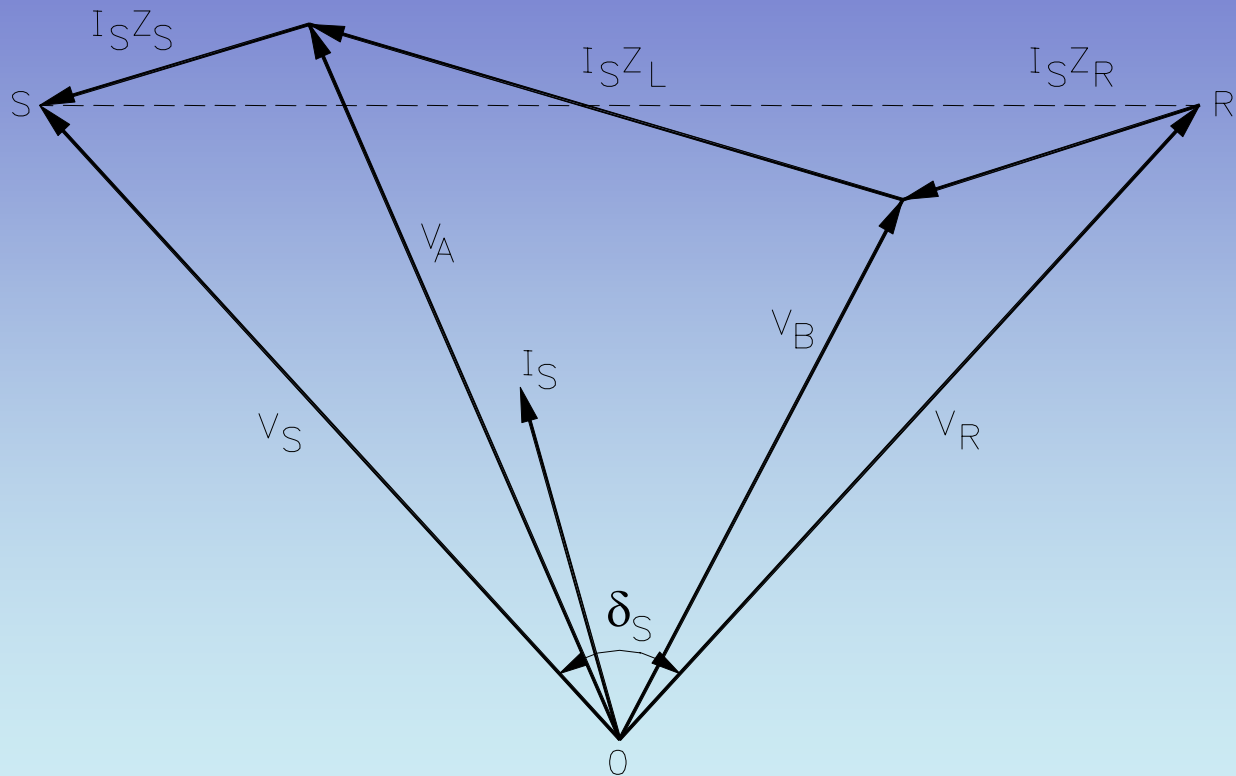
Power Transfer Curve

- Ways the protection system can mitigate the affect of the fault on power swings.
- Fast clearing
- Pilot systems
- Breaker failure systems
- Single pole tripping
- High speed reclosing
- Load shedding

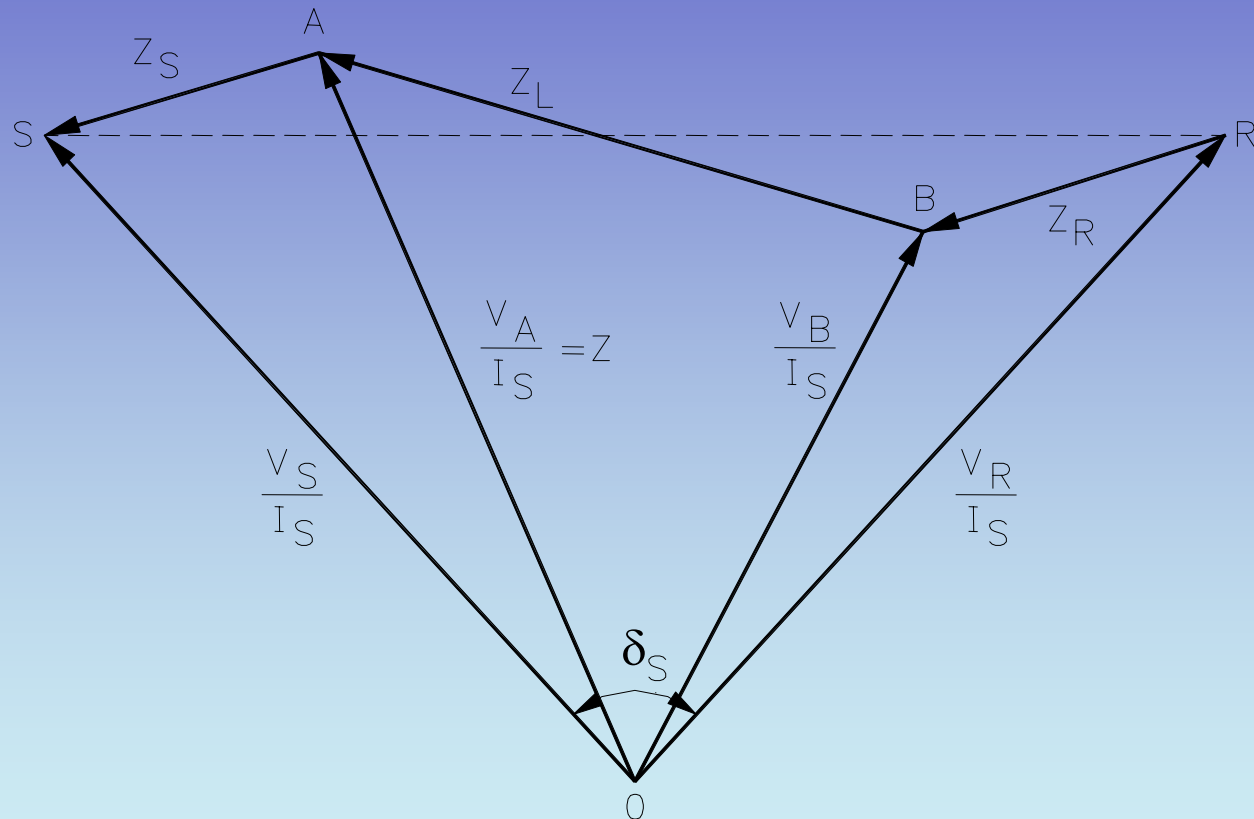
Impedances Seen by Relays



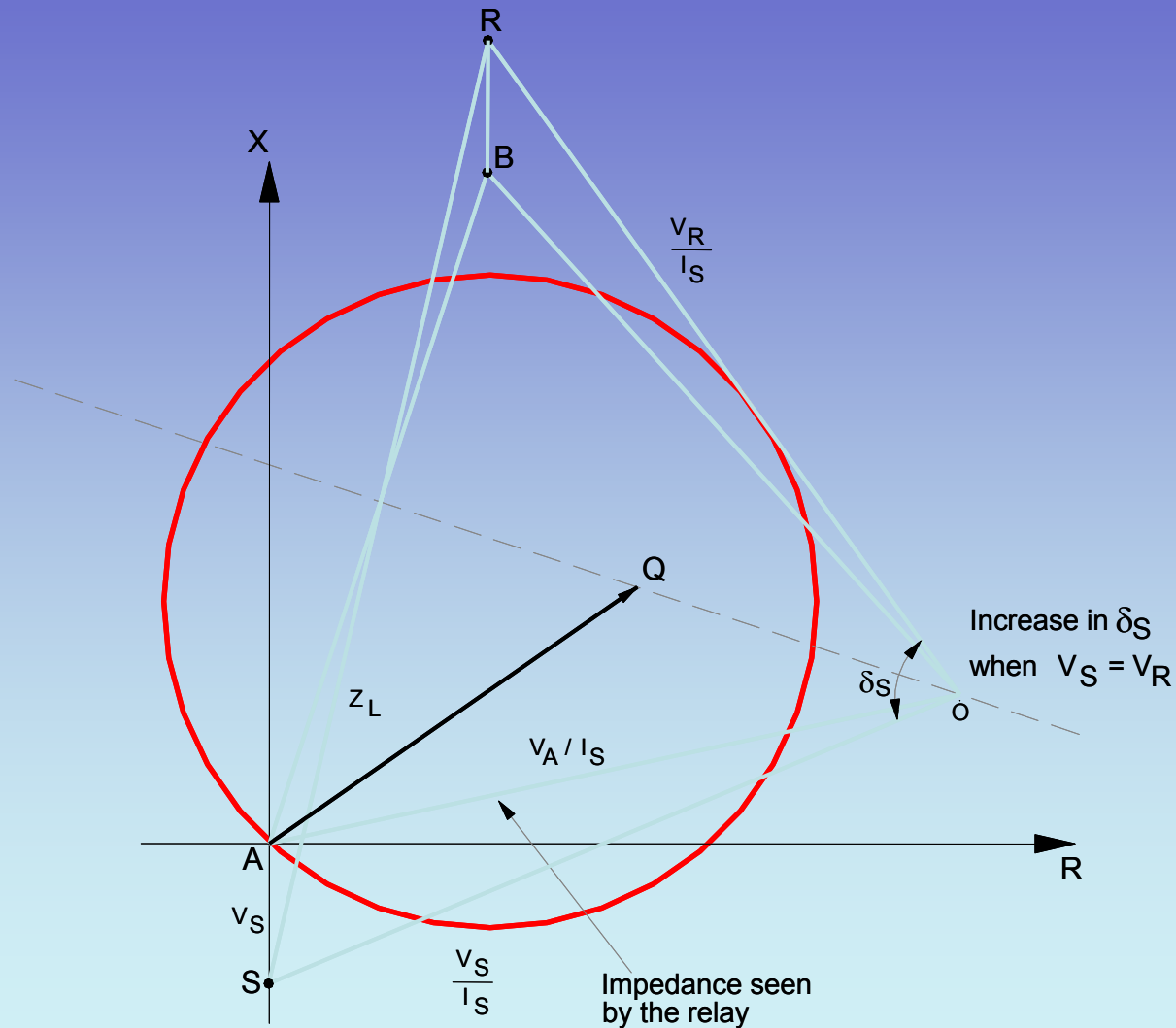
Impedances Seen by Relays



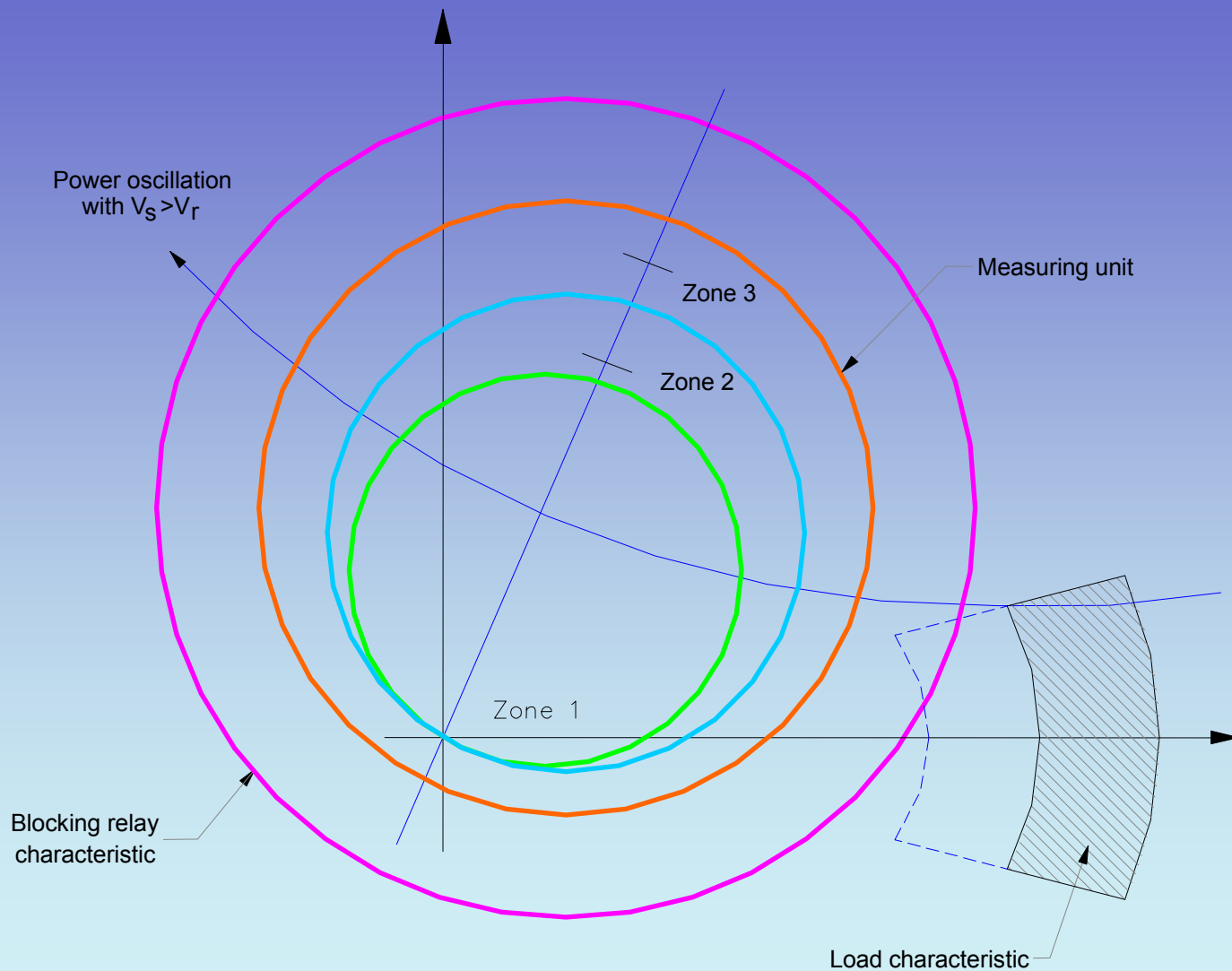
Impedances Seen by Relays



Basics of Power Swing Blocking



Basics of Power Swing Blocking

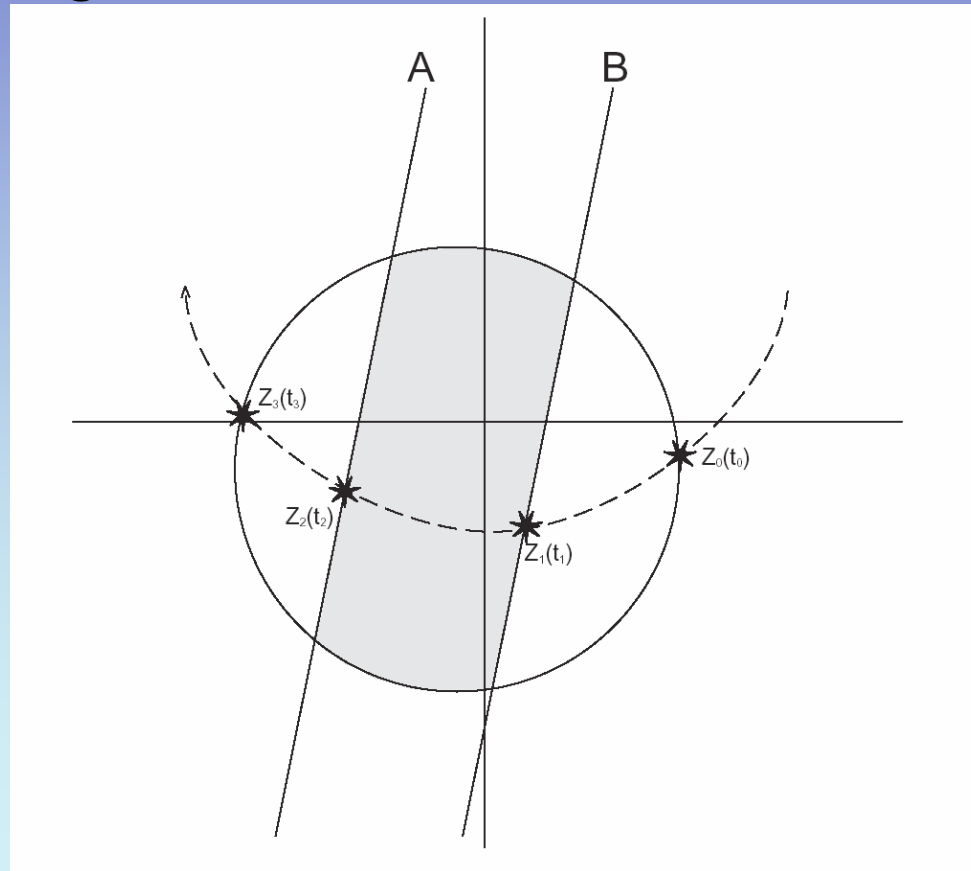


Basics of Out of Step Protection

- The Out-of-Step function (78) is used to protect the generator from out-of-step or pole slip conditions.
- There are different ways to implement Out of Step Protection.
- One of the commonest types uses one set of blinders, along with a supervisory MHO element.

Basics of Out of Step Protection

- The pickup area is restricted to the shaded area, defined by the inner region of the MHO circle, the region to the right of the blinder A and the region to the left of blinder B.



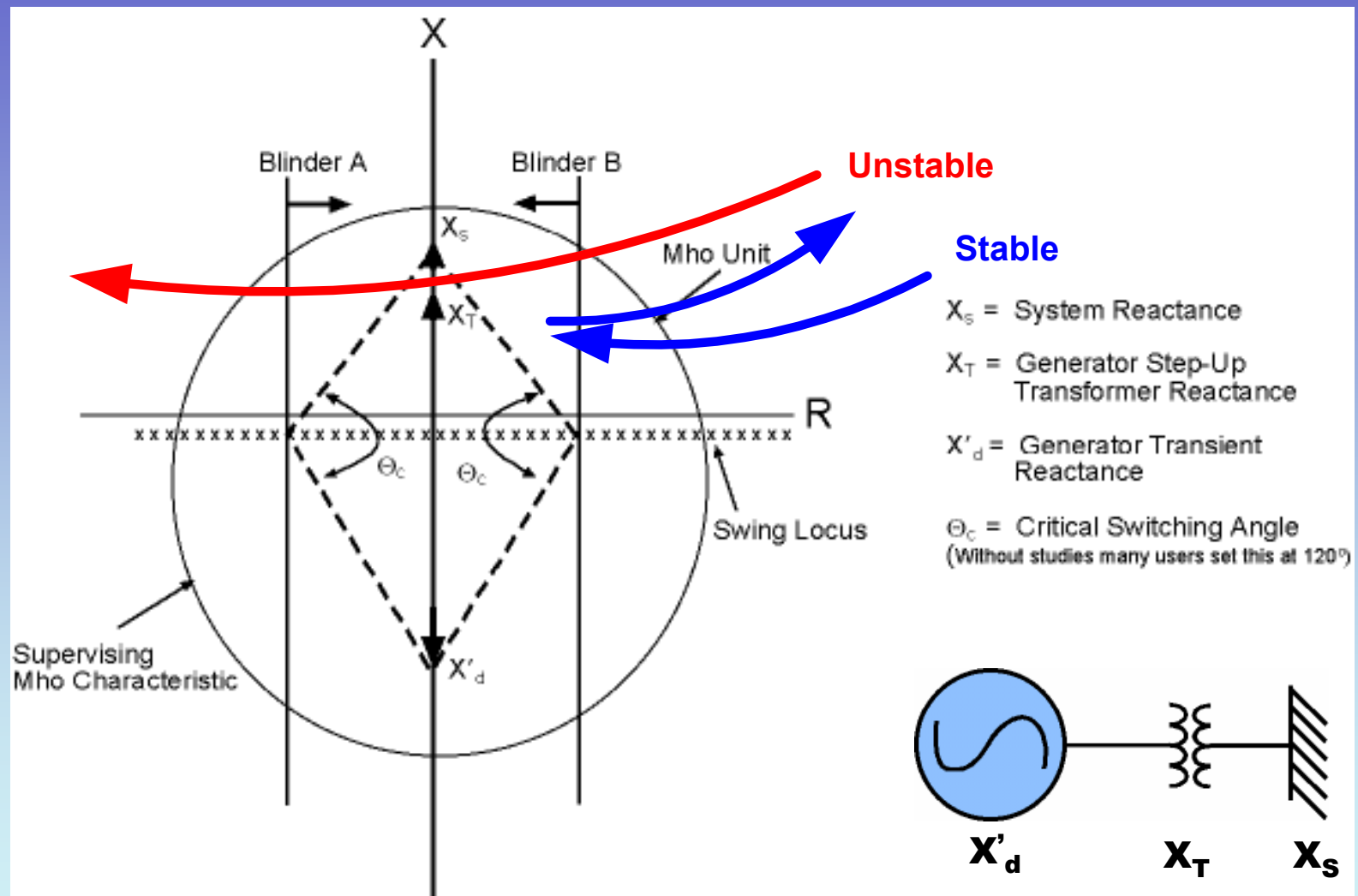
Basics of Out of Step Protection

For operation of the blinder scheme :

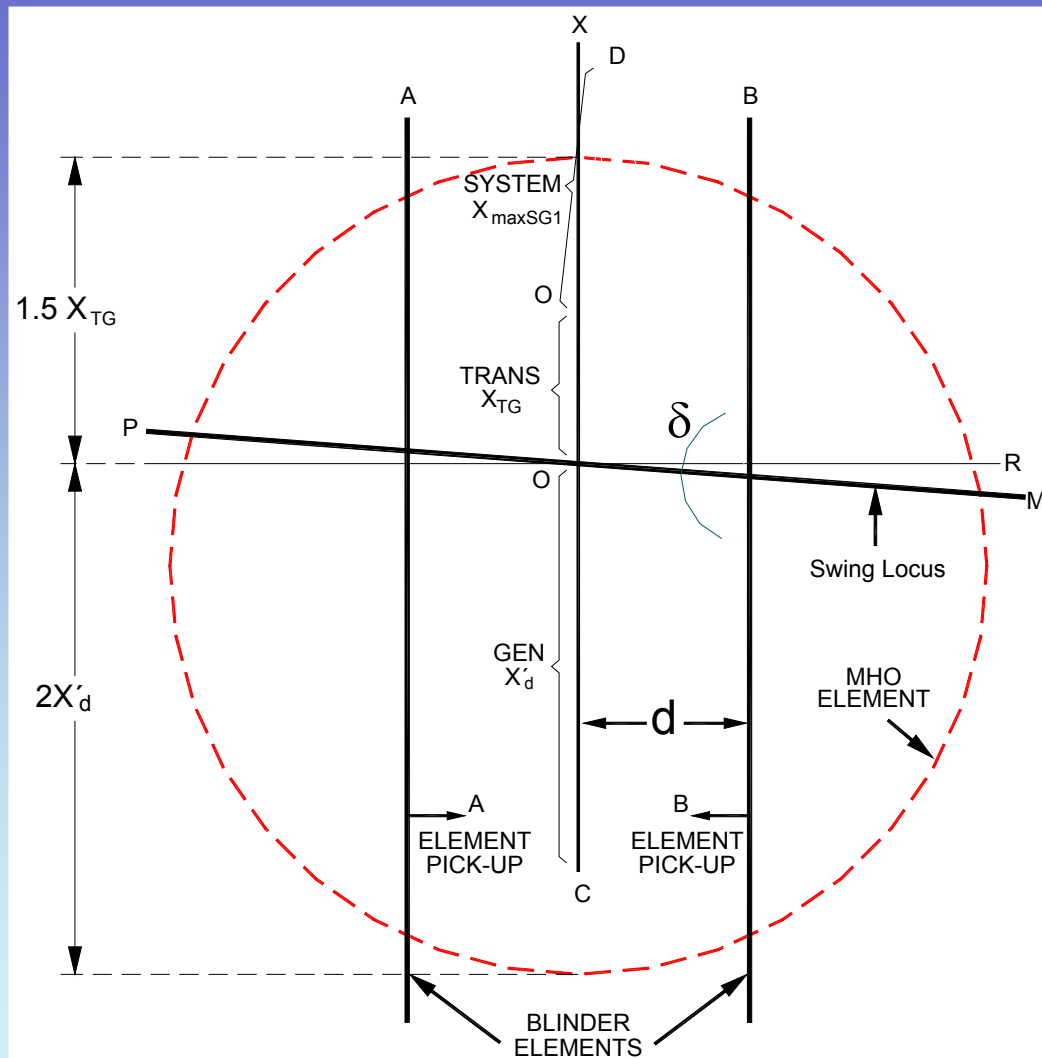
- ❖ The positive sequence impedance must originate outside either blinder A or B,
- ❖ It should swing through the pickup area and progress to the opposite blinder from where the swing had originated.
- ❖ The swing time should be greater than the time delay setting

When this scenario happens, the tripping circuit is complete. The contact will remain closed for the amount of time set by the seal-in timer delay.

Generator Out-of-Step Protection (OSP)



Setting of 78 Relays



Setting of 78 Relays

Settings summary per IEEE C37.102-2005

➤ Mho Diameter : $2X'_d + 1.5 X_{TG}$

➤ $d = ((X'_d + X_{TG} + X_{\max SG1})/2) \times \tan (90-(\delta/2))$

where d: Blinder distance

δ : angular separation between generator and the system which the relay determines instability.

If there is not stability study available $\delta = 120^\circ$

➤ t = as per transient stability study

typically $40 < t < 100$ ms

Frequency (81)

Frequency

- The operation of generators at abnormal frequencies (either overfrequency or underfrequency) generally results from full or partial load rejection or from overloading of the generator.
- Load rejection will cause the generator to overspeed and operate at some frequency above normal
- Steam and gas turbines are more limited or restrictive to abnormal frequency than hydrogenerators.
- At some point abnormal frequency may impact turbine blades and result in damage to the bearings due to vibration.

Frequency

Settings summary per IEEE C37.102

It is important to consult turbine manufacturer and get turbine off frequency operating curves or limits

Under frequency:

➤ 81U ALARM: 59.5 Hz time: 10 s

➤ 81U TRIP :

The generator 81U relay should be set below the pick-up of under frequency load shedding relay set-point and above the off frequency operating limits of steam turbine.

Over frequency:

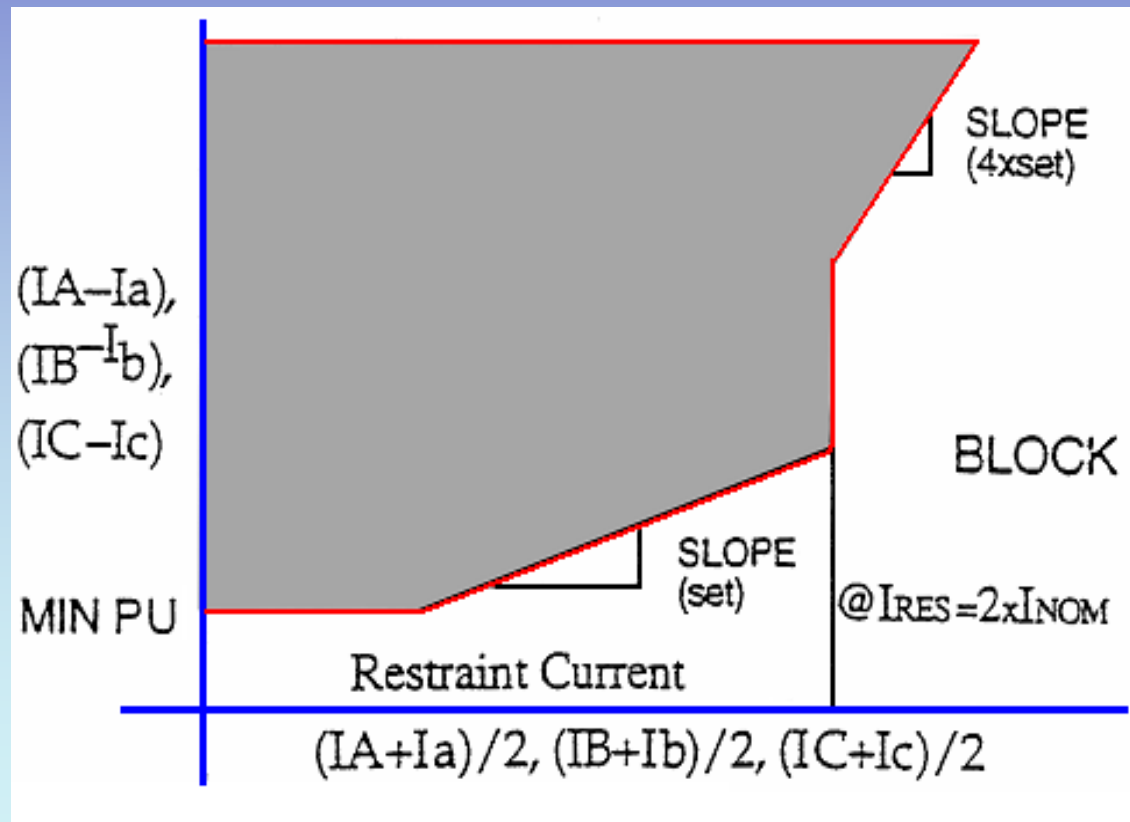
➤ 81O ALARM

Pick-up: 60.6 Hz, Time Delay 5 sec.

Phase Differential (87)

Phase Differential

- ❖ Fast response time (under 1 – ½ cycle)
- ❖ Percentage differential with adjustable slope



Phase Differential

Settings summary per IEEE C37.102

- PU : 0.3 A
- Slope1 : 10%
- time: Instantaneous

Typical Settings of Generator Relays

Table 1 - Recommended Settings

IEEE No.	FUNCTION	Per IEEE C37.102	
		SECTION	DESCRIPTION
21	Distance	A.2.3	Zone-1 = smaller of the two following criteria: 1. 120% of unit transformer 2. 80% of Zone 1 reach setting of the line relay on the shortest line (neglecting in-feed); time = 0.5 s Zone-2 = the smaller of the three following criteria: A. 120% of longest line (with in-feed). If the unit is connected to a breaker and a half bus, this would be the length of the adjacent line. B. 50% to 66.7% of load impedance (200% to 150% of the generator capability curve) at the RPFA C. 80% to 90% of load impedance (125% to 111% of the generator capability curve) at the maximum torque angle; time > 60 cycles Zone-2 < 2Z maxload @ RPF
24	Overexcitation	4.5.4.2	Single relay: PU = 110% p.u. time = 6 s Two stages relay: alarm pu = 110%; 45 < t < 60 s trip pu = 118% - 120%, 2 < t < 6s
25	Sync-check	5.7	Breaker closing angle: within ± 10 elect. Degrees Voltage matching: 0 to +5% Frequency difference < 0.067 Hz
27	Undervoltage	A.2.13	<u>Relays with inverse time charac and instantaneous</u> PU : 90%Vn; t= 9.0 s at 90% of PU setting Inst : 80% Vn <u>Relays with definite time charac and 2 stages</u> Alarm PU : 90%Vn; 10 < t < 15 s Trip PU : 80% Vn; time: 2s

Typical Settings of Generator Relays

Table 1 - Recommended Settings

IEEE No.	FUNCTION	Per IEEE C37.102	
		SECTION	DESCRIPTION
32	Reverse Power	4.5.5.3 & A.2.9	Pickup setting should be below the following motoring limits: Gas : 50% rated power; time < 60 s Diesel : 25% rated power; time < 60 s Hydro turbines : 0.2% - 2% rated power; time < 60 s Steam turbines : 0.5% - 3% rated power; time < 30 s
40	Loss-of-field	4.5.1.3	UNIT 1 Offset: $X'd/2$; Diameter: 1.0 pu; time: 0.1 s UNIT 2 Offset: $X'd/2$; Diameter: X_d ; time: 0.5 to 0.6 s
46	Negative Sequence Overcurrent	4.5.2	<u>Pickup setting should be below the permissible I_2 percent expressed in percent of rated current, which are indicated below:</u> Salient pole w/connected amortisseur windings: 10% Salient pole non-connected amortisseur windings: 5% Cylindrical rotor indirectly cooled: 10% Directly cooled up to 960 MVA: 8% Directly cooled 961 to 1200 MVA: 6% Directly cooled 961 to 1200 MVA: 6% Directly cooled 1201 to 1500 MVA: 5% <u>Permissible K ($I_{22} \times t$)</u> Salient pole generator: 40 Synchronous condenser: 30 Cylindrical rotor indirectly cooled: 30 Directly cooled: 10

Typical Settings of Generator Relays

Table 1 - Recommended Settings

IEEE No.	FUNCTION	Per IEEE C37.102	
		SECTION	DESCRIPTION
50/87	Differential via flux summation CTs or split-phase protection	4.3.2.5.1	The pickup of the instantaneous unit must be set above CT error currents that may occur during external faults.
50/27	Inadvertent Energization Overcurrent with 27, 81 Supervision	A.2.4	50: P.U $\leq$ 50% of the worst-case current value and should be $<$ 125% generator rated current. 27: 70% Vn, time: 1.5 s
50 BF	Generator Breaker Failure Protection	A.2.11	Current detector PU: should be more sensitive than the lowest current present during fault involving currents. Timer $>$ Gen breaker int time + Curr det. dropout time + safety margin
51N	Stator Ground Over-current (Low, Med Z Gnd, Phase CT Residual)		
50/51N	Stator Ground Over-current (Low, Med Z Gnd, Neutral CT or Flux Summation CT)		
51GN, 51N	Stator Ground Over-current (High Z Gnd)		

Typical Settings of Generator Relays

Table 1 - Recommended Settings

IEEE No.	FUNCTION	Per IEEE C37.102	
		SECTION	DESCRIPTION
50/51	Time overcurrent protection (against overloads)	4.1.1.2	51PU: 75-100% FLC, time: 7 s at 226% FLC. FLC means full load current. 50PU: 115% FLC, time: instantaneous
51VC	Voltage Controlled Overcurrent	A.2.6	Overcurrent PU: 50% FLC Control voltage: 75%VNOM. Inverse time curve and dial settings should be set to coordinate with system line relays for close-in faults on the transmission lines at the plant.
51VR	Voltage Restrained Overcurrent	A.2.6	Overcurrent PU: 150% FLC at rated voltage Inverse time curve and dial settings should be set to coordinate with system line relays for close-in faults on the transmission lines at the plant.
59	Overvoltage	4.5.6. & A.2.12	<u>Relays with inverse time charac and instantaneous</u> PU : 110%Vn; t= 2.5 s at 140% of PU setting Inst : 130 - 150% Vn <u>Relays with definite time charac and 2 stages</u> Alarm PU : 110%Vn; 10< t < 15 s Trip PU : 150% Vn; time: 2s
59N, 27-TH, 59P	100% Stator Ground protection (for high impedance grounding generators)	4.3.3.1.1 & A.2.7	59G element: Pickup = 5 V; t = 5 s Time setting must be selected to provide coordination with other system protective devices. 27TH element: Pickup = 50% of minimum normal generator 3rd harmonic, time = 5 s

Typical Settings of Generator Relays

Table 1 - Recommended Settings			
IEEE No.	FUNCTION	Per IEEE C37.102	
		SECTION	DESCRIPTION
64F	Generator Rotor Field protection (rotor ground faults)	4.4	<u>Field ground detection using DC a source:</u> $1 < t < 3$ s <u>Field ground detection for Brushless Machines with infrared LED communications:</u> time up to 10 s <u>Field ground detection using low frequency square wave voltage injection:</u> ALARM = 20 kOhm TRIP = 5 kOhm
67IE	Directional O/C for Inadvertent Energization		
78	Out of Step	A.2.2	Mho Diameter : $2X'd + 1.5 \text{ XTG}$ Blinder distance $(d) = ((X'd + \text{XTG} + X_{\text{maxSG1}})/2) \times \tan(90-(d/2))$; d: angular separation between generator and the system which the relay determines instability. <u>If there is not stability study available</u> $d = 120^\circ$ $t =$ as per transient stability study Typically $40 < t < 100$ ms
81	Over/under frequency (60 Hz systems)	A.2.14	81U ALARM: 59.5 Hz time: 10 s 81U TRIP: The generator 81U relay should be set below the pick-up of underfrequency load shedding relay set-point and above the off frequency operating limits of steam turbine. 81O ALARM: Pick-up: 60.6 Hz, Time Delay 5 sec.

Typical Settings of Generator Relays

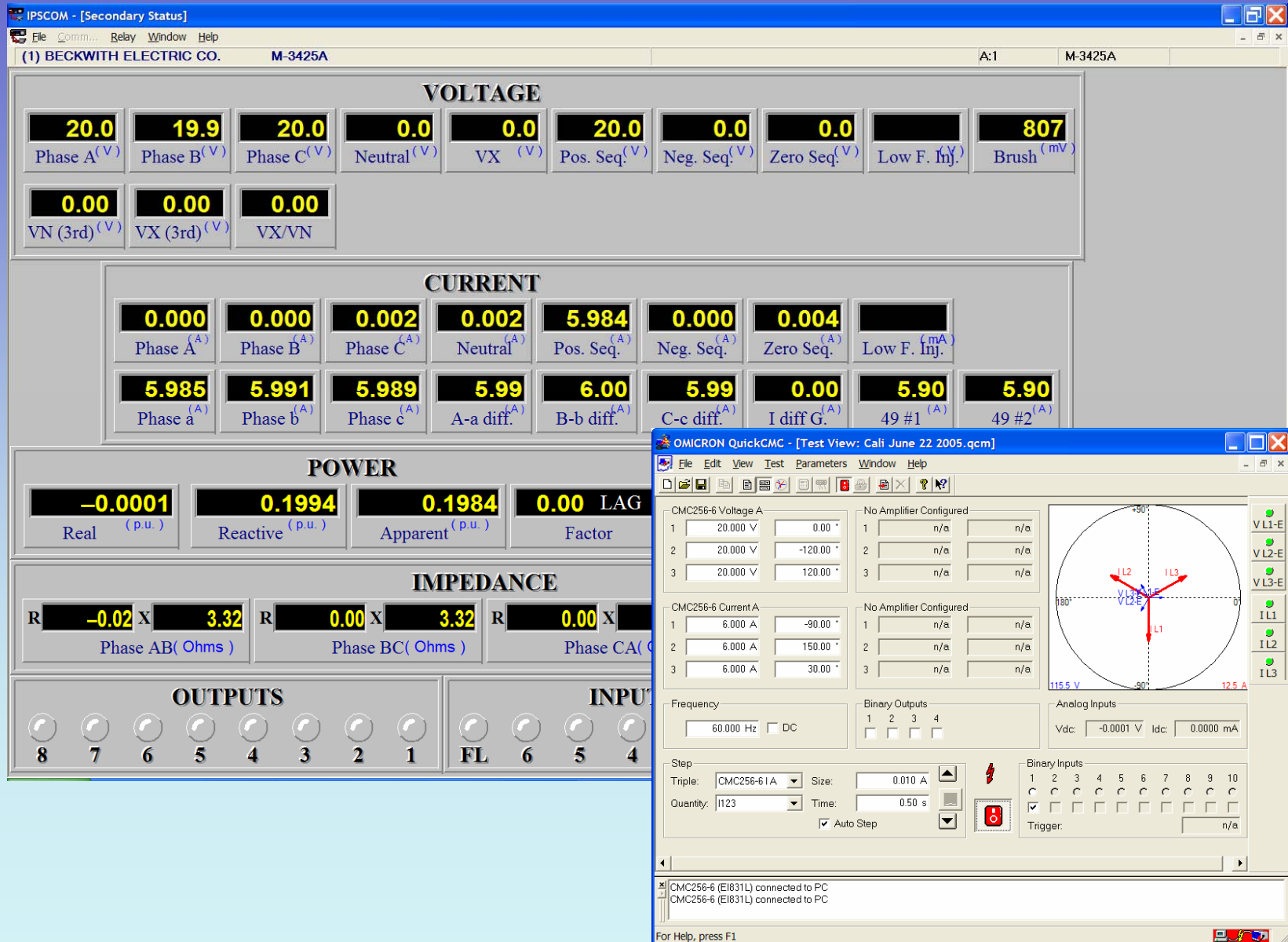
Table 1 - Recommended Settings

IEEE No.	FUNCTION	Per IEEE C37.102	
		SECTION	DESCRIPTION
87G	Generator Phase Differential	A.2.5	PU : 0.3 A Slope : 10% time: instantaneous
87GN	Generator Ground Differential		
87UD	Unit Differential		

Types Of Data

- Metering
- Function Status
- Breaker Monitoring
- Fault Reporting
- Oscillography
- Testing

Metering



Function Status

IPSCOM

File Comm... Relay Window Help

(1) BECKWITH ELECTRIC CO. M-3425A A:1 M-3425A

Function Status

P	T	P: Pickup	T: Tripped	T	P
☐	☐	(21) #1 Phase Distance		☐	☐
☐	☐	(21) #2 Phase Distance		☐	☐
☐	☐	(21) #3 Phase Distance		☐	☐
☐	☐	(24DT) #1 Volts/Hz DEF		☐	☐
☐	☐	(24DT) #2 Volts/Hz DEF		☐	☐
☐	☐	(24T) Volts/Hz INV		☐	☐
☐	☐	(27) #1 Phase Undervoltage		☐	☐
☐	☐	(27) #2 Phase Undervoltage		☐	☐
☐	☐	(27) #3 Phase Undervoltage		☐	☐
☐	☐	(25S) Sync Check		☐	☐
☐	☐	(25D) Dead Check		☐	☐
☐	☐	(27TN) #1 Third Harmonic Undervoltage		☐	☐
☐	☐	(27TN) #2 Third Harmonic Undervoltage		☐	☐
☐	☐	(32) #1 Directional Power		☐	☐
☐	☐	(32) #2 Directional Power		☐	☐
☐	☐	(32) #3 Directional Power		☐	☐
☐	☐	(40) #1 Loss Of Field		☐	☐
☐	☐	(40) #2 Loss Of Field		☐	☐
☐	☐	(40VC) #1 Loss Of Field with VC		☐	☐
☐	☐	(40VC) #2 Loss Of Field with VC		☐	☐
☐	☐	(46DT) Neg. Seq. Overcurrent DEF		☐	☐
☐	☐	(48T) Neg. Seq. Overcurrent INV		☐	☐
☐	☐	(49) #1 Stator Overload		☐	☐
☐	☐	(49) #2 Stator Overload		☐	☐
☐	☐	(50BF) Breaker Failure		☐	☐
☐	☐	(50) #1 Phase Overcurrent		☐	☐
☐	☐	(50) #2 Phase Overcurrent		☐	☐
☐	☐	(50N) Instant. Neutral Overcurrent		☐	☐
☐	☐	(50DT) #1 Time Overcurrent		☐	☐
☐	☐	(50DT) #2 Time Overcurrent		☐	☐
☐	☐	(50/27) Inadvertent Energizing		☐	☐
☐	☐	(51N) Inv. Time Neu. Overcurrent		☐	☐
☐	☐	(51V) Inv. Time Phase Overcurrent		☐	☐
☐	☐	(59) #1 Phase Overvoltage		☐	☐
☐	☐	(59) #2 Phase Overvoltage		☐	☐
☐	☐	(59) #3 Phase Overvoltage		☐	☐
☐	☐	(59D) Third Harmonic Voltage Differential		☐	☐
			(59N) #1 Neutral Overvoltage	☐	☐
			(59N) #2 Neutral Overvoltage	☐	☐
			(59N) #3 Neutral Overvoltage	☐	☐
			(59X) #1 Multi-purpose OverVoltage	☐	☐
			(59X) #2 Multi-purpose OverVoltage	☐	☐
			(60FL) VT Fuse Loss	☐	☐
			(64F) #1 Field Ground	☐	☐
			(64F) #2 Field Ground	☐	☐
			(64B) Brush Lift Off	☐	☐
			(64S) 100% Stator Ground	☐	☐
			(67NDT) Residual Definite Overcurrent	☐	☐
			(67NIT) Residual Inverse Overcurrent	☐	☐
			(78) Out of Step	☐	☐
			(81) #1 Frequency	☐	☐
			(81) #2 Frequency	☐	☐
			(81) #3 Frequency	☐	☐
			(81) #4 Frequency	☐	☐
			(81A) #1 Freq. Accumulator	☐	☐
			(81A) #2 Freq. Accumulator	☐	☐
			(81A) #3 Freq. Accumulator	☐	☐
			(81A) #4 Freq. Accumulator	☐	☐
			(81A) #5 Freq. Accumulator	☐	☐
			(81A) #6 Freq. Accumulator	☐	☐
			(81R) #1 Rate of Change of Frequency	☐	☐
			(81R) #2 Rate of Change of Frequency	☐	☐
			(87) #1 Phase Differential Current	☐	☐
			(87) #2 Phase Differential Current	☐	☐
			(87GD) Ground Differential Current	☐	☐
			(IPSL) #1 IPS LOGIC	☐	☐
			(IPSL) #2 IPS LOGIC	☐	☐
			(IPSL) #3 IPS LOGIC	☐	☐
			(IPSL) #4 IPS LOGIC	☐	☐
			(IPSL) #5 IPS LOGIC	☐	☐
			(IPSL) #6 IPS LOGIC	☐	☐
			(BM) Breaker Monitor	☐	☐
			(TC) Trip Circuit Monitor	☐	☐

OUTPUT 1 ☐

OUTPUT 2 ☐

OUTPUT 3 ☐

OUTPUT 4 ☐

OUTPUT 5 ☐

OUTPUT 6 ☐

OUTPUT 7 ☐

OUTPUT 8 ☐

INPUT 1 ☐

INPUT 2 ☐

INPUT 3 ☐

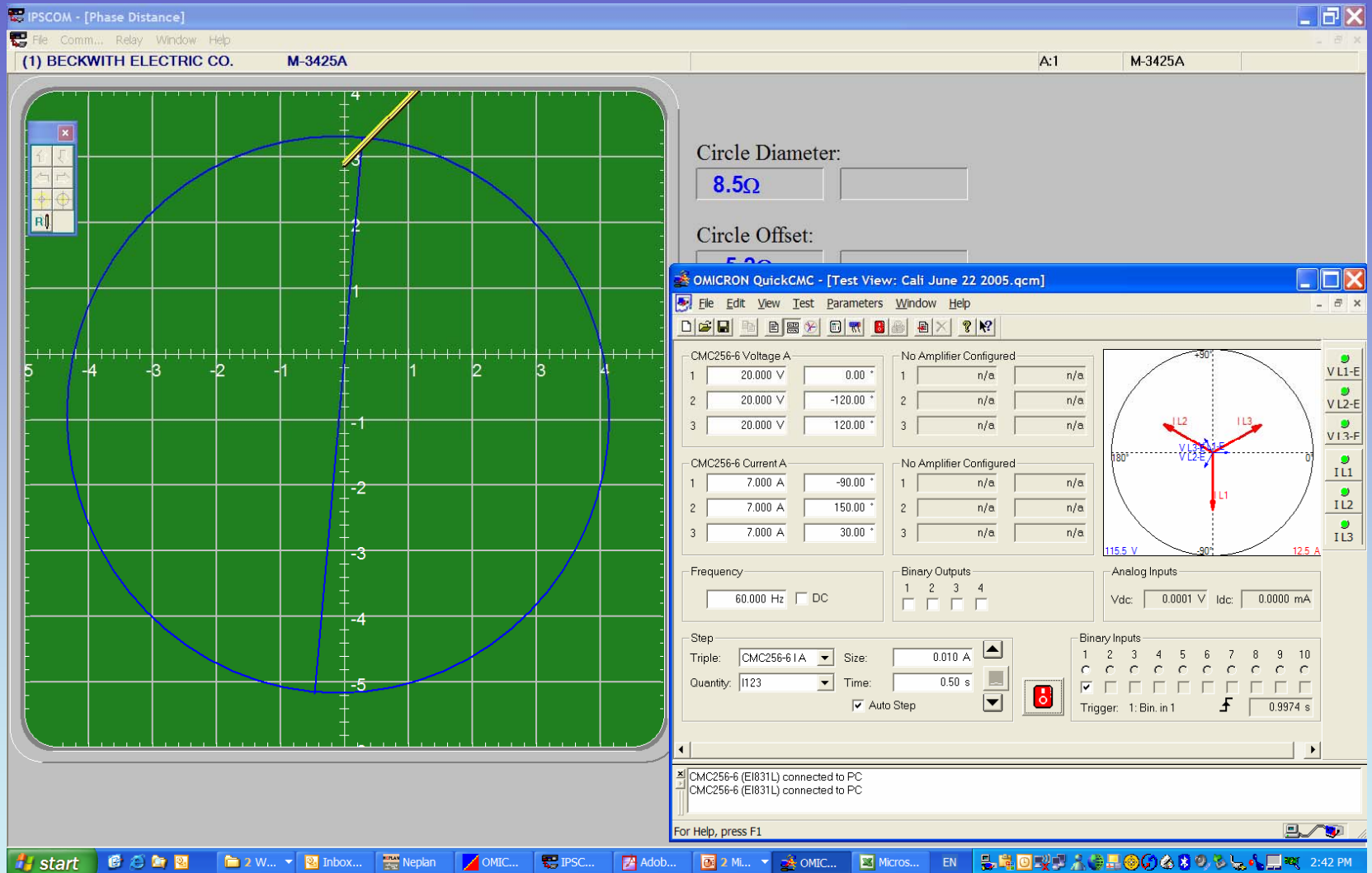
INPUT 4 ☐

INPUT 5 ☐

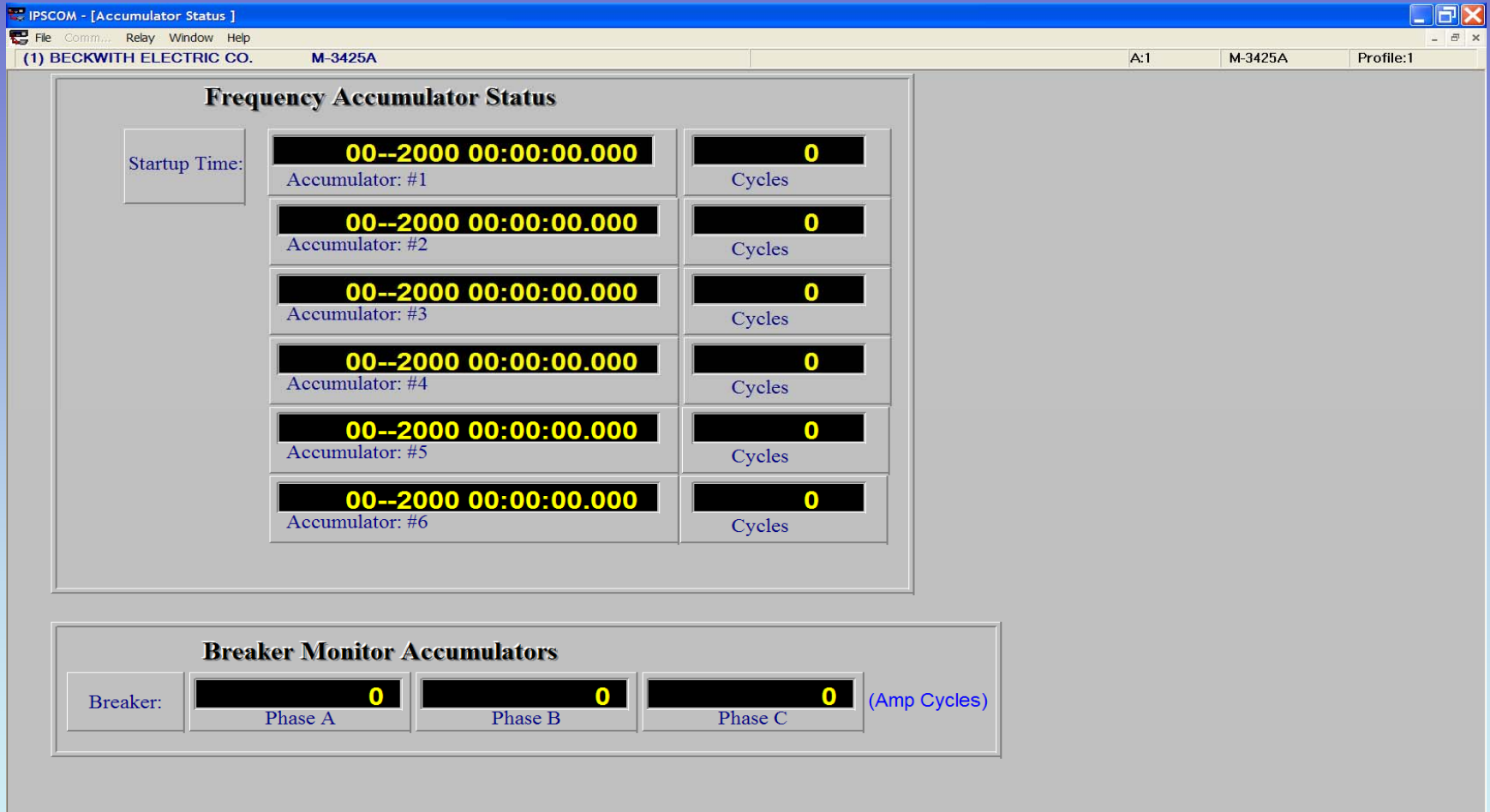
INPUT 6 ☐

INPUT FL ☐

Phase Distance Monitor



Breaker Monitoring



Fault Reporting

The screenshot displays the IPSCOM software interface. The main window title is "IPSCOM". The menu bar includes "File", "Comm...", "Relay", "Window", and "Help". The main display area shows "(1) BECKWITH ELECTRIC CO." and "M-3425A". The right side of the main display area shows "A:1" and "M-3425A".

A "Target List" dialog box is open, showing a list of targets with columns for "Function #" and "Status". The list includes:

Function #	Status
(21) Phase Distance #1 - Phase AB	Picked Up
(21) Phase Distance #1 - Phase BC	Operated
(21) Phase Distance #1 - Phase CA	Operated

The "Target #1" dialog box is also open, showing configuration details for the selected target. It includes sections for "Inputs", "Outputs", "Currents", and "Function #".

Inputs: FL, 6, 5, 4, 3, 2, 1

Outputs: 8, 7, 6, 5, 4, 3, 2, 1 (1 is selected)

Currents:

Phase a	Phase b	Phase c	Phase N
7.00 A	7.01 A	6.98 A	0.00 A

Function #

Function #	Status
(21) Phase Distance #1 - Phase AB	Picked Up
(21) Phase Distance #1 - Phase BC	Operated
(21) Phase Distance #1 - Phase CA	Operated

The "Target List" dialog box has a "Comment" field and "Print", "Save", and "Close" buttons.

The Windows taskbar at the bottom shows the "start" button and several open applications: "2 W...", "Inbox...", "Neplan", "OMIC...", "IPSCOM", "Adob...", "2 ML...", "OMIC...", "Micros...", and "EN". The system clock shows "2:48 PM".

Fault Reporting

Target List			
01.	25-Jun-2005	14:43:59.635	^
02.	25-Jun-2005	14:42:59.619	
03.	25-Jun-2005	14:42:54.019	
04.	25-Jun-2005	14:26:04.502	
05.	25-Jun-2005	14:24:27.869	
06.	25-Jun-2005	14:24:10.652	
07.	25-Jun-2005	14:23:57.152	
08.	25-Jun-2005	14:21:56.302	
09.	25-Jun-2005	14:20:16.835	
10.	25-Jun-2005	14:15:55.352	
11.	25-Jun-2005	14:14:25.302	
12.	25-Jun-2005	14:13:07.719	
13.	22-Jun-2005	10:18:21.102	
14.	19-Jun-2005	13:11:40.974	
15.	19-Jun-2005	13:11:22.857	
16.	19-Jun-2005	12:45:31.835	
17.	18-Jun-2005	13:41:33.669	
18.	13-Apr-2005	10:37:41.524	
19.	13-Apr-2005	10:30:52.707	
20.	13-Apr-2005	10:27:45.624	
21.	07-Apr-2005	17:02:52.724	
22.	26-Feb-2004	14:05:80.000	
23.	26-Feb-2004	14:05:80.000	v
Comment			Print
Save			Close

Fault Reporting

Target #1

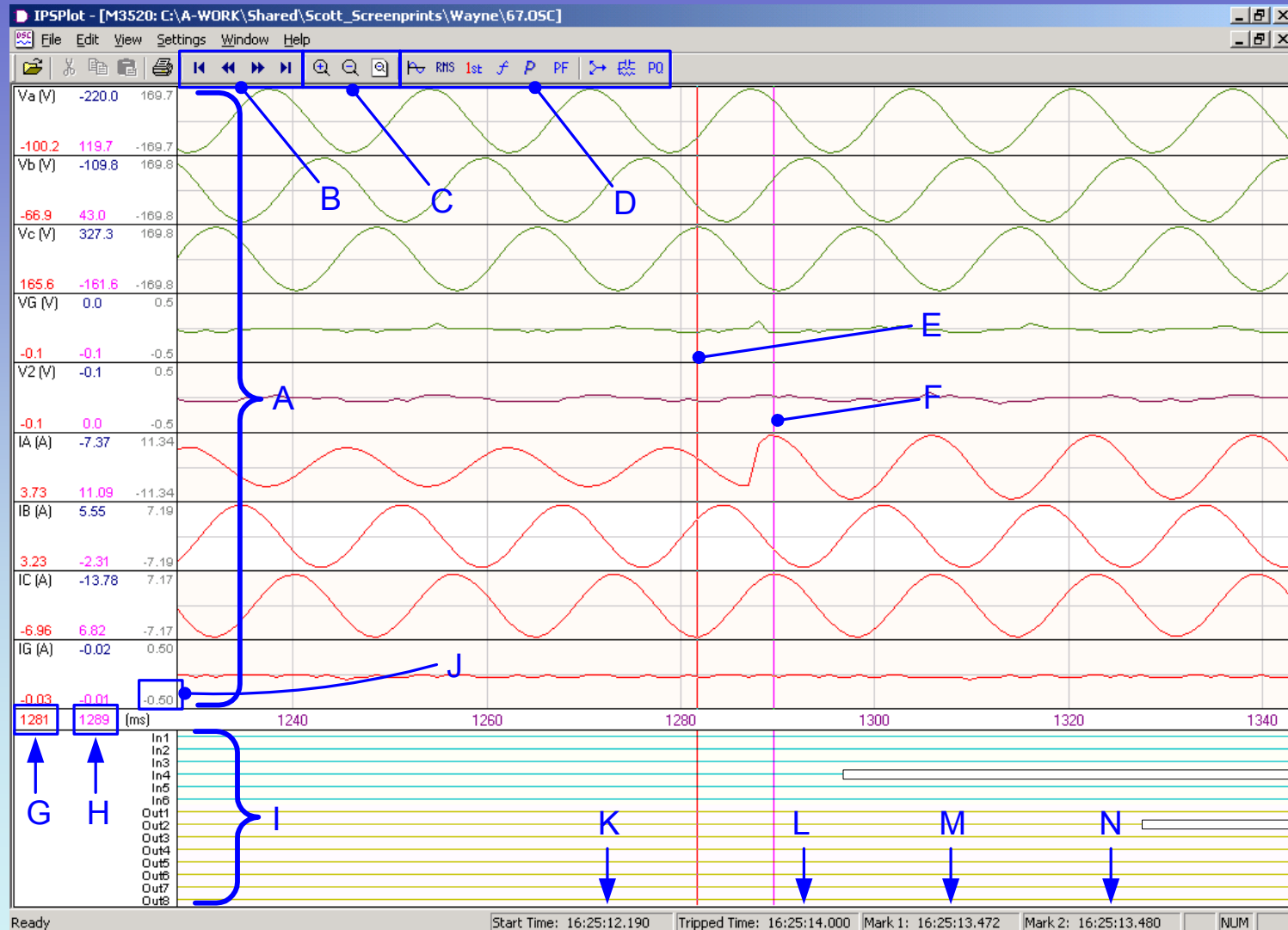
Inputs: ☐ FL ☐ 6 ☐ 5 ☐ 4 ☐ 3 ☐ 2 ☐ 1

Outputs: ☐ 8 ☐ 7 ☐ 6 ☐ 5 ☐ 4 ☐ 3 ☐ 2 ☒ 1

	Phase a	Phase b	Phase c	Phase N
Currents:	7.00 A	7.01 A	6.98 A	0.00 A

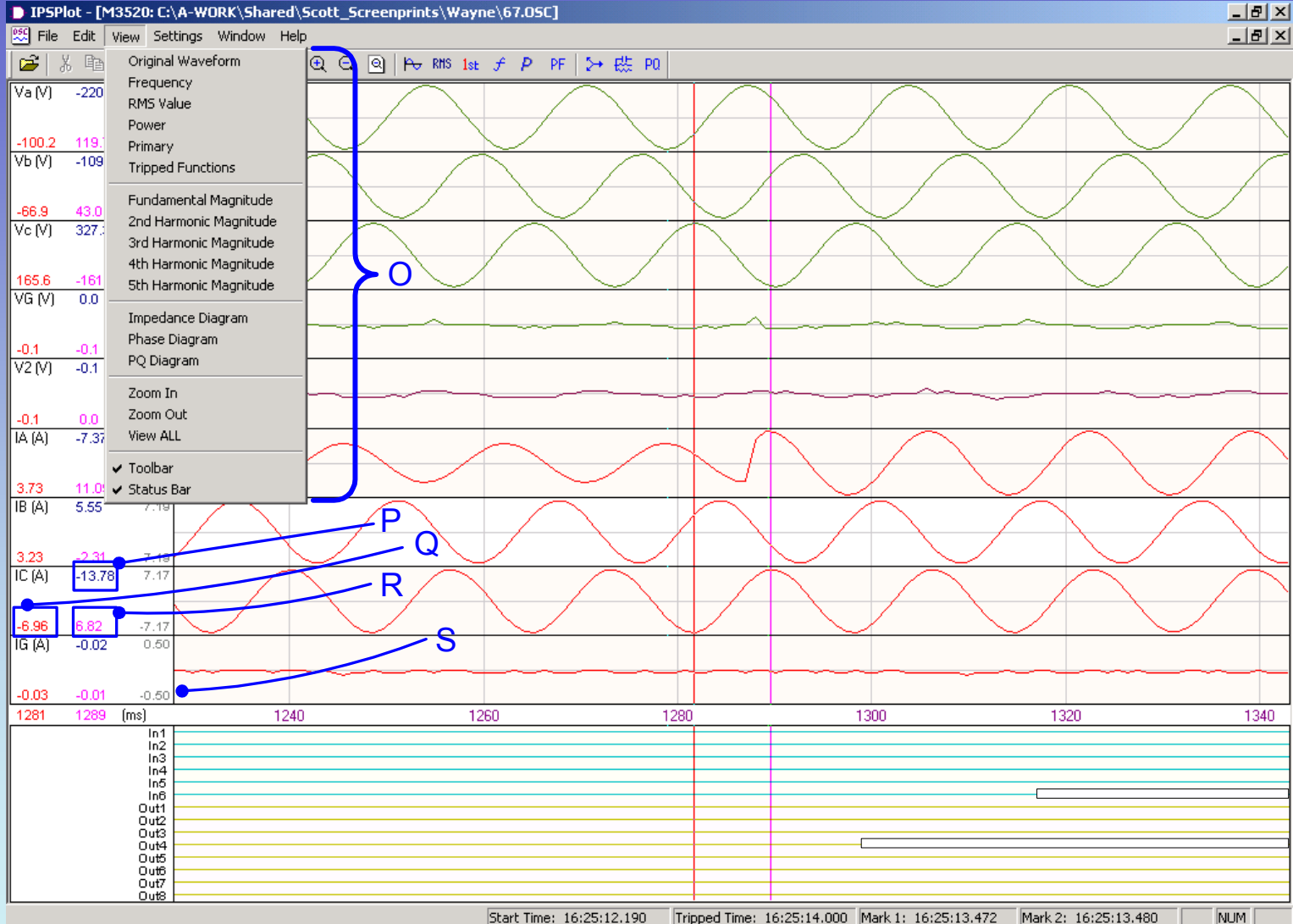
Function #	Status
(21) Phase Distance #1 - Phase AB	Picked Up
(21) Phase Distance #1 - Phase BC	Operated
(21) Phase Distance #1 - Phase CA	Operated

Oscillography



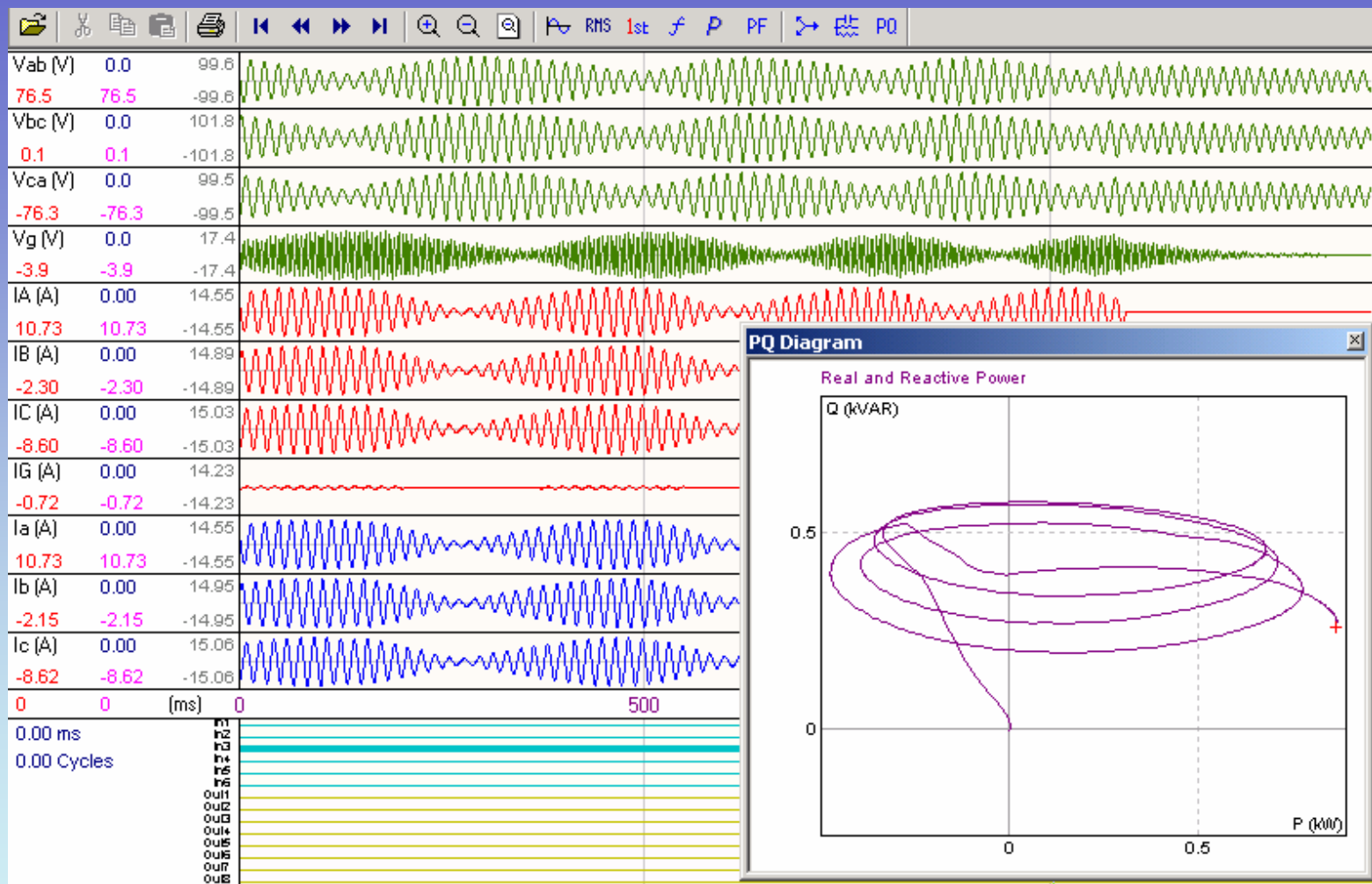
- A. All analog traces. This view shows peak values. RMS values may also be displayed.**
- B. Controls for going to the beginning or end of a record, as well as nudging forward or backward in time in a record**
- C. Zoom controls**
- D. Display controls for analog traces, RMS traces, fundamental waveform display, frequency trace, power trace, power factor trace, phasor diagram, impedance diagram and power diagram**
- E. Marker #1**
- F. Marker #2**
- G. Time at Marker #1**
- H. Time at Marker #2**
- I. Control status input and contact output traces (discrete I/O)**
- J. Scaling for each analog trace. This can be set automatically or manually adjusted.**
- K. Date and timestamp for record**
- L. Time of trip command**
- M. Time at Marker #1**
- N. Time at Marker #2**

Oscillography



- O. Drop down window for view selection, diagram selection and zoom**
- P. Delta value between Marker #1 and Marker #2**
- Q. Value at Marker #1**
- R. Value at Marker #2**
- S. Scaling for each analog trace. This can be set automatically or manually adjusted.**

Waveform Capture: PQ Plot



Ready

Date: 05/23/2001

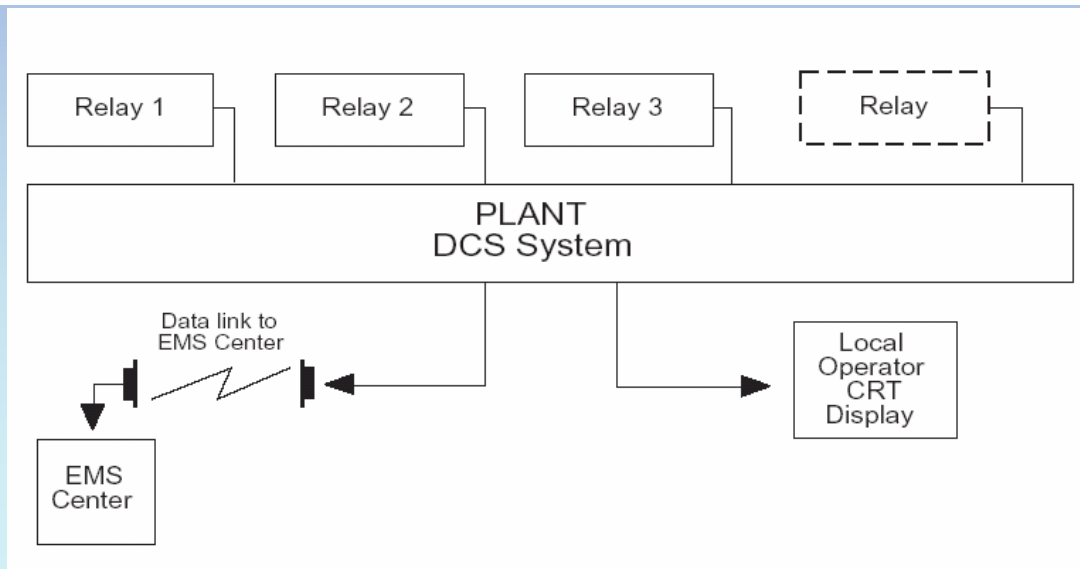
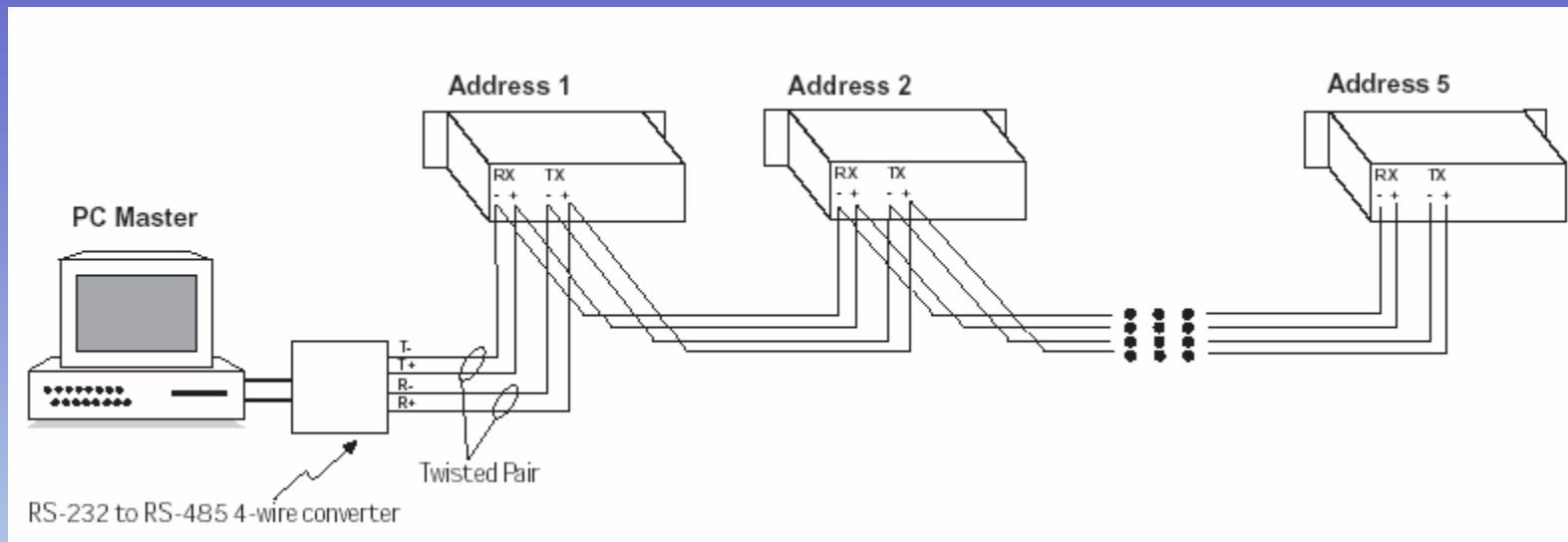
Start Time: 14:37:19.993

Tripped Time: 14:37:21.000

Mark 1: 14:37:19.993

Mark 2: 14:

Communications



Test Report

GERS CONSULTING ENGINEERS PROJECT : Meter and relay test at Spalding Energy Project MANUFACTURER : BECKWITH TYPE: M-3425	BECHTEL LIMITED			DATE FEBRUARY 26 / 2004
	TEST REPORT			TESTED BY: R. Bravo - C. Quintero
	GENERATOR PROTECTION			APPROVED BY: A.Tasama - G. Williams
	PANEL TAG: GPR	LOCATION : STG ELECT BUILDING	SERIAL NUMBER : 1815	CIRCUIT : STG PROT. A SYSTEM: AC01

1. GENERAL SETTINGS

Parameter	Value	Parameter	Value
Nominal Voltage [V]	120	V.T. Configuration	L-G to L-L
Nominal Current [A]	3.98	Relay Seal-in Time [Cycles]	300
Nominal Frequency [Hz]	50	V.T. Phase Ratio	200
Phase Rotation	ABC	V.T. Neutral Ratio	100
C.T. Secondary Rating [A]	5	C.T. Phase Ratio	2600
Delta - Y Transformer	Enable	C.T. Neutral Ratio	25

2. READINGS CHECK

Description	Injected	Theoretical Value	Obtained Read	% Error
V RY [V]	120.0	24000	23960	-0.17%
V YB [V]	120.0	24000	23940	-0.25%
V BR [V]	120.0	24000	24020	0.08%
I R [A]	5.0	13000	13005	0.04%
I Y [A]	5.0	13000	13021	0.16%
I B [A]	5.0	13000	13013	0.10%
I r [A]	5.0	13000	13018	0.14%
I y [A]	5.0	13000	13013	0.10%
I b [A]	5.0	13000	13000	0.00%
Active Power [W/MW]	900.0	468.00	466.36	-0.35%
Reactive Power [VAr/MVAr]	519.6	270.20	275.21	1.85%
Power Factor	0.87	0.87	0.86	-1.15%
Frequency [Hz]	50.000	50.00	50.00	0.00%

Note: IR, IY, IB = line side currents / Ir, Iy, Ib = generator side currents

Test Report

16. FUNCTION 87. PHASE DIFFERENTIAL PROTECTION

16.1. Settings

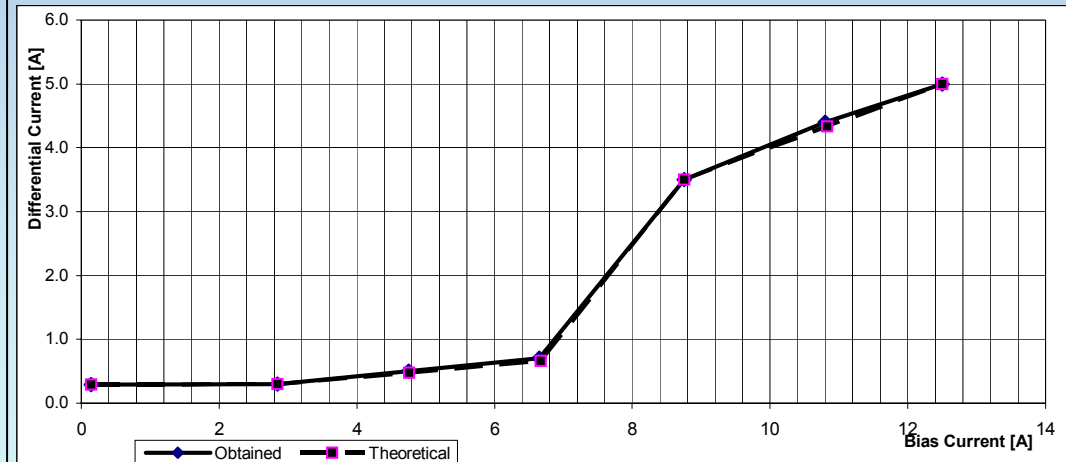
Parameter	Value		
Minimum Operation current [A]	0.3		
Slope	10%	Trip output	1
Time Delay [Cycles]	1	Blocking input	-

16.2 Function Test

Parameter	Theoretical Value	Result		% Error
Minimum current for operation [A]	0.30	IR	0.29	3.33%
		IY	0.29	3.33%
		IB	0.29	3.33%
Slope 1	10.00%	10.53%		0.53%
Slope 2	40.00%	40.00%		0.00%
Operation Time [ms]	20.00	19.00		-5.00%

Differential Characteristic Test

Line current [A] - Fixed	IR	0.29	3.00	5.00	7.00	10.50	13.00	15.00
Theoretical Values	I _r	0.00	2.70	4.52	6.33	7.00	8.67	10.00
	I _{diff} = (I _R -I _r)	0.29	0.30	0.48	0.67	3.50	4.33	5.00
	I _{bias} = (I _R +I _r)/2	0.15	2.85	4.76	6.67	8.75	10.83	12.50
Obtained values	I _r	0.00	2.70	4.50	6.30	7.00	8.60	10.00
	I _{diff} = (I _R -I _r)	0.29	0.30	0.50	0.70	3.50	4.40	5.00
	I _{bias} = (I _R +I _r)/2	0.15	2.85	4.75	6.65	8.75	10.80	12.50



Test Report

3. FUNCTION 21. DISTANCE PROTECTION

3.1. Settings

Parameter	Value		
Diameter [Ohms]	8.50		
Offset [Ohms]	-5.2		
Impedance Angle [Degrees]	85	Trip output	1
Time delay [cycles]	50	Blocking input	1 & FL

3.2 Function Test

Parameter		Theoretical Value	Result	% Error
Voltage [V LN]	Fixed	20	-	-
Current [A]	Varied	6.06	5.99	1.17%
Impedance [Ohms]	Calculated	3.30	3.34	1.18%
Operation time [s]		1.00	1.01	0.50%

Questions?

jmgers@gersusa.com